Using Polar Plots Of Velocity and Acceleration Data

We have been discussing balancing using shaft relative, displacement vibration data. However, on many small machines, the only available transducer is a casing-mounted velocity or acceleration transducer. Often these small machines have rolling element bearings with good transmissibility to the casing, and the machine bearing caps move more or less in phase with the rotor at that location.

The rotor model we developed was based on displacement, and the high spot/heavy spot relationships are *displacement* relationships. The behavior of the model is different for velocity and acceleration.

The rotating, synchronous rotor response vector is

$$\boldsymbol{r} = Ae^{j(\Omega t+\alpha)} \tag{16-5}$$

where A is the amplitude, Ω is the rotor speed, t is time, and α is the phase of the response. We find the velocity, $\dot{\boldsymbol{r}}$, and acceleration, $\ddot{\boldsymbol{r}}$, by differentiating the displacement with respect to time,

$$\begin{aligned} \dot{\boldsymbol{r}} &= j\Omega Ae^{j(\Omega t+\alpha)} \\ \ddot{\boldsymbol{r}} &= -\Omega^2 Ae^{j(\Omega t+\alpha)} \end{aligned} \tag{16-6}$$

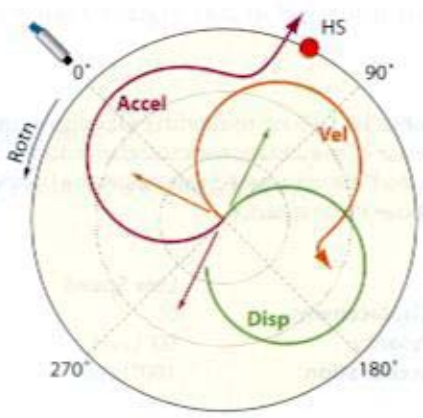


Figure 16-4. Comparison of displacement (green), velocity (orange), and acceleration (violet) data on a polar plot. Data shown is based on the rotor model of Chapter 10. The actual heavy spot orientation is shown as a red dot. Arrows show the low-speed phase direction for each measurement. The displacement data converges to a point (constant amplitude) at high speed, while velocity and acceleration plots increase in amplitude. Note that the direction of the acceleration vector at high speed is the same as the heavy spot and the opposite of the displacement vector.

The j at the beginning of the velocity term tells us that the phase of the velocity is 90° ahead of (leads) the displacement. The minus sign in the acceleration, the result of j times j, tells us that the phase of acceleration is 90° ahead of the velocity, or 180° ahead of the displacement. Thus, *the phase relationships we have used to locate the heavy spot on a displacement vibration polar plot do not apply for velocity and acceleration.* All of the relationships are shifted by 90° and 180° for velocity and acceleration, respectively. Note that integrating velocity data to displacement shifts the phase 90° in a *lag* direction, making integrated velocity equivalent to displacement.

Figure 16-4 shows polar plots of displacement, velocity, and acceleration, based on the rotor model of Chapter 10, and the direction of their low-speed vibration vectors. The actual heavy spot orientation is shown as a red dot. *When using velocity or acceleration data to locate the heavy spot, you must allow for the phase shifts shown in these plots.* These changes are summarized in Table 16-1.

Selecting the Calibration Weight

If we have polar plots, we will place a balance weight *opposite* the heavy spot indicated on the plot. If we have only steady state data, and no previous information about the machine, then we will have to guess where to put it. If we have the balance history of the machine, including the influence vectors (see below), then we can directly calculate the balance solution in one step.

The calibration weight, also called a trial weight, will show us how the rotor responds to unbalance at this speed, allowing us to "calibrate" the machine's

Table 16-1. Phase relationships for displacement, velocity, and acceleration. The table lists the phase of the vibration vector relative to the heavy spot for three operating conditions. "Low Speed" means a speed well below the balance resonance; "High Speed" means a speed well above the resonance.

	Low Speed	Resonance	High Speed
Displacement	0°	90° Lag	180° Lag
Velocity	90° Lead	0°	90° Lag
Acceleration	180° Lead	90° Lead	0°

response. We will use that information to calculate exactly how much weight to add and where to put it to balance the machine.

But, when little is known about a machine's response, how do you know how much weight to add? If you add too much weight, it could damage the machine; if too little, then the change in vibration might be too small to detect. A good, conservative rule is to install an amount of weight that will produce a force equal to no more than 10% of the rotor weight, *at the highest anticipated operating speed*:

$$F = mr_u\Omega^2 = 0.10W \tag{16-7}$$

where W is the total rotor *weight* (N or lb) produced by the rotor mass: $W = Mg$, where M is the rotor mass (kg or lb · s^2/in), and g is the acceleration of gravity. We can solve Equation 16-7 for the mass of the balance weight,

$$m = \frac{0.10W}{r_u\Omega^2} = \frac{0.10Mg}{r_u\Omega^2} \tag{16-8}$$

Example 2
Given a rotor mass of 1000 kg (2200 lb) that turns at 3000 rpm, find the calibration weight mass, in grams and ounces, that satisfies the 10% rule. The weight will be installed at a radius of 250 mm (9.8 in).

Solution
Because the rotor weight is given as a mass, we use the right side of Equation 16-8,

$$m = \frac{0.10\left[(1000\text{ kg})(9.81\text{ m/s}^2)\right]}{(250 \times 10^{-3}\text{ m})\left[(3000\text{ rev/min})(1\text{ min}/60\text{ s})(2\pi\text{ rad/rev})\right]^2}$$

$$m = 40 \times 10^{-3}\text{ kg} = 40\text{ g}$$

Converting to ounces,

$$m = 40 \times 10^{-3}\text{ kg}(2.2\text{ lb/kg})(16\text{ oz/lb})$$
$$m = 1.4\text{ oz}$$

Example 3

Given a rotor weight of 1250 lb (550 kg) that turns at 7800 rpm, find the trial weight, in ounces, that satisfies the 10% rule. The trial weight will be installed at a radius of 6.5 in (170 mm).

Solution

The rotor weight is given in units of force, so we use the left side of Equation 16-8,

$$m = \frac{0.10(1250 \text{ lb})}{(6.5 \text{ in})\left[(7800 \text{ rev/min})(1 \text{ min}/60 \text{ s})(2\pi \text{ rad/rev})\right]^2}$$

$$m = 28.9 \times 10^{-6} \frac{\text{lb} \cdot \text{s}^2}{\text{in}}$$

The units of the balance weight are U.S. customary *mass* units. We convert to force by multiplying by the acceleration of gravity ($F = mg$),

$$m = \left(28.9 \times 10^{-6} \frac{\text{lb} \cdot \text{s}^2}{\text{in}}\right)\left(386 \frac{\text{in}}{\text{s}^2}\right)\left(16 \frac{\text{oz}}{\text{lb}}\right)$$

$$m = 0.18 \text{ oz}$$

Converting to metric mass,

$$m = (0.18 \text{ oz})\left(\frac{1 \text{ lb}}{16 \text{ oz}}\right)\left(\frac{1000 \text{ g}}{2.2 \text{ lb}}\right)$$

$$m = 5.1 \text{ g}$$

Relating Balance Ring Location To Polar Plot Location

The next step is to locate the physical position on the balance ring that corresponds to the angular location on the polar plot where we wish to add weight. We must be able to tell the people who will actually install the weight exactly where on the rotor that weight must go.

This can be a challenging problem in the field because of the many different phase and angular measurement conventions that exist. Be sure you understand the correspondence (mapping) between the phase indications on a vector plot and any markings on the machine. Verify that the direction of rotation indicated on the plot matches the actual machine rotation, and that you are looking down the machine train in the correct direction.

On horizontal machines, the external reference direction is usually toward the top of the machine, "Up." But on vertical machines, the external reference direction can be any convenient direction, such as "North" or "East," or a significant feature of the machine or building. Be sure you have identified the correct external reference for the machine in question. Verify that the probe mounting orientations relative to this reference are as shown on the data plot.

On Bently Nevada polar plots, the phase lag angles start at 0° at the transducer location and increase in a direction opposite to rotation. Also, the plots are oriented so that the 0° mark is oriented at the same orientation on the paper as the actual physical transducer is on the machine. For example, if the transducer is located at 45° R from the physical reference, usually "Up," then the polar plot will show the 0° mark at 45° R from the top of the page. This provides the same perspective on the heavy spot location as an observer would have looking down the machine train.

On the rotor, however, things are different. Some rotors can have continuous circumferential slots where weights can be installed, moved to any desired angular position, and locked into place. Other rotors have only discrete holes available for weight addition. In either case, reference marks on machines will usually conform to a different measurement system unrelated to the vibration transducers. Degree marks may increase either with or against rotation, and the 0° mark on the shaft is unlikely to have any relationship to the installed transducers.

Often, rotors have discrete weight mounting holes. These holes (typically 30 or so) are equally spaced around the circumference of the rotor and are sequentially numbered; the numbers may increase in a direction with or opposite to rotation. The person balancing the machine must clearly communicate to the plant technicians how much weight to add and which balancing hole to install it in. To identify the correct hole, a *weight map* should be created that illustrates

the relationship between the instrumentation phase measurements and the physical hole numbers on the machine.

The Keyphasor notch and Keyphasor probe provide the link between the physical shaft and the polar plot. To establish this link, we must identify which balancing hole is aligned with the Keyphasor notch. Usually, there will be a machine drawing at the plant that shows balance hole orientations relative to some physical reference on the machine. The Keyphasor notch angular location must be established relative to that physical reference. Once this is done, the weight map can be completed.

Figure 16-5 shows an example of a typical weight map. In this example, rotation is X to Y when viewed from the driver looking toward the driven machine, and the vibration transducer is mounted at 45° L from the physical reference, in this case, "Up." The transducer phase lag angles are indicated on the outer perimeter. This balance plane, or *weight plane,* has 40 available holes, typical of the turbine and compressor balance planes of a Westinghouse 501 gas turbine. *The hole numbers on the map correspond to the hole number markings on the machine, and the holes are shown at the orientation when the Keyphasor notch triggers the Keyphasor probe* (bottom). The access port for installing weights is also shown where it exists on the machine. The balance weight radius is clearly indicated, and all installed weights are shown, with existing weights in brown and new (for the current balancing attempt) weights in red. (Any convenient symbols can be used for this purpose.) The table, right, shows the amount of weight that is installed in each hole.

Once the weight map is completed, the weight location that is indicated by the polar plot can be easily converted to a hole number that can be used by plant personnel. If at all possible, you should observe the weight installation process to confirm that things are happening the way you think they are.

Single Plane Balancing With Calibration Weights

In the following discussion, we will discuss balancing in a single plane using data from a single vibration transducer. If XY transducers are available, a good cross-check is to perform the calculations using data from both transducers. We will also assume that the balancing speed is the normal operating speed of the machine.

The calibration weight technique can be performed graphically or numerically. *We highly recommend that both methods be used.* At every step in the process, calculations should be checked for reasonableness. When using a calculator, data entry blunders or procedural errors can produce numbers that are incorrect and could even result in damage to the machine. Sometimes, it can be difficult to tell if a number is reasonable, and when the graphical solution is

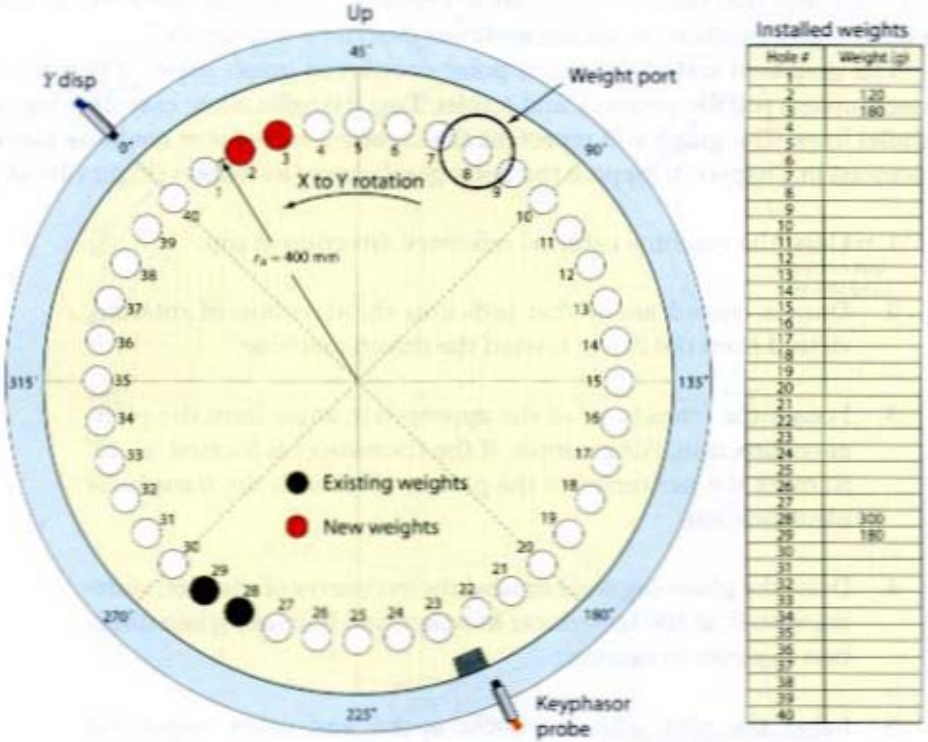


Figure 16-5. A typical weight map. This example shows 40 available balance hole locations and existing (brown) and new (red) balance weights. The holes are numbered to match the markings on the rotor when the Keyphasor probe and notch leading edge are aligned. The table, right, shows the amount of weight that is installed in each hole. Additional information includes the balance hole radius; direction of shaft rotation, viewed from the driver end of the machine; the vibration transducer location, 45° L; the physical reference, "Up;" the phase lag angles for this transducer; the Keyphasor probe location; and the location of the weight access port.

done in parallel, it is often much easier to interpret and set a level of confidence in the results.

Balancing calculations involve manipulation of vibration vectors, which are complex numbers. For this reason, numerical techniques are most easily done on a calculator that has complex number capability. Vector calculations can also be done algebraically by using the methods described in Chapter 3.

The graphical method requires polar coordinate graph paper, a pencil, an eraser (unless you are perfect), and a ruler. Two triangles allow easy drawing of parallel lines. The graph will represent the complex transducer response plane discussed in Chapter 3. Prepare the polar graph paper as follows (Figure 16-6):

1. Label the machine external reference direction at top.

2. Draw a curved arrow that indicates the direction of rotation, viewed from the driver toward the driven machine.

3. Locate the transducer at the appropriate angle from the reference direction. For example, if the transducer is located at 45° R, mark the perimeter of the plot at 45° R with the transducer identification.

4. Draw the phase lag scale around the perimeter of the plot, starting with 0° at the transducer location, and increasing in a direction *opposite* to rotation.

5. Label the plot with the rotor speed and other important machine variables, such as temperature, load, etc., that might affect the repeatability of the data.

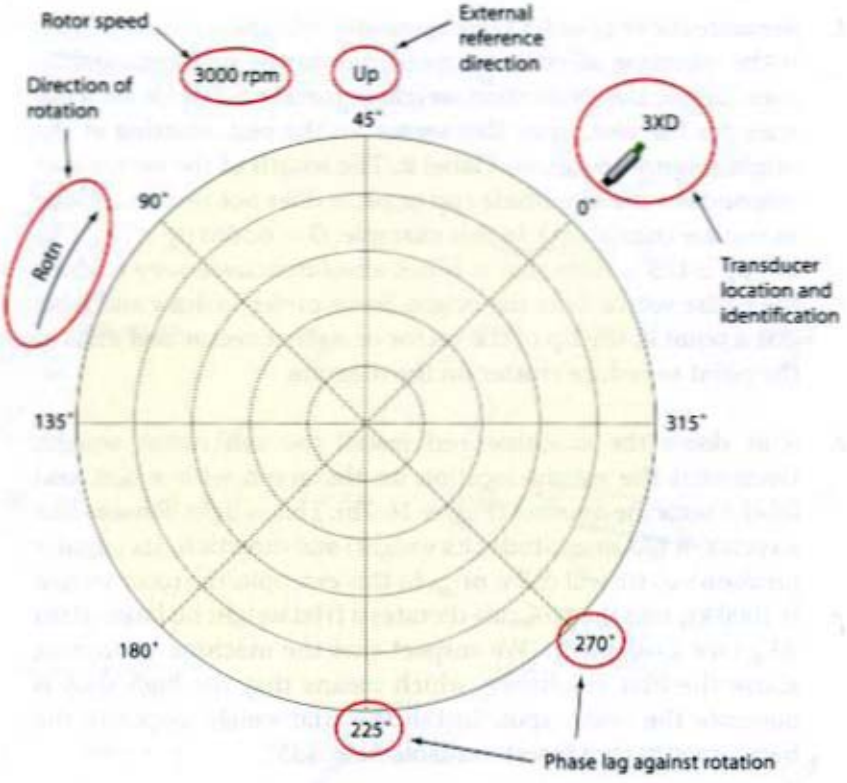


Figure 16-6. Polar graph prepared for balancing calculations. Information documented includes the transducer location and identification, phase lag angles referenced to the transducer and increasing opposite to rotation, the physical reference, direction of rotation viewed from the driver end, and machine speed.

We will show how to perform the graphical and then the mathematical techniques of calibration weight balancing. For the graphical calibration weight balancing technique (refer to Figure 16-7):

1. Measure the *Original*, 1X, compensated vibration vector, $\boldsymbol{O}$. This is the vibration at running speed due to the unknown unbalance, before the calibration weight is installed. Decide on a full scale for the plot, draw this vector on the plot, starting at the origin (Figure 16-7a), and label it. The length of the vector corresponds to the amplitude (pp or pk, it does not matter, as long as you are consistent). In this example, $\boldsymbol{O} = 80$ μm pp ∠125° (3.2 mil pp ∠125°). Note that it is not absolutely necessary to draw the entire vector from the origin. Some prefer to draw and label just a point at the tip of the vector or a short vector that ends at the point to reduce clutter on the diagram.

2. Shut down the machine and install the calibration weight. Document the weight location on the graph with a box, and label it with the amount (Figure 16-7b). This weight behaves like a vector; it has magnitude (its weight) and direction (its angular position), so we will call it $\boldsymbol{m}_{cal}$. In this example, the rotor weight is 1000 kg, and the 10% rule dictates a trial weight no larger than 40 g (see *Example 2*). We suspect that the machine is running above the first resonance, which means that the high spot is opposite the heavy spot. Install the trial weight opposite the heavy spot in the closest available hole, 135°.

3. Run the machine up to operating speed and measure the new vibration vector. This vector represents the machine's *Total* response, $\boldsymbol{T}$, to the original, still unknown unbalance plus the response, $\boldsymbol{C}$, due to $\boldsymbol{m}_{cal}$. Draw $\boldsymbol{T}$ on the graph paper, starting at the origin, and label it (Figure 16-7c). In our example, $\boldsymbol{T} = 50$ μm pp ∠155° (2.0 mil pp ∠155°).

4. Find $\boldsymbol{C}$, the response due to the calibration weight, $\boldsymbol{m}_{cal}$. The total response is

$$\boldsymbol{T} = \boldsymbol{O} + \boldsymbol{C} \tag{16-9}$$

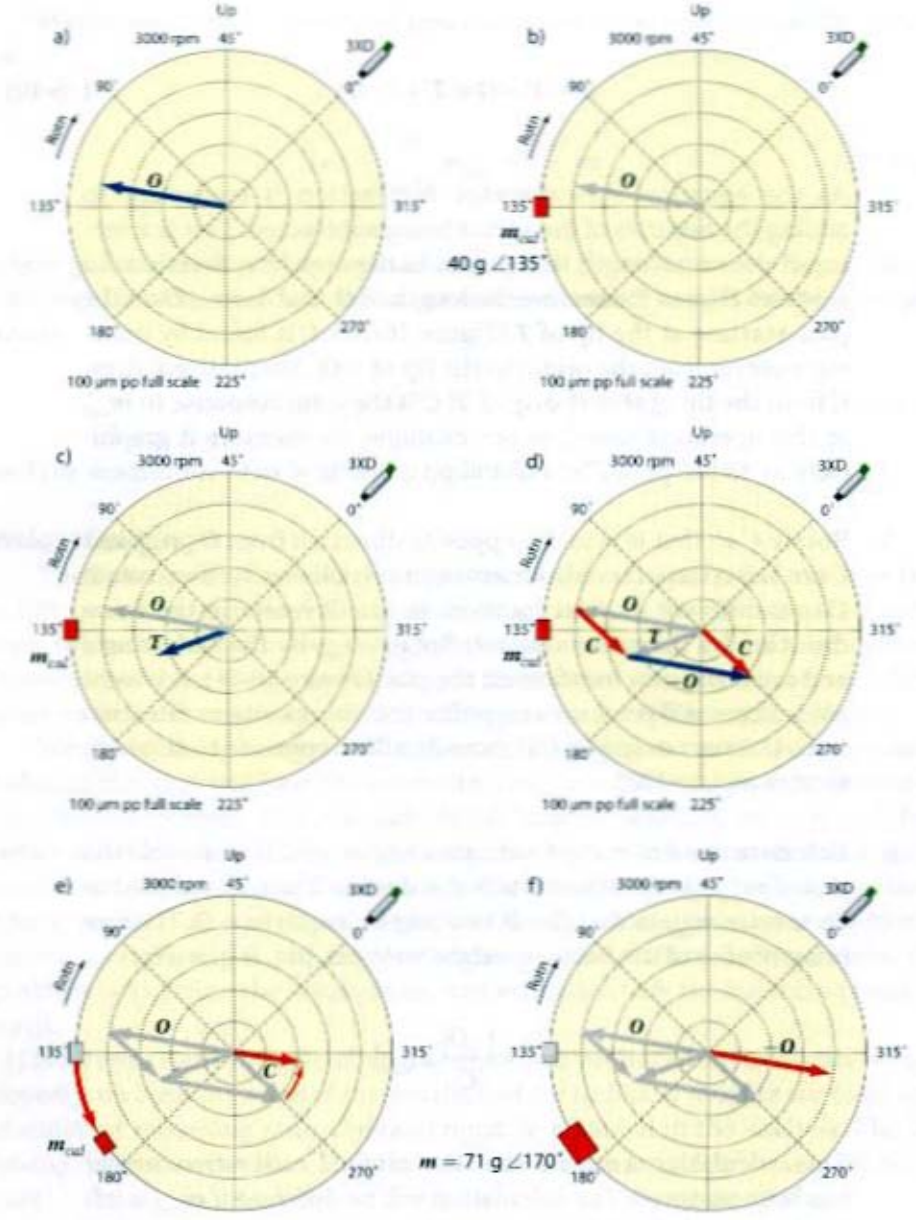


Figure 16-7. Single plane, graphical, calibration weight balancing. The sequence illustrates how the response to a calibration weight can be identified and used to graphically determine a balancing solution. See the text for details.

Thus,

$$\boldsymbol{C} = \boldsymbol{T} - \boldsymbol{O} = \boldsymbol{T} + (-\boldsymbol{O}) \tag{16-10}$$

As the equation shows, vector subtraction is equivalent to adding the *negative* of the vector being subtracted. This is a vector of the same length that points in the opposite direction. To subtract $\boldsymbol{O}$ from $\boldsymbol{T}$, measure the length of $\boldsymbol{O}$, and draw $-\boldsymbol{O}$ on the plot, starting at the tip of $\boldsymbol{T}$ (Figure 16-7d). $\boldsymbol{C}$ is found by drawing a vector from the origin to the tip of $-\boldsymbol{O}$. Alternatively, draw $\boldsymbol{C}$ from the tip of $\boldsymbol{O}$ to the tip of $\boldsymbol{T}$. $\boldsymbol{C}$ is the rotor response to $\boldsymbol{m}_{cal}$ at this operating speed. In our example, we measure it graphically as 45 μm pp ∠270° (1.8 mil pp ∠270°).

5. Rotate $\boldsymbol{C}$ so that it is in the opposite direction from $\boldsymbol{O}$. $\boldsymbol{m}_{cal}$ and $\boldsymbol{C}$ are locked together by our assumption of linearity; if we rotate $\boldsymbol{C}$ to a different angular location, $\boldsymbol{m}_{cal}$ will rotate in the same direction by the same amount. Rotate $\boldsymbol{m}_{cal}$ by the same angle and mark the new location on the plot (Figure 16-7e). A weight placed here will produce a response that is opposite in direction from $\boldsymbol{O}$. In our example, $\boldsymbol{C}$ is moved to 305°, opposite to $\boldsymbol{O}$, which rotates $\boldsymbol{m}_{cal}$ to 170°.

6. Calculate the size of the final balancing weight. If we double the size of $\boldsymbol{m}_{cal}$, the length of $\boldsymbol{C}$ will also double. Thus, all we need to do is scale $\boldsymbol{m}_{cal}$ so that $\boldsymbol{C}$ will be equal in length to $-\boldsymbol{O}$. The size (magnitude) of the balance weight we need, $|\boldsymbol{m}|$, is given by

$$|\boldsymbol{m}| = \frac{|-\boldsymbol{O}|}{|\boldsymbol{C}|}|\boldsymbol{m}_{cal}| \tag{16-11}$$

This calculation assumes that the original calibration weight has been removed. The calculation will be different if $\boldsymbol{m}_{cal}$ is left in place (see below). In our example, we increase $\boldsymbol{m}_{cal}$ by 80/45 to produce $\boldsymbol{m}$ = 71 g ∠170° (Figure 16-7f).

Mathematically, this graphical procedure can be represented, using vectors, as

$$\boldsymbol{m} = \frac{-\boldsymbol{O}}{\boldsymbol{T}-\boldsymbol{O}}\boldsymbol{m}_{cal} = \frac{-\boldsymbol{O}}{\boldsymbol{C}}\boldsymbol{m}_{cal} \tag{16-12}$$

where phase is measured using instrumentation phase lag angles (Appendix 1). The angular location of $\boldsymbol{m}$ is the location of the balance weight on the polar plot and is

$$\angle\boldsymbol{m} = (\angle\boldsymbol{O} + 180^\circ) - \angle\boldsymbol{C} + \angle\boldsymbol{m}_{cal} \tag{16-13}$$

and the magnitude of $\boldsymbol{m}$ is given by Equation 16-11).

Weight Splitting

The balance solution may point between balance holes, point to a hole that is full, or call for more weight than will fit in one hole. Or you may wish to (or have to) leave the calibration weight in place and install a final trim balance. When any of these happen, you can split the balance solution between available holes so that the result is equivalent. This process is called *weight splitting.*

Weight splitting combines the effects of two weights, mounted in available holes, so that they will have the same effect as the balance solution. In the example shown in Figure 16-8, the calculated balance solution, $\boldsymbol{m} = m\ \angle\theta$, falls between balance holes. Two weights, m_1, and m_2 are to be mounted at angular locations θ_1 and θ_2, producing the mass vectors $\boldsymbol{m}_1$, and $\boldsymbol{m}_2$. These vectors must add to produce an equivalent weight, $m\ \angle\theta$. Our calculated balance solution is known, as are θ_1 and θ_2, the angular locations of the available balance holes. The problem is to define the values of m_1 and m_2 that satisfy the balancing requirement.

It is important to remember that the force produced by a balance weight depends on both the mass of the weight *and the radius.* If weights are to be split at different mounting radii, the radii must be included in the analysis. The following derivations assume that the radii are the same, which is most often the case.

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The masses and angles can be treated as vectors (complex numbers), where the requirement becomes

$$\begin{aligned} \boldsymbol{m}_1 + \boldsymbol{m}_2 &= \boldsymbol{m} \\ m_1 e^{j\theta_1} + m_2 e^{j\theta_2} &= m e^{j\theta} \end{aligned} \tag{16-14}$$

where the phase angles in this case are instrumentation convention phase lag angles. Equating real and imaginary parts, we obtain a system of equations,

$$\begin{bmatrix} \cos\theta_1 & \cos\theta_2 \\ \sin\theta_1 & \sin\theta_2 \end{bmatrix} \begin{Bmatrix} m_1 \\ m_2 \end{Bmatrix} = \begin{Bmatrix} m\cos\theta \\ m\sin\theta \end{Bmatrix} \tag{16-15}$$

which can be solved numerically. Or, we can use explicit functions:

$$\begin{aligned} m_1 &= m\left[\frac{\sin(\theta_2 - \theta)}{\sin(\theta_2 - \theta_1)}\right] \\ m_2 &= m\left[\frac{\sin(\theta - \theta_1)}{\sin(\theta_2 - \theta_1)}\right] \end{aligned} \tag{16-16}$$

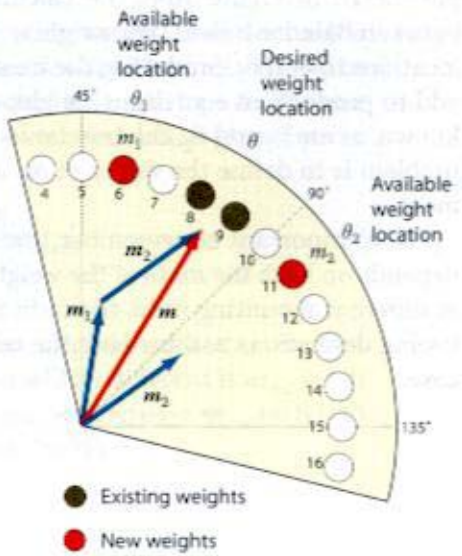


Figure 16-8. Weight splitting. Since the location where we want to install $\boldsymbol{m}$ is already occupied, two weights, m_1 and m_2, are mounted at angular locations θ_1 and θ_2 to produce the equivalent weight, $\boldsymbol{m} = m\angle\theta$.

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Example 4
Calculations indicate that a balance weight of 5.0 g must be installed at 42°. Balance holes are available at 20° and 90°. The balance ring radius for all holes is 100 mm. Find the amount of weight to add in the two available holes.

Solution
Because the balance hole radius is the same for all holes, we can use Equation 16-16. $\boldsymbol{m} = 5.0\ \text{g}\ \angle 42°$. Thus, $m = 5$ g and $\theta = 42°$. Let $\theta_1 = 20°$ and $\theta_2 = 90°$. Applying Equations 16-16,

$$m_1 = 5\ \text{g}\left[\frac{\sin(90° - 42°)}{\sin(90° - 20°)}\right]$$

$$m_1 = 4\ \text{g}$$

$$m_2 = 5\ \text{g}\left[\frac{\sin(42° - 20°)}{\sin(90° - 20°)}\right]$$

$$m_2 = 2\ \text{g}$$

Thus, $\boldsymbol{m}_1 = 4\ \text{g}\ \angle 20°$ and $\boldsymbol{m}_2 = 2\ \text{g}\ \angle 90°$.

A different case exists when the calibration weight, $\boldsymbol{m}_{cal}$, has already been installed, and we wish to leave it in place. A final trim weight needs to be added. $\boldsymbol{m}_{cal}$ effectively takes the place of $\boldsymbol{m}_1$ above. Now the problem is to find an $\boldsymbol{m}_2$ that, together with $\boldsymbol{m}_{cal}$, satisfies the balancing requirement, $\boldsymbol{m}$. Thus, we require that

$$\boldsymbol{m}_{cal} + \boldsymbol{m}_2 = \boldsymbol{m} \tag{16-17}$$

or, in exponential form,

$$m_{cal}e^{j\theta_{cal}} + m_2 e^{j\theta_2} = me^{j\theta} \tag{16-18}$$

Thus,

$$\boldsymbol{m}_2 = m_2 e^{j\theta_2} = m e^{j\theta} - m_{cal} e^{j\theta_{cal}} \tag{16-19}$$

or, in amplitude and phase notation,

$$\boldsymbol{m}_2 = m_2 \angle\theta_2 = (m \angle\theta) - (m_{cal} \angle\theta_{cal}) \tag{16-20}$$

If you have a calculator that has complex number capability, this equation can be used directly. Otherwise, we can find algebraic expressions for m_2 and θ_2 by breaking Equation 16-19 into real and imaginary parts, which leads to

$$\begin{aligned} \theta_2 &= \arctan 2\left[\frac{m\sin\theta - m_{cal}\sin\theta_{cal}}{m\cos\theta - m_{cal}\cos\theta_{cal}}\right] \\ m_2 &= \sqrt{m^2 + m_{cal}^2 - 2mm_{cal}\cos(\theta - \theta_{cal})} \end{aligned} \tag{16-21}$$

where the arctangent2 function is used to return a result between $\pm$ 180°. If the result is negative, add 360°. Note that this solution can also be split using Equations 16-16 if necessary.

Example 5
Given the situation of the previous example, assume that a calibration weight, 10 g $\angle$120°, has already been installed. Find $\boldsymbol{m}_2$ such that, together with $\boldsymbol{m}_{cal}$, will produce $\boldsymbol{m}$ = 5 g $\angle$42°.

Solution
$\boldsymbol{m}_{cal}$ = 10 g $\angle$120°, and $\boldsymbol{m}$ = 5 g $\angle$42°. Applying Equations 16-21,

$$\theta_2 = \arctan 2\left[\frac{5\text{ g}\sin 42^\circ - 10\text{ g}\sin 120^\circ}{5\text{ g}\cos 42^\circ - 10\text{ g}\cos 120^\circ}\right]$$

$$\theta_2 = -31^\circ = 329^\circ$$

$$m_2 = \sqrt{(5\text{ g})^2 + (10\text{ g})^2 - 2(5\text{ g})(10\text{ g})\cos(42^\circ - 120^\circ)}$$

$$m_2 = 10\text{ g}$$

The final trim weight is $\boldsymbol{m}_2 = 10\text{ g} \angle 330^\circ$.

The final trim weight, $\boldsymbol{m}_2$, can also be obtained graphically by adding a new weight, $\boldsymbol{m}_{cal}$, with a new associated vector, $\boldsymbol{C}$, and positioning and adjusting the size of this weight until it produces a response equal to $-\boldsymbol{T}$.

The Influence Vector

An *influence vector* is a complex number that describes how the 1X vibration of a machine will change when a balance weight is added to the machine; in other words, how a balance weight will *influence* both the amplitude and phase of the vibration of the machine. An influence vector is sometimes called an influence *coefficient*; we prefere the term *vector* because it contains both amplitude and phase information.

The influence vector describes the effect of a weight in a single balance plane (weight plane) on the vibration in a measurement plane. The measurement plane can be nearby, or it can be far away from the weight plane; in this case, the influence vector is sometimes called a *longitudinal influence vector.* Each influence vector is specific to a particular combination of balance plane and measurement point (a particular transducer in a particular measurement plane) and to a particular set of operating conditions, including speed, load, temperature, and alignment state.

These characteristics follow from the fact that the influence vector is very closely related to the Dynamic Stiffness that exists between the weight plane and the measurement plane. Dynamic Stiffness and the influence vector are actually different expressions of the same thing, and, as we will show, it is easy to calculate one from the other. When balancing, it is important to remember that a change in the machine since the last balancing operation may have changed the Dynamic Stiffness and the influence vector. For example, a machine that has been overhauled may have a change in alignment that affects the Dynamic Stiffness and the influence vectors.

The influence vector is easily calculated using the results of calibration weight balancing (see above). Given a calibration weight, $\boldsymbol{m}_{cal}$, the influence vector, $\boldsymbol{H}$, produces an associated response vector, $\boldsymbol{C}$, such that

$$\boldsymbol{C} = \boldsymbol{H}\boldsymbol{m}_{cal} \tag{16-22}$$

Thus, the influence vector, $\boldsymbol{H}$, is

$$\boldsymbol{H} = \frac{\boldsymbol{C}}{\boldsymbol{m}_{cal}} = \frac{(\boldsymbol{T} - \boldsymbol{O})}{\boldsymbol{m}_{cal}} \tag{16-23}$$

where $\boldsymbol{O}$ is the original unbalance response vector, and $\boldsymbol{T}$ is the total response, including the original unbalance response and the calibration weight response. The magnitude of $\boldsymbol{H}$ is found by taking the ratio of the magnitudes of $\boldsymbol{C}$ and $\boldsymbol{m}_{cal}$:

$$|\boldsymbol{H}| = \frac{|\boldsymbol{C}|}{|\boldsymbol{m}_{cal}|} \tag{16-24}$$

The angle of $\boldsymbol{H}$, a lag angle measured against rotation, is found by subtracting the lag angle of the calibration weight from the lag angle of the response due to that weight:

$$\theta_H = \angle \boldsymbol{H} = \angle \boldsymbol{C} - \angle \boldsymbol{m}_{cal} \tag{16-25}$$

What is the meaning of the influence vector? Going back to Equation 16-23, note that the difference, $\boldsymbol{T} - \boldsymbol{O} = \boldsymbol{C}$, tells us that $\boldsymbol{H}$ measures the *change* in vibration that occurs when a weight is added to the machine. Because $\boldsymbol{H}$ is a vector, it tells us how both the *amplitude* and the *phase* of 1X vibration will change with the weight addition.

The equation also tells us that the influence vector is the ratio of the change in vibration, $\boldsymbol{C}$, to the mass vector that produced that change, $\boldsymbol{m}_{cal}$. Thus, the units of $\boldsymbol{H}$ are vibration amplitude per unit mass (or weight): for example, μm pp/g or mil pp/oz.

Suppose that we measure an influence vector on a machine running at 3600 rpm as $\boldsymbol{H}$ = 5.7 μm pp/g ∠135°. If we assume that the machine is *perfectly balanced* and the 1X vibration is zero, $\boldsymbol{H}$ tells us that, at 3600 rpm, each gram of weight we add in the balance plane will, at the measurement plane, produce a 1X vibration of 5.7 μm pp at 135° lag relative to the weight location (Figure 16-9).

If the original vibration was *not* zero, then the vibration vector produced by the weight addition will add to the existing vibration vector. For our example, the *change* in vibration will be 5.7 μm pp ∠135° per gram of weight added.

The phase angle of the influence vector is the same as the heavy spot/high spot relationship that exists in the machine at that speed. If a single mode machine is running well below a resonance, the influence vector phase should be close to 0°, indicating that the high spot and heavy spot are in phase. At the resonance, the phase should be about 90°, and well above the resonance, the phase should be closer to 180°. The presence of higher modes can complicate these simple relationships.

The influence vector allows us to balance easily and quickly. If we have a valid influence vector, we can calculate the required trim balance shot in one step. If $\boldsymbol{O}$ is the original unbalance response, then we want to add an amount of weight in a location that will produce a response exactly *opposite* to the original response. We require a balance shot, $\boldsymbol{m}$, such that

$$\boldsymbol{Hm} = -\boldsymbol{O} \tag{16-26}$$

Solving for $\boldsymbol{m}$,

$$\boldsymbol{m} = \frac{-\boldsymbol{O}}{\boldsymbol{H}} \tag{16-27}$$

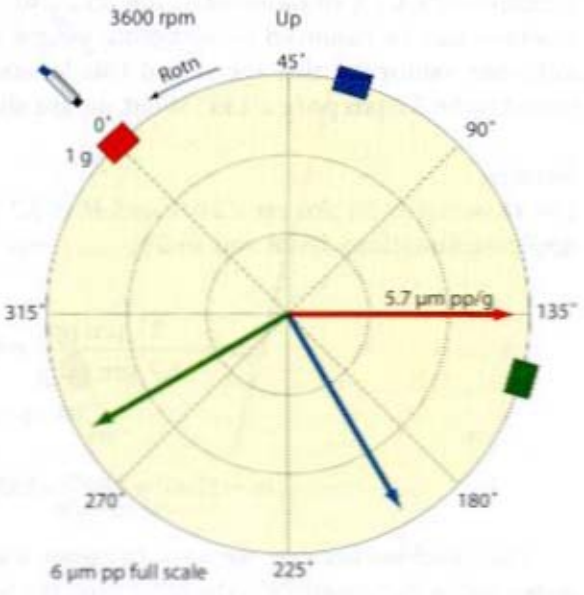


Figure 16-9. Example of an influence vector, $\boldsymbol{H}$ = 5.7 µm pp/g ∠135°. For this machine at 3600 rpm, *each gram* of weight will produce a change in 1X vibration of 5.7 µm pp at 135° phase lag relative to the weight. No matter where the mass is located, the influence vector and change in vibration will have the same magnitude and relative orientation. Doubling the weight will double the magnitude of the change in vibration to 11.4 µm pp ∠135°. The figure shows three positions of the weight; for each position, the influence maintains the same relative angular orientation, in this case, 135°.

where the magnitude of $\boldsymbol{m}$ is

$$|\boldsymbol{m}| = \frac{|-\boldsymbol{O}|}{|\boldsymbol{H}|} \tag{16-28}$$

and the angle of $\boldsymbol{m}$ is

$$\angle \boldsymbol{m} = \angle(-\boldsymbol{O}) - \angle \boldsymbol{H} \tag{16-29}$$

If an influence vector is known for a particular machine, it is possible to calculate a balance shot using steady state data at running speed, before shutting down the machine. Then, the machine can be shut down, and the balance weight installed. After startup, the response can be compared to what was predicted, and a new influence vector can be calculated if the machine response has changed significantly. For most machines, the influence vector will provide an excellent initial estimate of the balance solution, even if the machine has changed and an additional trim balance is required. A change in the influence vector can draw attention to changes in the machine that could be important.

Example 6
A steam turbine is running at 3600 rpm. A shaft relative probe at 45° L measures a compensated, 1X vibration of 35 μm pp ∠240°. Experience has shown that this machine can be balanced by installing weight in a nearby balance plane. The influence vector for this speed and this location have been previously determined to be 5.7 μm pp/g ∠135°. What weight should be installed?

Solution
The $\boldsymbol{O}$ vector is 35 μm pp ∠240°, and $\boldsymbol{H}$ is 5.7 μm pp/g ∠135° (Figure 16-10). Applying Equations 16-28 and 16-29,

$$|\boldsymbol{m}| = \frac{35 \text{ μm pp}}{5.7 \text{ μm pp/g}} = 6.1 \text{ g}$$

$$\angle \boldsymbol{m} = (240° + 180°) - 135° = 285°$$

180° is added because we want to create a vector equivalent to $-\boldsymbol{O}$. Where do we install the weight? On the polar plot, the 6 gram weight would be installed

285° from the measurement transducer in a lag (opposite to rotation) direction. This location must then be translated through the weight map for the balance plane.

The influence vector we have been using is also called the *direct* influence vector. Its use requires that we perform a division, which can be awkward to do mentally. Instead, you can use the *inverse* influence vector to calculate a balance solution using multiplication. For this calculation, we use only the magnitude; the angle is calculated as before. The magnitude of the inverse influence vector is the reciprocal of the magnitude of the direct influence vector.

$$|H^{-1}| = \frac{1}{|H|} \tag{16-30}$$

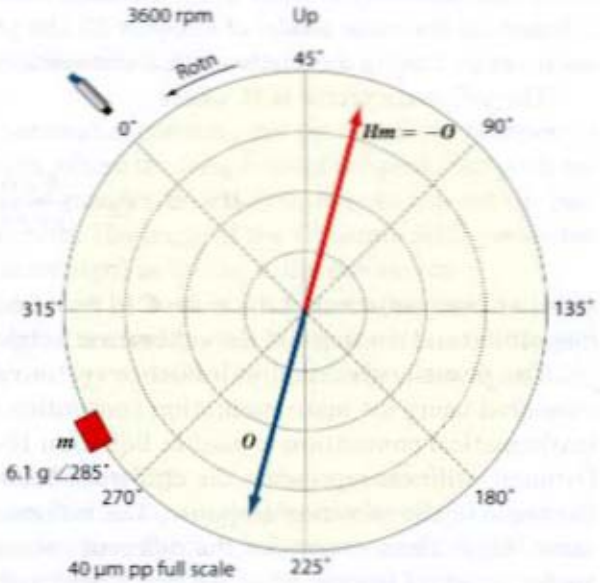


Figure 16-10. Balancing with an influence vector. A machine, running at 3600 rpm, has a 1X compensated vibration, $\boldsymbol{O}$ = 35 µm pp ∠240°. The influence vector, $\boldsymbol{H}$ = 5.7 µm pp/g ∠135°, is used to calculate the balance solution, 6.1 g ∠285°, that produces a response exactly equal and opposite to $\boldsymbol{O}$, balancing the machine.

The inverse influence vector has units of weight (or mass) per unit vibration, g/μm pp or oz/mil pp. The magnitude of the balance solution is now,

$$\boldsymbol{m}| = |\boldsymbol{O}||\boldsymbol{H}^{-1}| \tag{16-31}$$

The Influence Vector And Dynamic Stiffness

The influence vector is the ratio of the change in vibration to the rotating unbalance *mass* that produced that change. Dynamic Stiffness is the ratio of the rotating unbalance *force* to the vibration produced by that force. Both involve rotor response to a rotating unbalance; thus, the Dynamic Stiffness and influence vector must be related.

The synchronous Dynamic Stiffness, $\boldsymbol{K}_S$, is

$$\boldsymbol{K}_S = K_S \angle\theta_K = \frac{mr_u\Omega^2 \angle\delta}{A \angle\alpha} \tag{16-32}$$

where m and r_u are the mass and radius of unbalance, Ω is the rotor speed in rad/s, δ is the phase of the unbalance, and A and α are the amplitude and phase of the vibration response due to the unbalance mass, m. Because this expression is based on the rotor model of Chapter 10, the phase angles in this expression are measured using the mathematical convention (positive phase lead).

The influence vector is $\boldsymbol{H}$, where

$$\boldsymbol{H} = H \angle\theta_H = \frac{A \angle\alpha}{m \angle\delta} \tag{16-33}$$

Here, we have substituted $A \angle\alpha$ for $\boldsymbol{C}$ in Equation 16-23; m and δ represent the magnitude and the angle of the calibration weight.

The phase angles used in influence vector calculations in this chapter are measured using the instrumentation convention (positive phase lag), while the mathematical convention is used in Equation 16-32. Note that the angle of the Dynamic Stiffness represents the difference between the angle of the force and the angle of the vibration response. The influence vector angle represents the same thing. Thus, except for the different measurement conventions, the two angles are equal to each other. All that is required to convert between the measurement conventions is to take the negative of the angle.

Solving both equations for $A \angle\alpha$, and setting the two equations equal to each other leads to

$$\boldsymbol{K}_S = \frac{r_u \Omega^2}{H} \angle\theta_H$$
$$\boldsymbol{H} = \frac{r_u \Omega^2}{K} \angle\theta_K \tag{16-34}$$

where the angles have been adjusted for the measurement conventions.

Example 7
A machine that rotates at 4850 rpm has an influence vector, $\boldsymbol{H}$ = 325 mil pp/oz ∠41° (290 μm pp/g ∠41°), where the angle is measured in degrees lag. The balance hole radius is 7.5 in (190 mm). Find the Synchronous Dynamic Stiffness.

Solution
Apply Equation 16-34,

$$\boldsymbol{K}_S = \frac{(7.5 \text{ in})\left[(4850 \text{ rev/min})(1 \text{ min/60 s})(2\pi \text{ rad/rev})\right]^2}{(1/2)(325 \text{ mil pp/oz})(1 \text{ in/1000 mil})(16 \text{ oz/lb})(386 \text{ in/s}^2)} \angle 41^\circ$$

$$\boldsymbol{K}_S = 1900 \text{ lb/in} \angle 41^\circ \text{ (337 kN/m} \angle 41^\circ)$$

The factor of ½ in the denominator converts mil pp to mil pk. Stiffness is usually measured as lb/in or N/m, where the length units are peak, not peak-to-peak. Note that the acceleration of gravity (386 in/s^2) is there to convert the balance weight unit (oz) to a mass unit. The angle of the Dynamic Stiffness vector is the same number (with the same sign) as for the influence vector.

The Dynamic Stiffness vector can be broken into direct (magnitude times the cosine of the angle) and quadrature (magnitude times the sine of the angle) parts if desired. For this machine, K_D = 1430 lb/in (250 kN/m) and K_Q = 1250 lb/in (219 kN/m).

Multiple Modes And Multiplane Balancing

Up to this point we have been discussing single-plane balancing. Most modern turbomachinery rotors operate above the first balance resonance, and many operate above the second, or even higher, mode. Machines with multiple modes often require more complex balancing, where two or more planes are balanced simultaneously. In this section, we will confine ourselves to a discussion of two-

plane balancing. Although software exists that can handle many simultaneous planes of balancing, single- and two-plane balancing techniques are adequate to handle most of the trim balance problems you are likely to encounter in the field.

The amount of vibration at a resonance will depend on the unbalance distribution at various axial locations along the rotor, how well the shape of that distribution matches the mode shape of the rotor system, and the modal damping (more precisely, modal quadrature stiffness) for the mode. *If the shape of the unbalance distribution is a poor match for a mode shape, then that mode will be poorly excited or not excited at all.* On the other hand, *an unbalance distribution that matches a mode shape will excite that mode relatively strongly.* We can exploit this behavior to separately balance two different modes, one at a time.

In general, at any speed, the deflection shape of a flexible rotor is the result of a mixture of several modes of vibration. At a resonance, one mode will usually dominate the response. A rotor operating between resonances will usually have residual effects of the lower modes and, at the same time, the early effects of the higher modes, depending on how the shape of the unbalance distribution corresponds to the modes. Thus, the rotor deflection shape will vary with speed and modal contribution; between resonances, the relative phase of two axially separated planes can vary considerably due to the mixing of the modes.

As shown in Figure 16-11 (left), the first mode of a flexible rotor between bearings is usually a simple, predominantly *in-phase*, bending mode (overhung rotors have a different shape; see Chapter 12). In this mode, most of the length of the rotor will be deflected more or less in the same radial direction; thus, vibration measurements taken at different axial positions along the rotor will have approximately the same phase. For this mode, two nodal points (points of zero or minimum vibration) will exist, one near each bearing. The nodal points can be inboard or outboard of the bearings, depending on the stiffness and mass distribution of the system. (Note that, if a probe is located outboard of a nodal point, the phase at that location will be opposite to the predominant phase of this first mode. In such cases, mode identification probes can be installed to help identify the mode shape.) This in-phase mode will respond to an unbalance mass distribution that is predominantly in the same radial direction along the length of the rotor.

The second mode of a flexible rotor is usually an *out-of-phase* bending mode (Figure 16-11, middle). The deflection shape of this mode is approximately S-shaped, with an additional nodal point near the midspan of the rotor. Approximately one-half of the rotor will be deflected in one radial direction, with the other half in the opposite direction. The details of nodal point locations and deflection amplitudes will depend on the stiffness and mass distribution of

the rotor. This out-of-phase mode responds well to an out-of-phase unbalance distribution, where the unbalance is predominantly in one radial direction for half of the rotor, and in the opposite direction for the other half of the rotor.

Higher modes have more complicated bending patterns, with one additional nodal point per mode. These modes cannot be described using the simple in-phase and out-of-phase terms we have been using. Trim balancing rotors which exhibit higher mode shapes can be challenging; fortunately, most machines operate below the first resonance or between the first and second resonances, and the influence of higher modes is limited. We will confine our discussion to the techniques involved in balancing the first two modes.

In Figure 16-11, the unbalance is assumed to be concentrated in the two masses shown. These masses are also assumed to be both the weight and measurement planes in the rotor. In general, this will not be the case; mass and unbalance will be axially distributed along the rotor.

The first mode will not respond to an out-of-phase unbalance distribution because, from the point of view of the first mode, the two unbalance forces can-

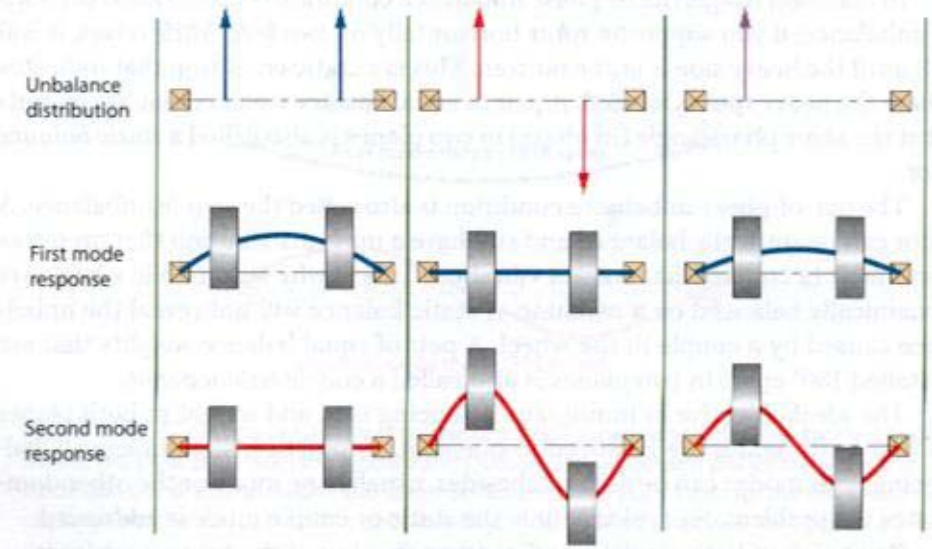


Figure 16-11. Unbalance distribution and mode excitation. The rotor shown has two modes: in phase and out of phase. The in-phase unbalance distribution (left) excites the first mode, but, since it does not fit the second mode shape, it does not excite the second mode. The out-of-phase unbalance distribution (middle) fits the second mode shape, but not the first. The asymmetric unbalance distribution (right) has characteristics of and excites both modes.

cel each other out. Similarly, the second mode will not respond to an in-phase unbalance distribution. Thus, if two identical weights are installed at the same angular location in both weight planes, the in-phase balancing force created will affect the first mode, *but will not excite the second mode.* Similarly, if two identical weights are installed 180° opposite each other in both weight planes, the out-of-phase balancing force created will excite the second mode, *but will not excite the first mode.*

If only one of the forces in the figure is active (right), the unbalance distribution will be able to excite *both* modes. It will be seen by the first mode as an unbalance acting in one direction and by the second mode as an axially asymmetric unbalance. A rotor with most of the unbalance concentrated at one end like this is said to have a *dominant end.*

Note that it is possible to apply these same principles to *rigid body* modes, where the rotor vibrates with little or no bending. Sometimes, two observable rigid body modes exist in a machine, an in-phase (*cylindrical*) mode and an out-of-phase (*pivotal*) mode. Often, these modes tend to be overdamped and are not observed.

In common usage, the in-phase unbalance condition is often called the *static* unbalance; if you support a rotor horizontally on two level knife edges, it will roll until the heavy side is at the bottom. This is a static condition that indicates where the heavy spot is located. A pair of equal balance weights that are mounted at the same phase angle (in phase) in two planes is also called a *static balance shot.*

The out-of-phase unbalance condition is also called the *couple* unbalance. A rotor can be statically balanced and still have a mass distribution that produces a moment (a couple) that creates vibration. This is why automobile wheels are dynamically balanced on a machine. A static balance will not reveal the unbalance caused by a couple in the wheel. A pair of equal balance weights that are installed 180° apart in two planes is also called a *couple balance shot.*

The ideal objective in multiplane balancing is to add weight in both planes so that both modes are balanced in one shot. In practice, simultaneously balancing both modes can be difficult; besides, usually one mode or the other dominates the problem. So, typically only the static or couple mode is addressed.

Balancing of both modes requires identification of the heavy spot location for each mode. Figure 16-12 shows polar plots of a startup of a small, experimental machine that experiences two modes. The polar plots show the response for the Inboard and Outboard measurement planes. Note that the polar loops for the first mode response (blue) are oriented in the same direction, indicating that vibration in both measurement planes is in phase, which identifies an in-phase

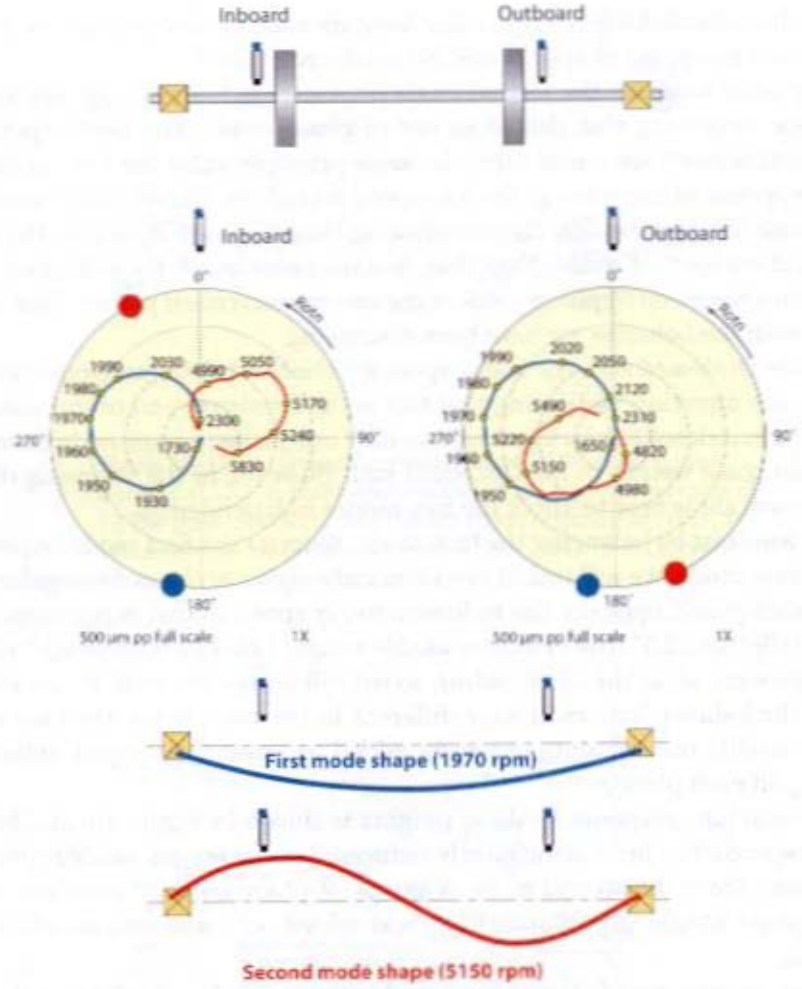


Figure 16-12. Two-plane unbalance response of a small, experimental machine that has two modes. The polar plots from the Inboard and Outboard measurement planes show the first mode (blue) and second mode (red) response. Note that the polar loops for the first mode response are oriented in the same direction, indicating that both measurement planes are deflected in the same direction, an in-phase mode. The indicated heavy spots (blue dots) for each plane are found by looking at the response phase at speeds well below the first resonance. The polar loops in the second mode are on *opposite* sides of the plots, indicating an out-of-phase mode. The heavy spots (red dots) for this mode can be found by looking at the response phase at the low end of the *second* mode resonance. Note that, like the polar loops, the indicated heavy spot locations are on opposite sides in the two measurement planes. At the Outboard end, the heavy spots for both modes are approximately in the same direction, indicating that this is the dominant end of the machine.

mode. The indicated heavy spots (blue dots) for each plane are found by looking at the response phase at speeds well below the resonance.

The polar loops for the second mode response (red) are on *opposite* sides of the plots, indicating that this is an out-of-phase mode. The heavy spots (red dots) for this mode are found using the same principle as for the first mode: look at the response of the rotor at the low-speed end of the *second* mode resonance (but above the first mode). The direction of that response identifies the heavy spot location for that mode. Note that, like the polar loops, the indicated heavy spot locations are on opposite sides in the two measurement planes. This is consistent with the behavior we have been discussing.

At the Outboard end, the heavy spots for the two modes are approximately in the same direction, indicating that this is the dominant end of this machine. If we were restricted to placing weight in only one plane to improve both modes, we would place weight in the Outboard end. However, in the following discussion, we will show how to affect the two modes independently.

We will start by balancing the first mode. Because the first mode response is an in-phase mode, we will install *two identical weights at the same angular location in each plane*, opposite the indicated heavy spots. In this experiment, 0.2 g was installed at 22.5° (the closest available weight hole) in both weight planes. The holes were all at the same radius, so we will ignore the radii in our discussion. If the balance hole radii were different in the two planes, then we would have to modify the amounts of weight added to produce an equal unbalance, $U = mr_u$, in each plane.

The machine response to these weights is shown in Figure 16-13. The first mode response has been significantly reduced, *but the second mode response is unchanged.* Since the second mode is an out-of-phase mode, it does not match the in-phase weight distribution that was added, so there was no change in response.

Next, we can modify the second mode response without affecting the first mode by adding a weight *couple*: two equal weights installed in opposite sides of the two planes. In this experiment, 0.4 g was installed at 135° in the Inboard plane, and 0.4 g was installed at 315° in the Outboard plane. These locations are approximately opposite to the indicated heavy spot for the second mode in each plane.

The machine response is shown in Figure 16-14. Now the second mode has been drastically reduced in size, but *the first mode is unchanged from the last run.* Since the first mode is an in-phase mode, it is insensitive to the out-of-phase weight distribution that was added.

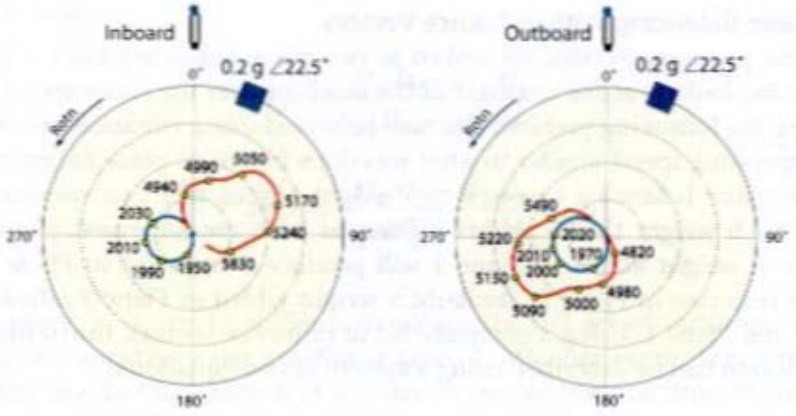


Figure 16-13. Response to in-phase weight addition. 0.2 g was installed at 22.5° in both weight planes of the machine in Figure 16-12. The polar plots show that the first mode response has been significantly reduced, but the second mode response is unchanged.

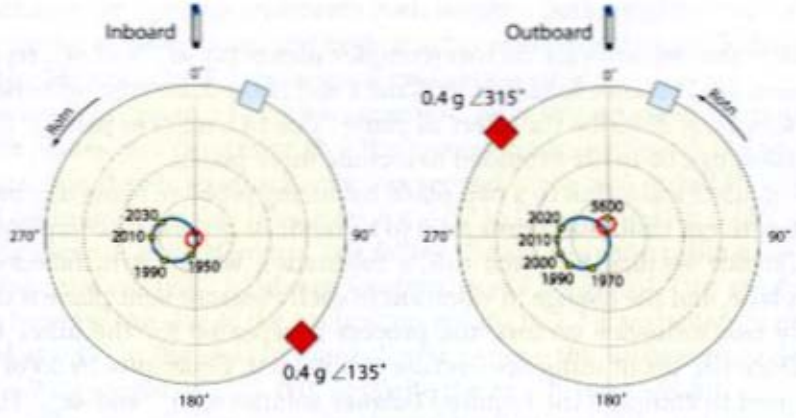


Figure 16-14. Response to out-of-phase weight addition. Without removing the previous weight correction, 0.4 g was installed at 135° in the Inboard plane, and 0.4 g was installed at 315° in the Outboard plane, approximately opposite to the indicated heavy spots for the second mode. The polar plots show that the second mode has been drastically reduced in size, but the first mode is unchanged from the last run. The first (in-phase) mode is insensitive to the out-of-phase weight correction.

Multiplane Balancing With Influence Vectors

In the previous section, we looked at two-plane balancing from a "global" perspective, looking at the response of the machine over the entire speed range. However, the balancing problem can also be solved using vibration vectors at a single operating speed, similar to what was done for single-plane balancing.

Two-plane balancing involves two weight planes and two measurement planes. Each weight that is added influences both measurement planes. For example, a weight added in Plane 1 will produce a response in Plane 1 and another response in Plane 2. Similarly, a weight added in Plane 2 affects both Plane 2 and Plane 1. Using a complete set of influence vectors, the desired balance solution can be described using a system of two equations,

$$\begin{aligned} -\boldsymbol{O}_1 &= \boldsymbol{H}_{11}\boldsymbol{m}_{p1} + \boldsymbol{H}_{12}\boldsymbol{m}_{p2} \\ -\boldsymbol{O}_2 &= \boldsymbol{H}_{21}\boldsymbol{m}_{p1} + \boldsymbol{H}_{22}\boldsymbol{m}_{p2} \end{aligned} \tag{16-35}$$

or, in matrix form,

$$\begin{Bmatrix} -\boldsymbol{O}_1 \\ -\boldsymbol{O}_2 \end{Bmatrix} = \begin{bmatrix} \boldsymbol{H}_{11} & \boldsymbol{H}_{12} \\ \boldsymbol{H}_{21} & \boldsymbol{H}_{22} \end{bmatrix} \begin{Bmatrix} \boldsymbol{m}_{p1} \\ \boldsymbol{m}_{p2} \end{Bmatrix} \tag{16-36}$$

where all of the elements are vectors (complex numbers); $\boldsymbol{m}_{p1}$ and $\boldsymbol{m}_{p2}$ represent the desired final balance solutions in Plane 1 and Plane 2, and the influence vector subscripts, *ij*, describe the effect in plane *i* due to a mass in plane *j*. This set of equations can be easily extended to include more planes.

To calculate a solution to a two-plane balancing problem using this method, two independent calibration runs have to be made to generate a complete set of four influence vectors. For each run, a calibration weight is installed in one weight plane, and the change in vibration in each measurement plane is used to calculate two influence vectors; the process is repeated for the other weight plane. Once the set of influence vectors is obtained, Equations 16-35 or 16-36 can be used to compute the required balance solutions, $\boldsymbol{m}_{p1}$ and $\boldsymbol{m}_{p2}$. The balance solution can be expressed in matrix form as

$$\begin{Bmatrix} \boldsymbol{m}_{p1} \\ \boldsymbol{m}_{p2} \end{Bmatrix} = \begin{bmatrix} \boldsymbol{H}_{11} & \boldsymbol{H}_{12} \\ \boldsymbol{H}_{21} & \boldsymbol{H}_{22} \end{bmatrix}^{-1} \begin{Bmatrix} -\boldsymbol{O}_1 \\ -\boldsymbol{O}_2 \end{Bmatrix} \tag{16-37}$$

or algebraically as

$$\boldsymbol{m}_{p1} = \frac{\boldsymbol{H}_{12}\boldsymbol{O}_2 - \boldsymbol{H}_{22}\boldsymbol{O}_1}{\boldsymbol{H}_{11}\boldsymbol{H}_{22} - \boldsymbol{H}_{21}\boldsymbol{H}_{12}}$$

$$\boldsymbol{m}_{p2} = \frac{\boldsymbol{H}_{21}\boldsymbol{O}_1 - \boldsymbol{H}_{11}\boldsymbol{O}_2}{\boldsymbol{H}_{11}\boldsymbol{H}_{22} - \boldsymbol{H}_{21}\boldsymbol{H}_{12}} \qquad (16\text{-}38)$$

In practice, it can be difficult to obtain the full set of influence vectors. Instead, we sometimes use a different type of influence vector that represents the effect due to the addition of a static or couple balance shot. In this technique, we decide which mode is most important to the balancing problem. If it is the first mode, then a best guess set of static (in-phase) calibration weights is installed in the two planes; if the second mode is deemed most important, then a best guess couple is installed. Once up to speed, the change in vibration in both planes is compared to the original. One plane is selected as a reference plane, and an influence vector is calculated using Equation 16-23.

This influence vector represents the effect of the pair of calibration weights that were installed, and the balancing operation can proceed as if it were a single-plane problem. The only difference (and an important one) is that the balance solution, $\boldsymbol{m}$, actually represents both weights. Both weights must be scaled in size by the same amount, and both weights must maintain the same relative angle to each other as did the original calibration weights.

For example, given a two-plane machine, after inspection of the polar plots for both planes, you determine that the second mode dominates the unbalance response. You decide to attack only this second mode and install the following pair of calibration weights (a couple): 5 g ∠45° in Plane 1, and 5 g ∠225° in Plane 2. Using Plane 1 as a reference, you follow the single-plane calibration weight procedure and calculate a balance solution, $\boldsymbol{m} = 7.2$ g ∠135°. This solution tells you to increase the size of the calibration weight to 7.2 g and rotate it until it is located at 135° in Plane 1. In Plane 2, you remove the calibration weight and install a 7.2 g weight at 315°, 180° opposite the new Plane 1 weight.

If you must, you can leave in the calibration weights, and calculate the required trim weight in Plane 1 using Equations 16-19 and 16-20. For this example, m_{p2} will be the same size as m_{p1}, and m_{p2} will be located 180° relative to m_{p1}. Note that the weights can be split independently in each plane as necessary.

The success of this technique depends on the machine having clearly defined, in- and out-of-phase behavior. It works best on axially symmetric rotors and is frequently used to balance dual flow LP rotors. Long rotors with a high

length-to-diameter ratio, such as power industry generators, often have balance planes located at the ends of the rotor and low sensitivity to static (in-phase) balance corrections. Some rotors may have a third, midspan balancing plane. This balancing plane can be used to balance the first, in-phase mode; the two outer planes can be used, with a couple correction, to balance the second mode.

When rotors are axially asymmetric (for example, have more mass near one end of the rotor), then the second mode nodal point will most likely be located nearer to one end of the rotor. This can produce higher vibration amplitude at the end of the rotor farthest from the nodal point; such a rotor may be more difficult to balance using a pure static or pure couple correction.

When balancing using this technique, there is no particular requirement that the two-plane calibration weights be exactly the same or exactly in or out of phase. The only requirement is that the *relative* masses and the *relative* angles remain the same. In practice, however, it is likely that a pair of weights at an arbitrary relative angle will affect both modes simultaneously, and the result of an unequal or arbitrary relative angle shot can be very difficult to predict. We recommend keeping the weights the same and either in phase or out of phase.

How Balancing Can Go Wrong

Balancing involves a complicated series of steps, from evaluation to weight installation to calculation and installation of a final shot. There are many ways a balancing job can go wrong. Here are a few things to think about.

You should always try to keep a global perspective on the problem when balancing. If at all possible, look at the data over the entire speed range, and plot data from both *XY* probes. Compare and cross-check the results of vector analysis with polar plots of the data and graphical solutions.

Make sure the polar plot and graphical plots have the correct rotation direction. When standing next to the machine, make sure you are looking down the machine train in the same direction indicated on your data. If you accidentally turn around without realizing it, you will install the weight at the wrong angular orientation.

In plants with multiple, identical machine trains, make sure you are working on the correct machine and your data comes from that machine.

Use hot shutdown data when possible. Axial growth can cause the target area of eddy current probes to change. This can change cold slow roll runout and adversely affect slow roll compensation. Shutdown data is generally preferred because the machine is hot, all the way down to slow roll speed. Polar plots are a superior tool for balancing partly because they will immediately reveal slow roll compensation errors, and they can be visually compensated.

Make sure the machine is not sensitive to changes in process conditions. Repeatability from run to run is essential for good results. Make sure the machine stabilizes at full load before judging its behavior.

Watch out for unbalance problems that are caused by other effects. Dirt or debris can shift inside a machine and cause the heavy spot to change from run to run. Balancing such a machine is futile until the situation is corrected. See the case history in Chapter 25.

Make sure the weight goes into the machine where you think it should. If at all possible, personally observe the weight installation.

Watch out for other personnel who might change something in your instrumentation setup while you are waiting. In one case, a jealous competitor deliberately switched probe leads in the middle of a balancing job. Fortunately, the change in data was noticed and allowed for.

Remember that, when trying to identify the heavy spot location, velocity and acceleration phase behavior is different from displacement behavior. Integrate velocity to displacement, or mentally adjust the heavy spot location. Calibration weight balancing will work fine with velocity or acceleration data.

Summary

Balancing is the act of adding or removing mass from a rotor in a way that reduces the unbalance-induced vibration below a minimum acceptable level. Balancing can be performed in a single axial plane, two planes, or it may involve simultaneously balancing several axial planes, which requires considerable knowledge of the machine and sophisticated balancing software. Most often, field balancing will involve only one or two planes.

Shop balancing is performed with the rotor removed from the machine and installed on a test stand; trim balancing is performed with the rotor in place in the machine. Trim balancing of shop balanced rotors may be required because of changes in Dynamic Stiffness of the assembled rotor system.

The unbalance force is proportional to the radius of unbalance and the square of the rotor speed. It acts through the Dynamic Stiffness to produce predominantly 1X vibration. The unbalance can be thought of as a small *heavy spot* that rotates with the rotor.

The polar plot is the best plot format for balancing. The vibration vectors at speeds well below the resonance point in the general direction of the heavy spot.

Calibration weight balancing is a multistep process that involves installing a known, *calibration weight* at a known angular location. The calibration weight and the change in vibration are used to calculate a final balance solution.

Weight splitting is a technique that is useful when the calculated location for the balance weight is not available. Two weights are installed in available

locations so that the effect is equivalent.

An *influence vector* describes how the addition of a balance weight changes, or influences, the amplitude and phase of machine vibration. It is calculated by taking the ratio of the vibration change to the balance weight that produced that change. Influence vectors are valid only for a particular speed, weight plane, measurement plane, and significant machine operating conditions.

The first mode of a flexible rotor is usually a simple, predominantly *in-phase*, bending mode (also called a *static* mode); the second mode of a flexible rotor is usually an *out-of-phase*, or S-shaped, bending mode (also called a *couple* mode). These modes respond to unbalance mass distributions that approximately match the mode shape of the rotor. Multiplane balancing techniques exploit this behavior to allow independent balancing of separate modes. Polar plots can be used to identify the heavy spot location for each mode.

Sets of influence vectors can be found that allow calculation of a complete, two-plane balance solution, but they require one calibration run for each weight plane. More often, identical pairs of weights are used to develop special static or couple influence vectors. These influence vectors are used to calculate a final balance solution for the particular mode being balanced.

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Malfunctions

Chapter 17

Introduction to Malfunctions

WITH OUR STUDY OF VIBRATION FUNDAMENTALS, data plots, and rotor dynamics theory completed, we are now ready to examine a complex topic, machine malfunctions and their detection. Each chapter of this section will deal with a specific type or family of malfunctions that are common to most rotating machinery.

This chapter will present some basic facts about malfunctions and their detection. We will begin with a definition of a machine malfunction and what it means in terms of machine behavior and performance. We will then discuss how malfunctions can be detected and identified using changes in many different vibration characteristics, and how Dynamic Stiffness is the important primary link between a malfunction and observed vibration.

What is a Malfunction?

We generally think of a malfunctioning machine as one that is broken or does not perform its intended task. But a breakdown is usually the result of a long period of accumulating damage. Most machines will continue to run and perform useful work even with an active or developing problem. However, machines and lost product are too expensive for us to wait until a machine breaks down before thinking about malfunctions. We do not want to restrict our malfunction definition too tightly, because we want to detect and *prevent* unplanned machine downtime.

Proactive machinery management requires us to detect problems as early as possible in order to schedule maintenance, make repairs, or adjust the process. It is an economic imperative to prevent unplanned downtime and reduced output. This requires a definition of a malfunction that is compatible with this goal.

A malfunction is an operating condition or mechanical problem that, if not treated, may cause a degradation in performance, an unplanned shutdown, or a catastrophic failure.

Continued operation of a machine with a malfunction may result in higher than normal stresses, fatigue, or even direct mechanical damage. A loss of efficiency, an indirect effect due to damage in the machine, is also possible. A good example is an unbalanced compressor that passes through a poorly damped (high SAF) resonance and wipes internal seals. The increased seal clearance allows excessive interstage flow, causing reduced interstage pressure and efficiency. In this case, the unbalance is the malfunction, which leads to degradation of performance. Additionally, the higher stresses associated with the higher vibration may lead to long term fatigue failure of some part of the machine.

Some malfunctions, called *secondary malfunctions*, appear as a result of other malfunctions. A *root cause* is the primary malfunction that triggers a secondary malfunction. In the example above, the high vibration at the balance resonance produces a seal rub, a secondary malfunction. The actual root cause is the unbalance. It is vital to identify the root cause of any machine problem, otherwise, a malfunction may seem to be corrected, only to reappear again at some later time. In our example, replacing the damaged seal does not address the root cause malfunction.

Assuming a good initial design, malfunctions almost always result from unplanned deviations from expected behavior. The beautiful, perfect machine on the drawing board has perfectly straight and round shafts with no gravity sag; is perfectly balanced; has perfect internal and external alignment; operates within design loads; operates within designed radial and axial clearances; has correctly sized bearings and seals; has low stress levels; and has no radial, torsional, or axial vibration.

Unfortunately, real machines can have bowed, out-of-round rotors that sag from gravity and are unbalanced; internal and external misalignment that cause excessive radial loads; clearance problems due to differential thermal expansion during startup; over- or undersized bearings and seals (or improperly installed bearings); high stresses; and radial, torsional, and axial vibration. Usually some combination of these things will be present in any machine at any given time.

The design must allow for these conditions in the manufacture, installation, and operation of the machine. If these imperfections are within the allowed tolerances, then the machine will operate without a problem. A malfunction will occur when one or more of these imperfections exceed these limits.

Detection of Malfunctions

Our machinery management objective is to detect and identify machine malfunctions at the earliest feasible time, given a balance of economic and technical considerations. Timely malfunction detection relies on using all the data that is available for the machine. There are two basic types of measurements that are used to detect and identify machine malfunctions: direct measurements and indirect measurements.

Direct measurements give us information about the physical state of machine components. Direct measurements include vibration and position measurements, rotor speed, and bearing temperature.

Indirect measurements tell us how the machine is behaving as part of a larger system. Indirect measurements include process data, such as power; working fluid temperature, pressure, and flow; and performance data, such as efficiency. These measurements are very important for correlation with direct measurements, but we must have direct measurements to diagnose malfunctions.

Malfunctions are most often detected through changes in vibration. In Chapter 11 we learned that vibration is the ratio of the force to the Dynamic Stiffness of the rotor system (Figure 17-1). Thus, a change in vibration indicates that either the force has changed, the Dynamic Stiffness has changed, or both. The force in rotor systems is most often rotating unbalance, which can change due to the erosion or loss of material or accumulation of foreign material on rotating parts. However, most often, a change in vibration is caused by a change in Dynamic Stiffness.

Dynamic Stiffness consists of Direct and Quadrature Dynamic Stiffnesses, which can be broken down into rotor parameter-related stiffnesses. Thus, changes in any of these parameters can produce a change in Dynamic Stiffness and a change in vibration. While a significant change in mass is unlikely, and

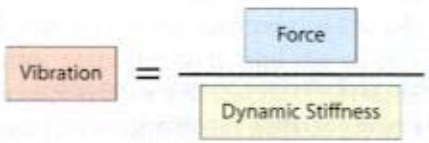


Figure 17-1. Vibration is a ratio. Changes in vibration can occur due to a change in force or Dynamic Stiffness. A malfunction can cause a change in any of the rotor parameters (most often spring stiffness) and cause a change (not always an increase) in vibration. Dynamic Stiffness consists of Direct Dynamic Stiffness and Quadrature Dynamic Stiffness. Dynamic Stiffness is a function of the rotor parameters of mass, M, spring stiffness, K, damping, D, lambda, λ, and rotor speed, Ω; changes in any of these parameters change vibration response.

changes in damping produce subtle effects, changes in lambda and spring stiffness are more likely and will produce significant vibration changes.

Malfunction diagnosis depends on the detection of changes in key signal characteristics (Table 17-1): direct vibration amplitude, nX amplitude and phase, frequency content, direct and nX orbit and timebase shapes, and rotor position, including eccentricity ratio and attitude angle. These vibration and position changes must be correlated with our knowledge of the physical mechanisms of malfunctions, how the malfunctions affect rotor parameters, and the resulting response characteristics.

Most malfunctions have overlapping symptoms. Because of this, a single piece of data is rarely adequate to make a clear diagnosis, and many types of data must be cross correlated. Thus, vibration analysis should be performed using data from all operating modes of the machine (Table 17-2): *steady state*, *transient*, *slow roll*, and *stopped*.

Steady state data (data taken during operation at constant speed) gives us information about changes in overall vibration levels, frequency content, nX amplitude and phase, position, and orbit and timebase shape during slowly changing, or static, rotor dynamic and process conditions. This is an extensive amount of information, but the single operating speed provides only a narrow window on machine behavior.

Transient data (data taken during startup or shutdown) gives us a much wider picture of the rotor dynamic behavior of the machine. It gives us information about heavy spot locations, balance resonance speeds, and how vibration,

Table 17-1. Vibration signal characteristics.

Vibration signal characteristics	Data plots
Direct vibration amplitude	Orbit, timebase
nX amplitude and phase	Bode, polar, APHT
Frequency	Full spectrum, half spectrum
Position	Average shaft centerline
Orbit and timebase shape	Orbit, timebase

Table 17-2. Machine operating conditions.

Machine operating conditions	Description
Steady state	Constant speed, load can vary
Transient	Startup, shutdown (changing speed)
Slow roll	Low speed, negligible unbalance response
Stopped	Stopped

frequency content, position, and orbit and timebase shape change versus rotor speed.

Transient data is especially important because it reveals changes in the balance resonance speed, Ω_{res},

$$\Omega_{res} = \sqrt{\frac{K}{M}} \tag{17-1}$$

where K is the rotor system spring stiffness, and M is the rotor mass. Many rotor system malfunctions produce changes in K and, therefore, the resonance speed. Also, changes in the shape of the resonance peak on a Bode plot can give us clues about the nature of the malfunction.

Slow roll data allows us to identify slow roll vectors, slow roll waveforms, slow roll shaft position, and mechanical and electrical runout under cold and hot conditions. Slow roll vectors and waveforms can be used to compensate transient vector plots, orbit plots, and timebase plots. Slow roll data can also reveal a bowed shaft.

When a machine is stopped, the shaft centerline position can be measured. This is most useful in horizontal machines, where the shaft should rest on the bottom of the bearing in a properly aligned machine. Shutdown or startup shaft centerline data can be plotted relative to the stopped readings to display the relative position of the shaft versus speed.

A stopped machine should not vibrate. This is the time to check vibration signals for the presence of adjacent or auxiliary machinery vibration, or for electrical noise in the transducer signal path. Adjacent machines can transmit vibration through the foundation or piping system and cause casing and shaft vibration in the stopped machine. When the stopped machine is restarted, the adjacent machinery vibration or noise signals will be present in the vibration signal. By knowing that another machine is the source of a particular vibration signal, a more effective diagnosis can be made.

To properly diagnose a machinery problem, you must understand the physical basis for a malfunction, the rotor dynamic response to it, and the vibration signal characteristics of it. In the following chapters, we will discuss the physical cause of each type of malfunction, note how the malfunction affects Dynamic Stiffness, and present typical symptoms and signal characteristics. Because there is so much overlap in malfunction symptoms, we will also list other malfunctions that have similar symptoms.

Chapter 18

Unbalance

UNBALANCE IS THE MOST COMMON ROTOR SYSTEM MALFUNCTION. Its primary symptom is 1X vibration, which, when excessive, can lead to fatigue of machine components. In extreme cases, it can cause wear in bearings or internal rubs that can damage seals and degrade machine performance. Usually, whenever increased 1X vibration is detected, the immediate suspect is unbalance. However, because there are many other malfunctions that produce 1X vibration, many machines have been balanced only to have the real root cause problem reemerge. Thus, to properly diagnose unbalance, it is important to understand both the mechanism that produces unbalance and the other malfunctions that produce 1X vibration.

The cause of unbalance and the techniques used to correct it have been discussed in Chapter 16. In this chapter, we will concentrate on the diagnosis of unbalance as a malfunction. We will discuss the effects of unbalance on rotor system behavior; how it is manifested in rotor and casing vibration, stresses, and secondary malfunctions, such as rub. We will list the other malfunctions that produce 1X vibration and mimic unbalance. Finally, we will discuss a special case of unbalance, the loose rotating part.

Rotor System Vibration Due To Unbalance

The 1X unbalance force acts through the Dynamic Stiffness of the system to cause 1X vibration:

$$\text{Vibration} = \frac{\text{Force}}{\text{Dynamic Stiffness}} \tag{18-1}$$

Because the unbalance is part of the rotor, it rotates at the same speed as the rotor. Thus, the force caused by unbalance is *synchronous* (1X). A linear system will produce only 1X vibration for a 1X force. However, rotor systems possess nonlinearities, such as strongly increasing fluid-film bearing stiffness at high eccentricity ratios. Another source of nonlinearity would be any sudden change in rotor system stiffness, such as due to a rotor-to-stator rub or looseness in the support system. These nonlinearities can generate harmonics of 1X vibration, which can sometimes be seen on a spectrum cascade plot when the rotor is at a resonance. During resonance, the higher 1X vibration amplitude can cause the rotor to pass through a higher eccentricity ratio region in a fluid-film bearing, and the sharp increase in stiffness can produce harmonics.

1X rotor vibration appears as a dynamic load in the bearings. The bearing stiffness and damping transmit this load into the bearing support structure and machine casing, which are part of the extended rotor system. This system will have vibration modes that include the rotor and the machine casing. These modes may be in phase, where rotor and casing move approximately together, or out of phase, where rotor and casing move approximately opposite to each other. The amount of casing vibration will depend on several factors, including the relative masses of the rotor and casing, the stiffness of the bearings, the stiffness of the casing, and the stiffness of the casing mounting and foundation (see Chapter 12).

The masses, damping, and stiffnesses of the bearing, casing, and mounting combine into the Dynamic Stiffness of the overall rotor support, and it is this Dynamic Stiffness that will determine the relative amounts of shaft and casing vibration excited by the unbalance.

A high ratio of casing mass to rotor mass will usually result in low casing vibration. High pressure compressors and the HP unit of steam turbines fall into this category. Because of the high casing mass, vibration in these machines is best measured using shaft relative transducers.

A lower ratio of casing mass to rotor mass, or a soft support, is likely to produce significant amounts of both shaft relative and casing vibration, which requires casing transducers in addition to shaft relative transducers for a complete vibration picture. Shaft relative transducers move with the casing or bearing housing; if the rotor and casing are vibrating in phase, shaft relative vibration may be low, even though shaft absolute vibration may be high. If the rotor and casing are vibrating out of phase, then measured shaft relative vibration may be high, even though shaft absolute vibration may be low. For this reason, casing measurements should be used with shaft relative measurements to provide a complete picture of system vibration.

Aeroderivative gas turbines have low casing mass to rotor mass ratios and relatively flexible casings and supports. They also have rolling element bearings that provide high transmissibility of rotor vibration to the casing. The casings on such machines, due to their flexibility, may possess several modes of casing vibration, and casing transducers must be carefully positioned to avoid nodal points.

When an unbalanced rotor is highly loaded in a fluid-film bearing (perhaps due to misalignment), it operates at a very high eccentricity ratio. The bearing stiffness can become quite high and transmit rotor vibration very easily to the machine casing. At the same time, the high bearing stiffness can suppress the vibration of the rotor. When this occurs, shaft relative rotor orbits will be small and 1X casing vibration will be large.

Similarly, rolling element bearings, because of their extremely high stiffness, strongly suppress shaft relative rotor vibration near the bearings (although *midspan* shaft relative vibration can be quite large) and transmit vibration directly to the casing. A rolling element bearing machine with a rigid rotor behaves as though the casing and rotor are combined into a large, lumped mass with an unbalance. The resulting casing vibration depends to a great deal on the stiffness of the casing structure and the stiffness of the machine mounting. Very stiffly mounted machines have low levels of measurable 1X casing vibration even though internal dynamic forces and stresses may be high.

Stress and Damage

If a rotor centerline moves in a 1X, circular orbit centered on the rotor system axis, the rotor will maintain a constant deflection shape as it rotates. To an observer rotating with the shaft, the rotor will appear to be statically deflected. Under this condition, all areas of the rotor surface will see no change in stress (Figure 18-1, top).

A static radial load (either from a process load or the gravity load on a horizontal rotor) will deflect the rotor and produce 1X stress cycling. The constant stress due to the 1X, circular orbit is added to the alternating stress due to the static radial load deflection. If the orbit is elliptical, an additional 2X component of stress may appear (Figure 18-1, bottom). As the size of the orbit increases, the alternating component of the stress also increases. Add any nonsynchronous vibration, and it is easy to see that a typical rotor operates in a very complicated stress environment.

The nominal stresses due to bending of the rotor are increased by various stress concentration factors, such as diameter changes, keyways, (drilled) holes, shrink fits, surface finish defects, slag inclusions, and corrosion. Thus, rotors

that experience large radial loads and high 1X vibration due to unbalance are at increased risk for crack initiation and fatigue failure.

Unbalance-induced vibration can damage couplings. The bending of the rotor due to unbalance increases stress on rigid, diaphragm, and disk-pack couplings, and increases wear in gear couplings.

1X vibration can also excite casing and piping resonances if their natural frequencies coincide with running speed. The high stresses caused by the resonance response can cause fatigue failure of either the casing or the piping. In one example, a poorly supported steam injection pipe attached to a gas turbine experienced large, 1X excitation of the resonance vibration during startups and, to a lesser degree, broadband rumble excitation of the resonance at running speed. The combination caused catastrophic fatigue failure of the pipe.

High 1X vibration can cause the rotor to contact stationary parts in the machine, a condition called rub. The rub is most likely to occur during a resonance and can damage seals, which can reduce efficiency.

Figure 18-2 shows a 1X Bode plot from a cold startup of a compressor with excessive unbalance. As the rotor approached the first resonance, a simple bending mode shape, it contacted a midspan seal, producing a rub. The effect of the rub is visible in the truncated resonance of the amplitude plot and the high rate of change of phase. The measurement probe was located near one of the bearings in this two bearing machine, so the midspan amplitude, where the rub occurred, was much larger than that shown on the plot.

Other Things That Can Look Like Unbalance

There is a temptation to view any high 1X vibration problem as an unbalance problem. Unfortunately, there are many conditions that can generate 1X vibration or indirectly cause an unbalance in the rotor. If the true root cause of the problem is not found, then balancing the machine may seem to be the only solution. But the symptoms will probably return, and the potentially damaging operating condition will continue. For example, a cracked rotor, because of crack-induced rotor bow, can look like an unbalance problem. But, balancing a cracked rotor will only temporarily address the symptoms of the problem, and the real root cause will remain.

If a machine does not balance or stay in balance after three (presumably correct) attempts at balancing, then the root problem is probably not simple unbalance. The problem lies elsewhere, and further investigation is warranted.

Many of the following malfunctions have one thing in common, a significant response at slow roll speed. *Any 1X response at slow roll speed must be due to something other than unbalance:* runout, a permanent bow in the rotor, severe misalignment, a coupling problem, etc.

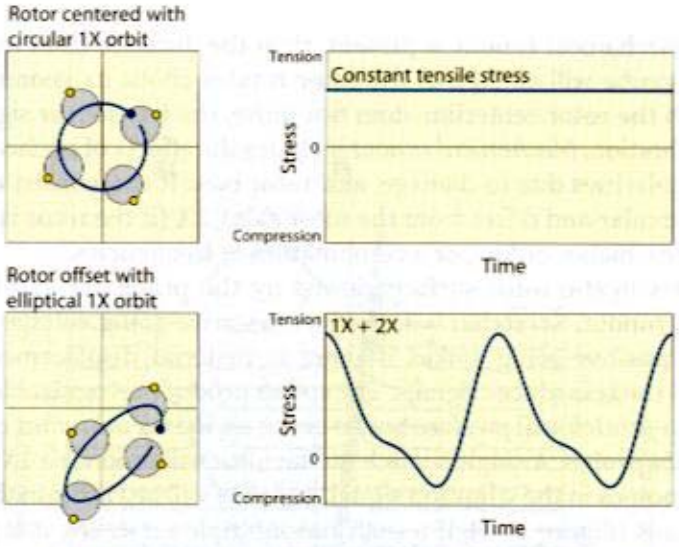


Figure 18-1. 1X vibration and stress cycling. A circular, 1X orbit, centered on the rotor system axis, produces a constant stress level (top). When the rotor is subjected to a static radial load, the constant stress due to the circular orbit is added to any alternating stress due to the static radial load deflection. If the orbit is elliptical, an additional 2X component of stress may appear (bottom).

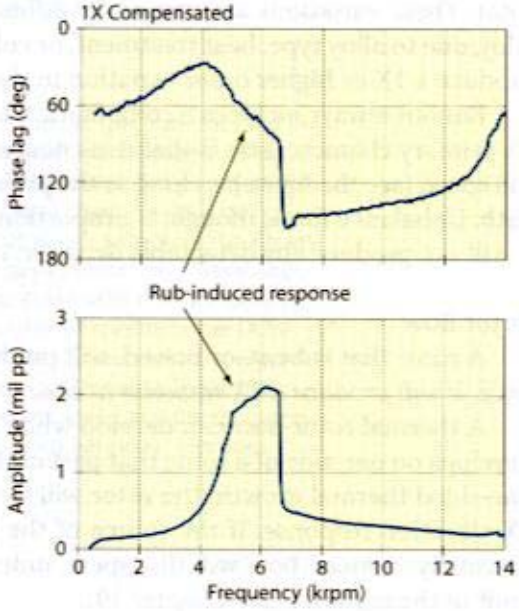


Figure 18-2. 1X startup data from a compressor with excessive unbalance. When the rotor passed through its first-bending-mode resonance, it contacted a midspan seal, producing a rub. The rub is visible in the resonance amplitude and phase response, which has been modified from its usual classical appearance. The vibration was measured near one of the bearings in this two bearing machine, and the midspan vibration amplitude, where the rub occurred, was much larger.

Runout

When mechanical runout is present, then the distance from the rotor surface to the probe will change as the rotor rotates about its geometric center; even though the rotor centerline does not move, the transducer signal is interpreted as vibration. *Mechanical runout* includes the effects of surface machining errors, irregularities due to damage, and rotor bow. It can consist of 1X (if the surface is circular and offset from the rotor axis), 2X (if the rotor is elliptical in cross section), higher orders, or a combination of frequencies.

Scratches in the rotor surface viewed by the probe are another form of mechanical runout. Scratches will produce negative-going voltage spikes (and sometimes positive-going spikes if there is material displacement, not just removal) on the transducer signals. The spikes produce recognizable orbit characteristics: *a scratch will produce spikes on an unfiltered orbit that usually point away from the probes.* A single scratch on the rotor will produce a 1X component and its harmonics in the vibration signal, and they will be visible in the spectrum over all speeds (Figure 18-3). If a shaft has multiple scratches, it is possible for two scratches to affect two probes simultaneously. When this happens, one spike will appear on the orbit that moves away from both probes (but not directly away from either), and two more will appear that move directly away from each probe. The three spikes will be spaced at 90° of rotation from each other.

Electrical runout occurs with eddy current transducers when the electrical conductivity or magnetic permeability varies around the circumference of the rotor. These variations are caused by differences in the microstructure of the alloy, due to alloy type, heat treatment, or cold working. Any of these effects can produce a 1X or higher order variation in the probe signal.

Runout always includes a combination of mechanical and electrical runout. Its primary characteristic is that it is constant in amplitude, even down to slow roll speed (see the figure), as long as the probe looks at the same circumferential path. Unbalance force, though, is proportional to the square of rotor speed, and it will not produce any detectable dynamic 1X rotor response at slow roll speed.

Rotor Bow

A rotor that is bent, or bowed, will produce 1X vibration, but unlike unbalance, *it will produce a 1X response at slow roll speed.*

A thermal rotor bow can develop while a machine is running. If a hot spot develops on one side of a rotor, that part of the rotor will expand. Because of the one-sided thermal growth, the rotor will develop a bow, which will change the 1X vibration response. If the source of the local heating is removed, then the thermally induced bow will disappear, unless the area has exceeded the yield limit of the material (see Chapter 19).

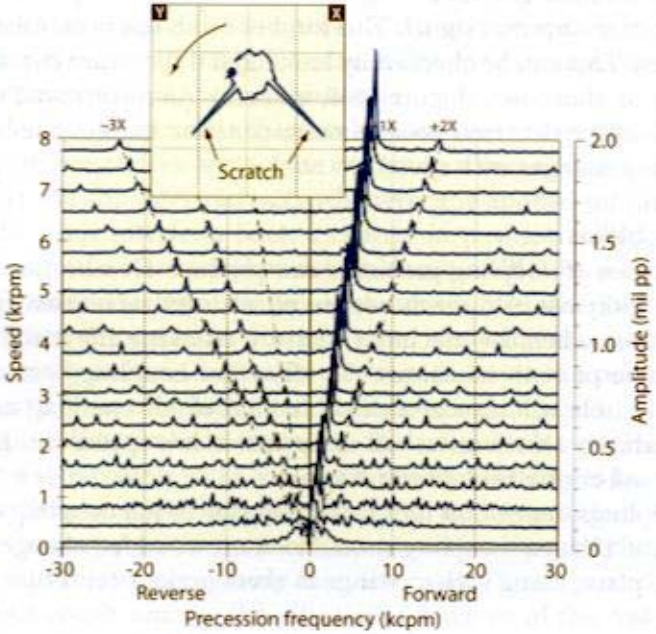


Figure 18-3. Direct orbit and full spectrum showing the effect of a scratch on the shaft in the probe viewing area. As the shaft rotates, the scratch first passes under the *X* transducer, producing a sharp, negative-going voltage spike in the orbit that points away from the *X* transducer. The scratch then passes under the *Y* transducer, producing another spike in the orbit that points away from the *Y* transducer. In the full spectrum, the 1X frequency consists of both the 1X rotor dynamic response and the once-per-turn scratch response. The scratch response persists to slow roll speed, and the sharpness of the scratch produces a rich, harmonic spectrum.

Electrical Noise in the Transducer System

Turbogenerator sets operate either at line frequency or at some submultiple of line frequency, depending on the number of poles in the generator. A 2-pole generator will operate at 3600 rpm (60 Hz) or 3000 rpm (50 Hz). If power line noise couples into the transducer signal line, then a line frequency component will appear in the vibration signal. This kind of noise has been mistaken for 1X rotor vibration. This can be checked by looking at a spectrum cascade plot during a startup or shutdown (Figure 18-5, bottom). As rotor speed changes, the spectrum line of the electrical noise remains constant in frequency while the 1X rotor frequency changes with speed.

Coupling Problems

Certain types of coupling problems can produce 1X vibration. If the rotor axes of two rigidly coupled machines are offset from each other (parallel misalignment), then, when the machines rotate, a cranking effect will produce 1X vibration in one or both machines. An off-center coupling bore or off-center coupling bolt circle will also produce this kind of 1X cranking action. While unbalance-induced 1X vibration will disappear at low speed, cranking-induced 1X vibration will continue at slow roll speed.

Gear couplings depend on lubricated slippage of the coupling elements for their operation. If a gear coupling should lock up, a sudden change in 1X vibration can take place, along with a change in average shaft centerline position.

Shaft Crack

As a shaft crack propagates across the rotor, the rotor bending stiffness decreases in the vicinity of the crack. Thus, the rotor is less able to resist the dynamic forces that try to deform the rotor and will usually bow as the crack develops. Because the bow moves rotor mass away from the rotor axis, the effective heavy spot of the rotor changes. The crack-induced bow changes the 1X vibration response of the machine.

Shaft cracks usually cause changes in 1X amplitude or phase over time. In the first weeks to months of crack propagation, the 1X response will usually change slowly. The changes in 1X vibration may be mistaken for simple unbalance. While balancing may reduce the vibration due to the crack-induced bow, the root cause problem still remains, and the 1X response will change again.

If the crack is well developed, the large amplitudes of vibration associated with resonance can plastically deform the rotor in the vicinity of the crack, suddenly changing the bow, 1X slow roll vectors, and the effective heavy spot. Balancing calculations based on shutdown data may not be correct after the

rotor experiences high amplitude resonances during startup. In this scenario, the rotor will have an erratic response to repeated balancing attempts.

Loose Part or Debris

If a part shifts position on the rotor, or debris shifts position in the rotor, the unbalance distribution and the resulting 1X vibration response of the rotor will change. Such a change can happen occasionally (for example, during a startup or shutdown), intermittently, or continuously.

A rotor disk or thrust collar that has become loose may rotate on the rotor. Under some circumstances, a machine component can also move axially. Most likely, the part will slip intermittently whenever the applied torque exceeds the friction at the rotor interface. Such a change in position will produce a step change in 1X vibration that could be detected on an APHT or acceptance region plot. If the part slips during a startup or shutdown, the observed vibration will be different when compared to previous data. Note that the movement of the part might actually *reduce* 1X vibration if the part moves to a position where its unbalance partially or completely cancels the rotor unbalance.

Catastrophic failure of a component, such as a broken turbine blade, will also produce a step change in 1X vibration, but that is likely to be of much larger magnitude than what would be produced by a shift in position of a rotor part.

If the friction between the part and the rotor is low enough, the part may slip on the rotor and rotate more slowly than the rotor. The unbalance of the loose part will continuously change the effective unbalance of the rest of the rotor. This case will be discussed in more detail below.

Fluid or debris can become trapped inside a rotor. When the rotor is shut down, the debris can shift position, changing the unbalance of the rotor. In one case, a 6 m (20 ft) diameter, 300 kW (400 hp), 880 rpm induced-draft fan exhibited a recurring unbalance problem. Over a period of two years, the fan was balanced 24 times! Finally, a new consultant was called in, and he discovered that dirt was trapped in the hub of the fan. Every time the fan was shut down for balancing, the dirt shifted position, rendering the balance shot useless. Once the fan was cleaned and sealed, the problem disappeared. See the case history in Chapter 25.

In another case, the LP shaft borehole plug of a 300 MW steam turbine generator came loose and settled inside the turning gear ring. Each time the turbine was stopped, it settled in a different position. It was only when the turning gear was removed to facilitate alignment that the troubling unbalance problem was solved.

Rub

If a rotor begins to rub lightly on the stator once per turn, the friction at the contact point will cause local heating and bow and transfer energy of rotation to lateral vibration. Thus, the 1X vibration amplitude and phase will change once the rub starts. Over time, the rub may wear away part of the contacting element, and the rub-induced 1X vibration changes may become smaller or disappear.

Under rare circumstances, it is possible to develop a light rub that produces a continuous change in amplitude and phase while the machine is operating in a steady state condition. This type of rub can produce circles on a polar trend plot, with a cycle time of minutes to hours. See Chapter 19 for an example of this behavior. Rub can also modify the spring stiffness of the rotor system and increase or decrease vibration.

Changes in Spring Stiffness

Changes in any of the rotor parameters (mass, damping, spring stiffness, lambda, and rotor speed) will change the Dynamic Stiffness and the 1X vibration response of the rotor. The term that is most likely to change, the spring stiffness term, K, can be affected by many different rotor malfunctions. Table 18-1 lists different operating conditions and malfunctions and their effects on spring stiffness.

When a rotor operates near a resonance, changing K can increase *or* decrease 1X vibration amplitude, depending on whether the rotor operates above or below resonance (Figure 18-4). However, a *decrease* in spring stiffness will always *increase the 1X phase lag* near a resonance, and vice versa.

Thus, an increase in 1X vibration amplitude can look like an unbalance problem when the real problem may be a reduction in rotor system spring stiffness due to a deteriorating foundation, a loose foundation bolt, a change in eccentricity ratio in a fluid-film bearing (possibly due to misalignment), or even a shaft crack. If a spring stiffness *increase* shifts a resonance closer to running speed, vibration may also increase (Ω_4 in Figure 18-4, left).

Electric Motor Related Problems

Electric motors, because of their construction, may become unbalanced in normal operation due to shifting windings and loosened wedges. However, various induction motor malfunctions can also produce 1X vibration and vibration at near 1X frequencies.

The rotor in an electric motor is built from a set of thin, insulated sheets of metal (laminations) that are stacked together along the rotor. If the insulation breaks down, or if the laminations become smeared due to rotor/stator contact, then adjacent laminations can come into electrical contact and larger eddy cur-

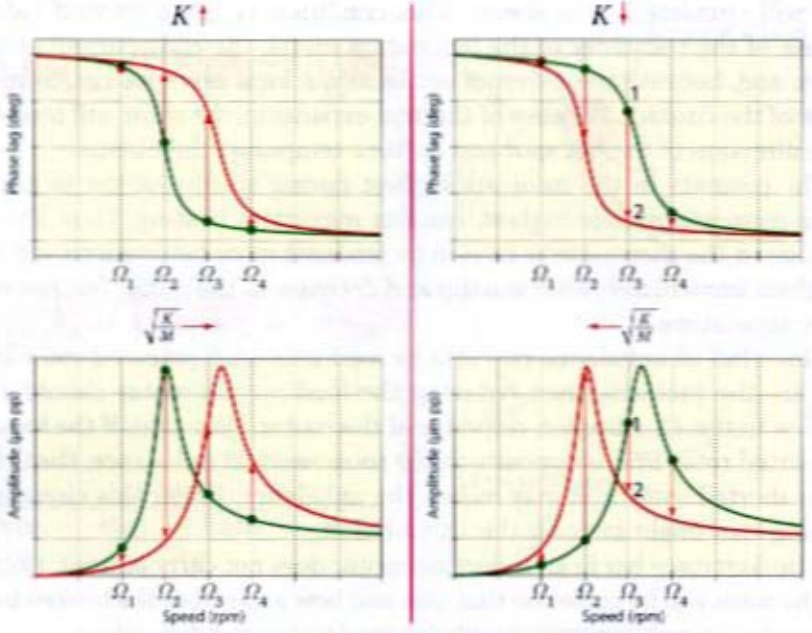


Figure 18-4. Changes in spring stiffness versus changes in vibration. In all plots, the original rotor response curves are green. On the left, the spring stiffness, K, has increased, causing the resonance to shift to a higher speed (red curves). On the right, K has decreased, shifting the resonance to a lower speed. The arrows show how the vibration amplitude and phase would change if the machine operated at speeds above and below the resonance.

Table 18-1. Causes of changes in spring stiffness, K.

Characteristic	Spring stiffness becomes
Shaft crack	Lower
Deteriorating foundation	Lower
Oversize bearing	Lower
Fluid-film bearing lower eccentricity	Lower
Fluid-film bearing higher eccentricity	Higher
Rub	Higher
Locked floating seal	Higher

rents will circulate in the sheets. This condition is called *shorted rotor iron.* Because of the resistivity of the lamination metal, the eddy currents dissipate energy, and, because the currents act locally, a local hot spot can form in the region of the contact. Because of thermal expansion, the rotor will tend to bow in the direction of the hot spot and create a temporary unbalance.

The currents in the rotor are highest during startup of the motor, when torque requirements are highest, causing maximum heating. Thus, the unbalance due to the thermal bow caused by smeared rotor laminations will usually be highest immediately after startup and decrease as the motor reaches equilibrium temperature.

This kind of unbalance can also be load related. If smeared rotor laminations are the problem, then reducing the load on the motor should cause a decrease in the 1X vibration response of the motor. Note that, if the location of the shorted rotor iron is opposite to the rotor residual unbalance, then the bow due to shorted rotor iron may *reduce* the unbalance. Under this circumstance, reducing load might *increase* the 1X vibration.

A broken rotor bar in an induction motor does not carry current. Because of this, the rotor will be cooler on that side and bow away from the broken bar. This bow creates an unbalance that will also tend to be load dependent.

In addition, the bow due to the broken rotor bar will create an uneven air gap between the rotor and the stator. The overall effect will be to modulate the amplitude of the 1X vibration of the rotor at a beat frequency equal to the number of poles of the motor times the slip frequency.

Two tests are helpful, but not conclusive, in diagnosing a broken rotor bar. Cutting power to the motor should cause the beat frequency to immediately disappear, and a load change should change the amplitude of the modulated 1X vibration.

Eccentric rotor iron will produce a 1X rotor vibration. But the interaction of the rotor with the rotating stator magnetic field will produce a modulation of 1X vibration similar to that produced by a broken rotor bar. Cutting power will cause the modulation to disappear, confirming that the problem is not a simple unbalance problem.

If line frequency electrical noise gets into a transducer system, then a 60 Hz (or 50 Hz) frequency will appear in the signal. If the machine train is driven by a 2-pole induction motor, then the line frequency will be only slightly above the 1X frequency. With poor spectral resolution, it may not be possible to separate the line frequency from the rotor frequency. Depending on the signal to noise ratio, the 1X vibration may appear to be modulated at a beat frequency equal to the slip frequency, which is the difference between the rotor speed and twice the line frequency divided by the number of poles.

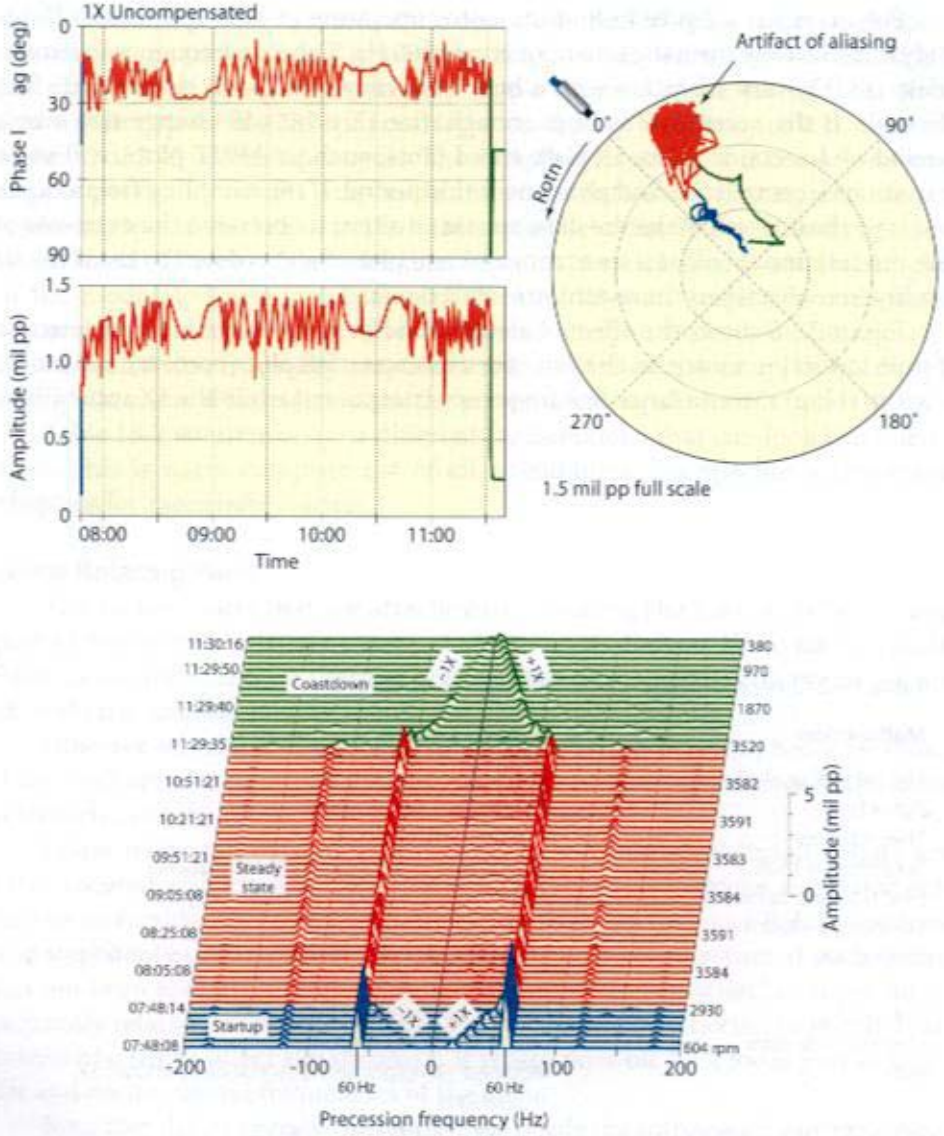


Figure 18-5. The effect of electrical noise in the transducer system of a 2-pole induction motor. In the full spectrum waterfall plot (bottom), the 1X rotor vibration and the large line frequency component can be seen during startup (blue). During steady state operation (red), the two components combine in one frequency bin, but the 1X APHT plots (top) show the modulation of the 1X vibration component by the line frequency component. (The red circular region is filled in because of aliasing. See the text.) Note that line frequency amplitude is highest during the high current demand during startup, drops during steady state, and disappears when the power is cut.

For example, a 2-pole induction motor operating at 3585 rpm (59.75 Hz) is only 0.25 Hz away from the line frequency of 60 Hz. This line frequency will modulate the 1X rotor vibration with a beat frequency of 0.25 Hz, or one cycle in 4 seconds. If the noise level is high enough, the 1X orbit will change size over a period of 4 seconds. Also, 1X data trend plots, such as APHT plots, will show variations in amplitude and phase over this period. If the sampling frequency is too low, aliasing will make the data appear to alternate between the extremes of the modulation. Look at a spectrum cascade plot of a shutdown to see if the 60 Hz line remains when the machine is shut down.

Figure 18-5 shows the effect of electrical noise in the transducer system of a 2-pole induction motor. In the full spectrum waterfall plot (bottom), the motor startup (blue) shows a large, line frequency component while the 1X rotor vibra-

Table 18-2. Other malfunctions that can produce 1X vibration.

Malfunction	Symptom or test
Mechanical or electrical runout	Significant 1X at slow roll
Rotor bow	Significant 1X at slow roll
Thermal bow	VAR test generators
Scratches in shaft	Orbit spikes point away from transducers
Electrical line noise in xdcr system	Always at line frequency, vary rotor speed
Coupling problems	Significant 1X at slow roll
Shaft crack	1X usually changes slowly over time, erratic balancing response
Loose part	Periodic or sudden change in 1X
Debris inside rotor	Erratic balancing response
Rub	Modified 1X response in resonances, periodic 1X changes
Spring stiffness changes	Resonance frequency changes
Induction motor problems	
Shorted rotor iron	Load change affects 1X
Broken rotor bar	1X modulated at poles times slip, power cut
Eccentric rotor iron	1X modulated, power cut

tion is increasing. During steady state operation (red), there is not enough spectral resolution to separate the 1X from the line frequency. But the 1X APHT plots (top) show disturbances in both amplitude and phase versus time. Note that a motor speed of 3584 rpm corresponds to a slip frequency of 0.27 Hz, with a period of 3.75 seconds per cycle. The sampling interval for this data was one sample per minute, so the amplitude and phase data show aliasing and are not representative of the real-time, instantaneous behavior. The solid red region in the polar APHT plot is filled with lines because of aliasing; consecutive samples of the modulated response have significantly different amplitude and phase. Without aliasing, the data would have formed an open circle in the polar plot.

Note that line frequency amplitude is highest during startup, drops during steady state, and disappears when the power is cut for shutdown (green).

Table 18-2 summarizes the different malfunctions that can look like unbalance. This is not a complete list of all possibilities. See specific malfunction chapters for more information.

Loose Rotating Parts

The various parts that are attached to a rotating shaft all contribute some part of the overall unbalance state of the rotor. As long as these parts remain fixed to the rotor, and no thermal bowing, deposition, or erosion processes are at work, the unbalance state should not change significantly.

However, when a disk that is normally attached to the rotor becomes loose, it can shift angular position relative to the shaft, producing a change in the mass distribution of the rotor.

Loose parts can occur for several reasons. Nonintegral thrust collars are often secured with a locknut that can loosen. When this happens, the thrust collar can intermittently or continuously move relative to the rotor. Balance pistons and impellers in compressors can also loosen. In stacked rotors, if an impeller has not been assembled properly onto the rotor, it can "ratchet," or move intermittently relative to the rotor shaft. This is most likely to occur during thermal transients, which affect the shrink fit. It is also possible for a loose part to chatter and excite natural frequencies of the rotor.

Note that debris or fluid that is trapped inside the rotor can change position, either during operation or, most likely, during startup and shutdown. The induced draft fan case history in Chapter 25 is an example of this kind of problem.

The continuous or intermittent change in angular position of the loose part will change the magnitude and direction of the overall unbalance vector, producing a change in 1X vibration, dependent on the amount of unbalance in the part.

Rarely, the part will move continuously, and the 1X vibration vector will change in a periodic way. When the unbalance vector of the part points in the same direction as the unbalance vector of the rest of the machine, the two vectors will add, increasing the 1X response. When the unbalance vector points in the opposite direction from the machine unbalance vector, then the two vectors will subtract.

The frequency of the vibration modulation (beat), f, in Hz is related to the difference between the rotor speed, Ω, and the speed of the loose part, ω (in rpm):

$$f(\text{Hz}) = (\Omega - \omega)\left(\frac{\text{rev}}{\text{min}}\right)\left(\frac{1 \text{ cycle}}{\text{rev}}\right)\left(\frac{1 \text{ min}}{60 \text{ s}}\right) \tag{18-2}$$

The speed of the loose part is determined by the combination of the hydrodynamic or aerodynamic drag forces that act on the part and the friction force between the part and the rotor surface.

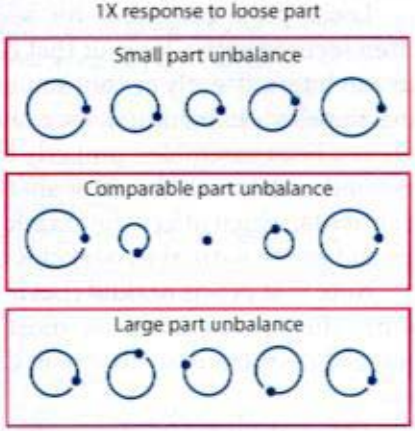


Figure 18-6. Behavior of 1X orbits with a loose rotating part. The top row of orbits shows orbit behavior when the ratio of part unbalance to rotor unbalance is low. The Keyphasor dot moves in a small circle, but remains in the same quadrant of the orbit. The middle row shows the behavior when the part unbalance to rotor unbalance ratio is comparable. In this case, the orbit size will change dramatically, shrinking to a point and reappearing, while the Keyphasor dot moves around the orbit. In both cases, the orbit appears to pulsate at a frequency defined by Equation 18-2. The bottom row shows the effect of a large, loose part unbalance combined with a rotor with little or no unbalance. The orbit remains open while the Keyphasor dot moves around the orbit in a direction opposite to rotation.

If the sample rate for the 1X vector plots is not at least twice the beat frequency, aliasing may produce strange results on the plots. For this reason, the modulation of 1X vibration is best seen in 1X timebase plots or on an oscilloscope. The series of 1X orbits (Figure 18-6 top) shows loose rotating part behavior when the ratio of part unbalance to rotor unbalance is low. The Keyphasor dot moves in a small circle, and remains in the same quadrant of the orbit, indicating a small variation in unbalance. The series of 1X orbits (middle) shows behavior when the ratio of loose part unbalance to rotor unbalance is comparable. In this case, the orbit size will change dramatically, collapsing to a point and reappearing, while the Keyphasor dot moves around the orbit in a direction opposite to rotation. In both cases, the orbit appears to pulsate at a frequency defined by Equation 18-2. When the rotor has very little unbalance, a loose, unbalanced part can produce an orbit that remains relatively constant in size, where the Keyphasor dot moves around the perimeter in a direction opposite to rotation (bottom).

Summary

The primary symptom of unbalance is 1X vibration. Unbalance can produce high rotor and casing vibration, and it can produce vibration in foundation and piping systems.

1X vibration can also contribute to stress cycling in rotors, which can lead to eventual fatigue failure. Unbalance-induced vibration can also cause internal rubs in machinery, especially when passing through balance resonances.

Diagnosis of unbalance can be complicated by the fact that many different malfunctions can produce 1X vibration. Mechanical and electrical runout, rotor bow, thermal bow, electrical noise, coupling problems, shaft cracks, loose rotating parts, trapped debris or fluids, rub, decreasing foundation spring stiffness, and various electric motor problems can all produce 1X or near 1X vibration. Because so many malfunctions can masquerade as unbalance, the machinery diagnostician should be careful to establish unbalance as the root cause before balancing a machine.

A loose rotating part can cause an intermittent or continuous change in 1X vibration.

Chapter 19

Rotor Bow

In the previous chapter, we studied unbalance as a malfunction of rotor systems. Rotor bow is another malfunction that can produce 1X vibration. Rotor bow has been known to cause expensive delays in the startup of large steam and gas turbines and has led to significant internal damage of machines. In severe cases, a rotor bow can produce such high levels of vibration that it becomes impossible to pass through a balance resonance. Thus, prevention of rotor bow is an important part of machinery management. Machinery operation guidelines should be strictly followed, supplemented by a thorough understanding of the physical causes of rotor bow.

We will start with a definition of rotor bow and how it is measured. Then we will discuss the underlying physical cause of rotor bow so we can understand why it occurs and how it can be prevented. We will follow with a discussion of the rotor dynamic behavior of rotor bow, its symptoms, and a list of other malfunctions that may exhibit similar symptoms. Finally, we will talk about how to cure rotor bow when it does occur.

What is Rotor Bow?

Rotor bow is a condition that results in a bent shaft centerline. The shaft centerline is most often bent in a single plane, but it can have a three-dimensional corkscrew shape. There are three general categories of bow: elastic, temporary, and permanent.

Elastic bow occurs because of static radial loads on the rotor which do not exceed the yield strength. They can be caused, for example, by process fluid loads in pumps, internal misalignment, and partial arc steam admission in steam turbines, but the most common static load is due to the weight of the

rotor in a horizontal machine. The stresses associated with the gravity load of a rotor (tensile on the bottom of the rotor and compressive on the top) are quite small. For example, a solid steel rotor that has a bearing span of 6 m (20 ft) and is 1 m (3 ft) in diameter will have a midspan elastic gravity sag of about 130 μm (5 mil). The maximum bending stress for this deflection is only 3 MPa (500 psi). Annealed 4140 steel has a yield strength of about 420 MPa (60 000 psi, or 60 ksi). Thus, the gravity load of the rotor produces stresses that are far below the lowest yield strength of a typical rotor steel. However, we will see later that these stresses can contribute indirectly to permanent rotor bow.

Temporary bow can occur due to uneven heating of the rotor surface or due to anisotropic thermal material properties. If a hot spot develops on one side of a rotor, it will thermally expand and become longer than the other. If the rotor is unconstrained, the resulting deflection of the rotor will not be permanent, and the rotor will return to its original shape when the temperature difference is eliminated.

Permanent bow results when the rotor has deformed to a condition that is not self-reversing without special intervention. For this to be true, the stresses in some part of the rotor must have exceeded the yield strength of the material.

Temporary and permanent rotor bows are almost always temperature-related problems; because of this, they are sometimes referred to as *thermal bow*.

All high temperature rotors are vulnerable to permanent bow. HP, IP, and LP steam turbine rotors, gas turbine rotors with shrunk on disks, and rotors built up with disks and tie bolts (typically gas turbine rotors) can experience bow problems.

Eccentricity describes the measurement (in μm pp or mil pp) of shaft bow at low rotor speeds. It is usually measured at the end of the exposed rotor shaft, well away from the bearing. For steam turbines, this is usually near the free end of the HP shaft. Eccentricity monitors are used to measure rotor bow during startup and to alarm when it exceeds an upper limit.

Causes of Rotor Bow

Rotor bow is a complex, often temperature-related problem. During startup and after shutdown, most rotors undergo significant temperature changes. If a hot rotor cools with a uniform surface temperature distribution, and if the material properties are uniform, then the rotor contracts uniformly and remains straight. Uneven cooling is more typical and, when combined with variations in geometry and material properties, causes the resulting rotor thermal behavior to be quite complex. There are several ways that an initially straight rotor can become thermally bowed.

An operating rotor can experience an uneven circumferential temperature distribution. This can be caused by uneven heating in generator and motor windings, shorted turns in generators, and broken rotor bars or shorted rotor iron in induction motors. It can also be caused by the local frictional heating from a rub or uneven heating in fluid-film bearings. The uneven heating causes the rotor to thermally expand on, and bow toward, the hot side. This is usually a temporary bow, which goes away when the uneven heating disappears. However, if the rub is intense and concentrated, the heat generated in the rub region can cause the material to yield in compression. When the rotor cools, the local, permanent plastic deformation will cause the rotor to be slightly shorter in that area, and a permanent bow will develop in the opposite direction from the rub.

Temporary bow forms as a result of differential thermal expansion of the rotor (Figure 19-1). At the top, a stopped, unconstrained rotor is at a uniform temperature in free space (no gravity load). When the top of the rotor is heated and the bottom is cooled, compressive stresses form inside the upper half of the rotor, and tensile stresses in the bottom half (middle). These stresses produce a bending moment that warps the rotor in the warm direction until the differential stresses disappear. After warping, the top surface is longer and the bottom surface is shorter, and no internal stresses exist in the rotor. (If they did, they would have to relieve themselves by further warping of the rotor.) When the source of differential heating is removed from the rotor, the process will reverse and return the rotor to its original shape.

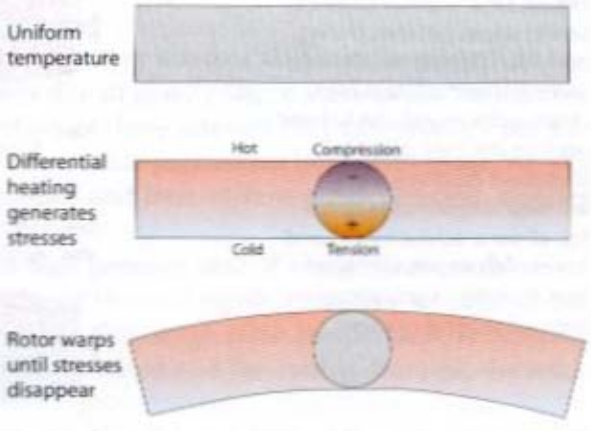


Figure 19-1. Thermal warping of a rotor. At top is a solid rotor in free space, at uniform temperature, with no gravity load. When the top of the rotor is heated and the bottom is cooled, compressive stresses (negative) form inside the upper half of the rotor, with tensile stresses (positive) in the bottom half. These stresses produce an internal bending moment that warps the rotor until the differential stresses disappear. After warping, the top surface is longer and the bottom surface is shorter, and no internal stresses exist in the rotor.

A rotor, stopped or not, can develop a temporary thermal bow if the material properties are nonuniform across the section. During manufacturing, rotors made from a solid forging that experience different rates of cooling (leading to differences in heat treatment and microstructure), can have a slightly different value of thermal conductivity or coefficient of thermal expansion in different regions in the rotor. Differences in thermal conductivity will contribute to an unequal temperature distribution during thermal transients. Differences in the coefficient of thermal expansion in different areas will cause variation in the thermal expansion. The combination of these factors can cause bow during both transient and uniform temperature conditions.

This mechanism is similar to the thermal changes in a bimetal strip, which consists of two, dissimilar metals bonded together. One of the metals has a thermal expansion coefficient higher than the other. When subjected to a uniform temperature increase, the metal with higher thermal expansion will expand more than the other, causing the bimetal strip to warp.

After shutdown, it is common for a *temperature stratification* to occur inside a steam or gas turbine machine casing (Figure 19-2), which produces a higher temperature at the top of a horizontal rotor. The cooler (bottom) side of the rotor will contract more than the warmer (top) side. Because the cool side becomes shorter than the warm side, the rotor will deflect in the warm direction, a thermal bow. As long as the rotor is unconstrained, the stresses that develop in the rotor will be small. However, if the upward-bowing rotor contacts

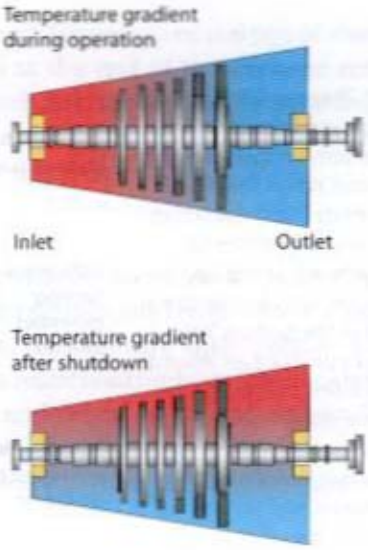


Figure 19-2 Thermal stratification in steam turbines. During operation (top), an axial temperature gradient exists in the steam turbine, with the hot end (red) on the inlet side. After shutdown (bottom), the warmest steam moves to the top of the machine casing, and the coolest moves to the bottom, forming a vertical (top-to-bottom) thermal gradient.

internal stator parts, it will become constrained, and the internal rotor stresses can build to high levels, with tension on the bottom and compression on the top. If the stresses exceed the yield limit, then, after reaching thermal equilibrium, the rotor will be shorter on top and longer on the bottom and will bow downward.

Immediately after shutdown, a stopped rotor will begin to cool at the outer surface. Because of the size and heat capacity of large turbine rotors, it can take hours for the rotor to reach thermal equilibrium. During this period, another thermal gradient exists between the outer surface and the center of the rotor (or the surface of the bore hole), and thermal stresses appear in the rotor that can reach a significant fraction of the yield strength. For a hot rotor that is cooling, the outer surface tries to shrink and forms tensile stresses, while compressive stresses develop at the bore hole surface. For a cold rotor that is heating, the stresses change sign: tension appears at the bore hole surface, with compression at the outer surface.

Thermal Shock

Thermal shock occurs whenever an object is subjected to a sudden temperature change on its surface. The stresses generated can be so high that they can cause fracture and catastrophic failure of the object.

For example, if you drop an ice cube (the colder the better) into warm water (sometimes even tap water will do), it is very likely to shatter *inside*, generating a network of cracks in the middle of the cube. This is thermal shock: the outer surface heats rapidly and tries to expand, but the surface material is constrained by its attachment to interior material. As a result, the interior material is put into tension. If the tensile stress exceeds the strength of the material, then the ice cube, a brittle material, fractures.

Thermal shock caused by rapid cooling of the surface of a hot object causes the outer surface to attempt to contract faster than the interior. As a result, the outer surface goes into tension, and the interior goes into compression. If the stresses are high enough, cracks can form on the surface of the object.

The thermal stresses add to any other stresses that may be present in the rotor, such as elastic bow stresses due to gravity sag or stresses due to thermal stratification. It is the combination of these stresses that can produce permanent bow of the rotor. To understand how permanent bow takes place, we will work through a shutdown sequence and look at how these different stresses interact.

Assume that the rotor in the high pressure case of a large steam turbine has been operating long enough to be at thermal equilibrium. For an inlet steam temperature of 540 °C (1000 °F), the first stage rotor temperature is typically 480 °C (900 °F). When shutdown is initiated and the load is dumped, the rotor coasts down.

At this point, good operating practice dictates putting the rotor on turning

gear when it reaches a low speed, to prevent the sequence we are going to describe. Let us assume that, for some reason, the rotor stops. The hot rotor is statically deflected downward by its own weight, producing low level, axial tensile stresses at the bottom and equal compressive stresses at the top.

As the rotor begins to cool, two things typically occur, temperature stratification in the machine casing and overall cooling of the rotor surface.

Temperature stratification inside the machine casing creates a *top-to-bottom temperature gradient* in the rotor, which causes the top of the rotor to expand relative to the bottom. The rotor deforms upward, producing, at this point, a temporary thermal bow. The thermal bow is easily able to overcome the gravity load of the rotor. If the rotor does not contact the stator, no new stress will appear due to this thermal gradient. However, because of the gravity load, the top of the rotor is still in compression, the bottom of the rotor is still in tension, and a low stress distribution exists across the entire rotor section (Figure 19-3a).

If the rotor contacts the stator and becomes constrained, the thermal expansion of the top surface will continue, but the rotor will be unable to move to relieve the situation. The axial compressive stress at the top of rotor, and the axial tensile stress at the bottom will both increase. Two sources of axial stress now exist, the gravity stress and the constrained thermal stress due to the top-to-bottom thermal gradient (Figure 19-3b). Both stresses have the same effect: compression on the top of the rotor, tension on the bottom.

In practice, when the temperature stratification occurs, the casing also tends to bow (*hog*), and the center of the casing rises, to some extent, along with the rotor. This actually helps delay contact between the rotor and the casing. There have been cases where, after the rotor failed to go on turning gear, operators tried to turn it manually, only to find that the rotor turned freely at first, but became tight after 180° of rotation, where the bow of the rotor was down and the bow of the case was up.

While temperature stratification creates a top-to-bottom thermal gradient, overall cooling of the rotor surface creates a *surface-to-core thermal gradient*, which produces another set of high stresses inside the rotor. These stresses are due to a situation similar to *thermal shock* (see Sidebar). This thermal gradient produces high tensile stresses near the outer surface of the rotor and compressive stresses in the center of the rotor or at the surface of the bore hole (Figure 19-3c). These stresses act in the circumferential and axial directions.

When all of the stresses are considered, the surface-to-core tensile stress at the rotor surface *subtracts* from the other compressive stresses at the top of the rotor, while it *adds* to the other tensile stresses at the bottom (Figure 19-3d). If the total tensile stress at the bottom is greater than the yield strength of the

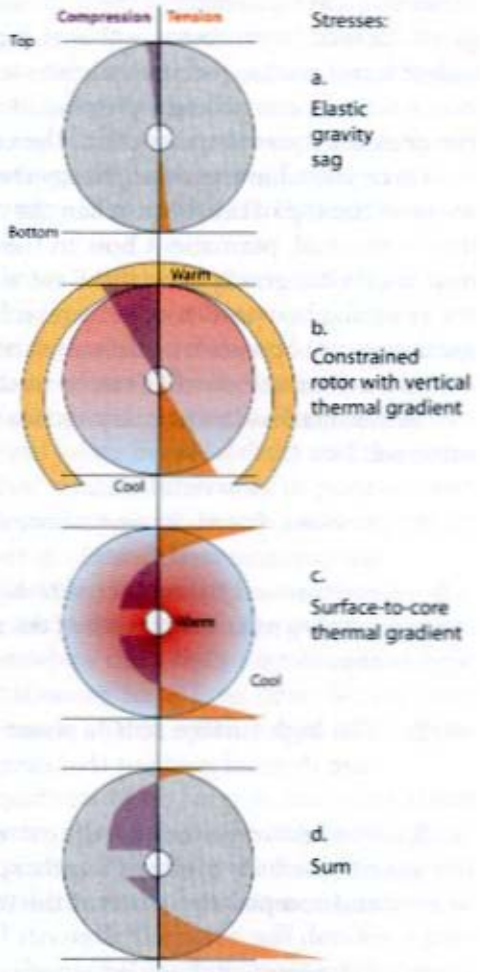


Figure 19-3. Stresses in a hot, stopped, horizontal rotor. The rotor, with a bore hole, sags due to its own weight. This produces low level tensile stress (orange) at the bottom of the rotor and compressive stress (violet) at the top (a).

After shutdown, thermal stratification often develops in the machine casing, and the resultant vertical thermal gradient in the rotor causes it to bow upward. If the rotor contacts the stator when it bows upward, then large stresses will develop at the rotor surface (b).

Another stress distribution forms as a result of the surface-to-core thermal gradient created as the entire rotor cools (c).

This produces large stresses that are tensile at the surface and compressive at the bore hole surface. All of these stresses add to produce the total stress distribution on the rotor (d).

If no rotor to stator contact occurs, then only the gravity sag and surface-to-core stresses will exist. With either scenario, the potential exists to develop very large tensile stresses at the bottom of the rotor. If these stresses exceed the yield strength, plastic deformation will occur, usually around stress concentrators such as inclusions and geometric discontinuities, and the rotor will bow downward when cool.

material, then gross plastic deformation will occur along the bottom of the rotor over a large area.

Even if the stress at the bottom remains below the yield strength, there are still several factors that can increase the stress. Slag inclusions in the rotor forging or geometric discontinuities (such as diameter changes) are localized stress concentrators that multiply the nominal stress. Thus, the combination of elastic gravity stress, top-to-bottom thermal gradient stress (which will be significant only if stator contact occurs), surface-to-core thermal gradient stress, and stress concentrators can produce yielding in areas of the bottom of the rotor, even if the nominal stress remains below the yield strength.

Once they have yielded, these areas become longer than corresponding areas on the top of the rotor. When the rotor cools to thermal equilibrium, it will have a residual, permanent bow in the *downward* direction, in addition to the original elastic gravity bow. If the rotor bow was constrained by stator contact, the resulting bow can be quite severe. If the yielding occurred only locally near inclusions or at geometric discontinuities, though, then the result is likely to be a mild permanent bow that can be worked out later during startup.

To summarize, the primary causes of permanent thermal bow in a stopped rotor are

1. Stresses due to a top-to-bottom temperature gradient due to temperature stratification in the machine casing. If the rotor is constrained by contact with the stator, high tensile stress can develop near the bottom of the rotor, with compressive stress on top.

2. The high surface tensile stress due to the powerful surface-to-core thermal gradient that develops as the rotor begins to cool.

3. The relatively weak bending stresses due to the gravity sag of the rotor, which produce tensile stress at the bottom of the rotor and compressive stress at the top.

4. The presence of any inclusions or geometric discontinuities that produce stress concentration factors.

Rotors can bow during startup if heat is applied too suddenly or unevenly. During a startup, as the rotor surface is heated, surface-to-core thermal gradient stresses appear again, but with reverse sign. The surface of the rotor becomes hotter than the interior, producing tension in the interior. For a typical

steam turbine startup, the outside surface temperature of the rotor can be 100 °C (180 °F) higher than the bore hole surface temperature, and this difference is maintained for hours. For our large, 1 m (3 ft) diameter rotor with a 130 mm (5 in) bore diameter, circumferential and axial stresses at the bore hole surface are about 240 MPa (35 ksi), a substantial fraction of the yield strength of rotor steel. Again, stress concentration factors can increase these nominal stresses significantly and can cause cracking of the bore hole surface or rotor interior, most likely near stress raisers, such as inclusions. Because of the potential for damage due to thermal stress, rotors must be heated very slowly.

Rotors that go through many startup and shutdown cycles experience cyclic thermal stresses that are capable of initiating and propagating fatigue cracks, a process called *thermal fatigue*. Thermal fatigue is most likely in HP and IP rotors that operate at 430 °C (800 °F) and higher.

Our discussion has focused primarily on forged and machined steam turbine rotors. However, both temporary and permanent bows can develop on "built-up" rotors, such as turbine rotors with shrunk on disks and those built up with disks and tie bolts (typically gas turbine rotors). Gravity loads, the temperature effects we have discussed, and *axial* thermal gradients all cause stresses that must be transmitted across mechanical joints between disks and the rotor. Any shifting of parts relative to each other creates differences in position that can lead to bow. This problem can develop during both heating and cooling of the rotor.

Another kind of permanent bow can develop in built-up rotors that are stationary for a long time. As we have seen, the stresses due to the gravity load are too low to cause a permanent bow by themselves, but a built-up rotor of any type may develop a "bow memory" if it is stationary for a long time. Rotors with shrunk-on disks are most vulnerable; because of this, spare rotors of this type are best stored in a vertical position.

Finally, shipping a rotor without adequate midspan support can lead to bow. In this case, dynamic shock loads while in transit can multiply the normal gravity bow many times. If the rotor ends are pushed up rapidly, the midspan sag will increase, producing large tensile stress at the bottom and compressive stress at the top. If these stresses exceed the yield strength, the rotor will develop a permanent, downward bow. Rotors should always be carefully supported during shipment.

Use of a torch on a local region of the rotor can alter the heat treatment properties of the area. Welding can change the properties of the metal and leave residual stresses that are different from those on the opposite side of the shaft, resulting in a permanent bow. For this reason, *the casual use of a torch is not recommended.*

Rotor Dynamic Effects of Rotor Bow

The elastic bow due to gravity or process-related, static radial loads *does not turn with the shaft*; it tends to remain at the same position angle while the rotor turns, producing a low level of 1X stress cycling that is usually not damaging. It is possible, however, for large cyclic stresses to develop as a result of radial side loads in vertical pumps, and, when combined with concentration factors, these stresses have led to fatigue failure of the shaft.

Elastic bow in built up rotors has also been known to cause *friction instability* in rotors, a condition that manifests itself as large amplitude, subsynchronous vibration. This was a serious problem in turbines built in the early 20th century, but newer designs rarely encounter it.

Temporary or permanent bows *turn with the shaft*, and the displaced mass of the rotor creates a new, rotating unbalance. If the bow changes during operation, as can happen with temporary bows, the result will be a change in the 1X amplitude or phase of the rotor response.

Permanent rotor bow creates an unbalance in the rotor that leads to 1X vibration, but there are fundamental differences between rotor response due to unbalance and rotor response due to bow. To illustrate this, we will use a variation of the simple rotor model that was developed in Chapter 10.

We start by assuming that a permanent rotor bow has displaced the rotor mass, M, a distance, r_e, from the rotor system centerline in a direction, δ. In reality, r_e will change in response to the dynamics, but we will assume that it is constant. This produces a measurable rotor bow response vector, $\boldsymbol{r}_1$, at low speed, where there is negligible dynamic response:

$$\boldsymbol{r}_1 = r_e e^{j\delta} \tag{19-1}$$

The bow also produces an effective unbalance of the rotor system. In the absence of any other source of unbalance, the bow-induced unbalance produces a synchronous rotor response, $\boldsymbol{r}_2$, similar to that of Equation 10-37:

$$\boldsymbol{r}_2 = Ae^{j\alpha} = \frac{Mr_e\Omega^2 e^{j\delta}}{\left[K - M\Omega^2 + jD(1-\lambda)\Omega\right]} \tag{19-2}$$

The only difference between this equation and Equation 10-37 is that the unbalance is due to the bow, not due to a conventional unbalance. The dynamic rotor response due to bow is the sum of the two responses,

$$\boldsymbol{r} = \boldsymbol{r}_1 + \boldsymbol{r}_2 = r_e e^{j\delta} + \frac{Mr_e\Omega^2 e^{j\delta}}{\left[K - M\Omega^2 + jD(1-\lambda)\Omega\right]} \tag{19-3}$$

At low speed, the unbalance force is very small, and the response is primarily due to the bow term, $\boldsymbol{r}_1$. At speeds well above resonance, where the spring stiffness term, K, and the quadrature stiffness term, $jD(1 - \lambda)\Omega$, can be neglected (see Chapter 11), the rotor response becomes

$$\boldsymbol{r} = r_e e^{j\delta} + \frac{Mr_e\Omega^2 e^{j\delta}}{-M\Omega^2} = r_e e^{j\delta} - r_e e^{j\delta} = 0 \tag{19-4}$$

The model tells us that *rotor response due to the bow is zero above the resonance.* This is why it is said that, once you get a machine above resonance, the bow will often "balance itself out." This self-balancing feature of the bow is helpful, but if the bow is severe enough, vibration levels may exceed allowable levels *before* reaching the resonance peak. In this case, it is not possible to pass through the

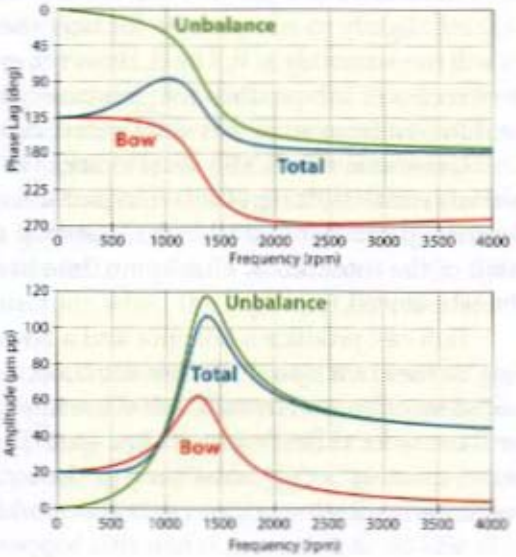


Figure 19-4. 1X Bode plots for a rotor system with bow and unbalance. The unbalance response (green) shows that, at zero speed, the amplitude is zero and the phase of the unbalance is 0°, while the bow response (red) has an eccentricity of 20 µm pp (0.8 mil pp) and a high spot phase of 135°. At low speed, the total response (blue) is dominated by the bow. Above the resonance, the bow begins to balance itself out and disappear, leaving only the residual unbalance response. Because of the phase relationship between the unbalance and the bow eccentricity in this example, the amplitude at resonance is lower than would be the case for unbalance alone.

resonance without risk of serious damage, and the bow must be removed before a startup is possible.

Equation 19-4 states that the vibration due to the bow is zero at speeds well above resonance. However, the total response of the rotor includes yet a third term, the ordinary response due to some *other* unbalance. The 1X Bode plots in Figure 19-4 show the response of a rotor system with a combination of bow and unbalance. The unbalance response (green) shows that, at zero speed, the amplitude is zero and the phase of the unbalance is 0°, while the bow response (red) has an eccentricity of 20 μm pp (0.8 mil pp) and a high spot phase of 135°. At low speed, the total response (blue) is dominated by the bow. Above the resonance, the bow begins to balance itself out and disappear, leaving only the residual unbalance response. Because of the phase relationship between the unbalance and the bow eccentricity in this example, the amplitude at resonance is lower than would be the case for unbalance alone.

Thermal Bow During Operation

Temporary thermal bow can increase or decrease the 1X vibration amplitude at operating speed depending on the phase relationship of the bow to the unbalance. Shorted turns in generators, and broken rotor bars or shorted rotor iron in induction motors, can produce local hot spots on the rotor that cause a temporary thermal bow.

Even generators in new condition can have small asymmetries in slots, windings, and cooling gas flow that produce uneven heating and thermal bow. Manufacturers will often set up *offset balancing* (the act of deliberately unbalancing slightly to compensate for later thermal changes) in a generator so that it will run smoothly at full load. However, even when this is done, changes in load or excitation can produce temperature changes that lead to a thermal bow that is different from what was anticipated, causing increased 1X vibration.

Generator rotors also deteriorate over time as repeated startups and shutdowns cause shifting of internal parts. As this happens, cooling passages may become partially or fully blocked, causing a change in the temperature distribution of the rotor. Thus, after some time in service, generator rotors may have to be rebalanced.

Rub can produce a hot spot and a bowed rotor. Imagine a rotor that is turning at constant speed. In this situation, a once-per-turn rub will occur at the same location each revolution of the shaft, and the friction at the contact area will cause local heating and a hot spot. The rotor will then bow toward the hot spot, creating a new unbalance in the rotor system. The new unbalance adds vectorially with the original unbalance, and, if the total unbalance increases, the rub will be maintained. When this happens, it is possible for the 1X vibration

response vector to move cyclically in a direction opposite to rotation. This behavior typically occurs with the following sequence (Figure 19-5):

1. Before the rub, the rotor response (high spot) lags the heavy spot by some amount, α, that is constant for the operating speed. A gentle, wiping, once-per-turn seal rub occurs at the high spot, which is the part of the rotor that is farthest from the rotor system center.

2. A hot spot develops at the rub location, and the hot spot causes a rotor bow in the direction of the rub.

3. The mass of the bowed rotor produces a new heavy spot in the direction of the high spot. The new heavy spot adds vectorially with the original (unbalance) heavy spot to produce a single, effective heavy spot, which is located between the original heavy spot and the bow direction.

4. The rotor dynamics dictate that the high spot must lag the new effective heavy spot by the same amount, α. Thus, the rotor high spot shifts to a new location that lags the effective heavy spot by α.

5. The cycle repeats as the rotor rubs at this new high spot location, creating a bow in this direction.

As this cycle continues, the 1X response vector slowly rotates in a direction opposite to rotation. The time to complete one cycle can be anywhere from several minutes to hours. The polar APHT plot (Figure 19-5, bottom) shows data from a small, high-speed turbine that encountered this type of rub. The rotation direction is Y to X (CW), and the 1X vibration vector slowly moves opposite to rotation, completing one full revolution of the plot in about 5 minutes. This behavior can occur on any type of machine when the high spot lags the heavy spot by less than 90° or so.

Other cases lead to different behavior. If the high spot (contact point) is significantly out of phase with the heavy spot, then the thermal bow will reduce the total unbalance and the vibration response, and the rub contact may cease or reappear periodically. If the contact point happens to *lead* the heavy spot, the resulting thermal bow will tend to shift the vibration vector in the *forward* direction.

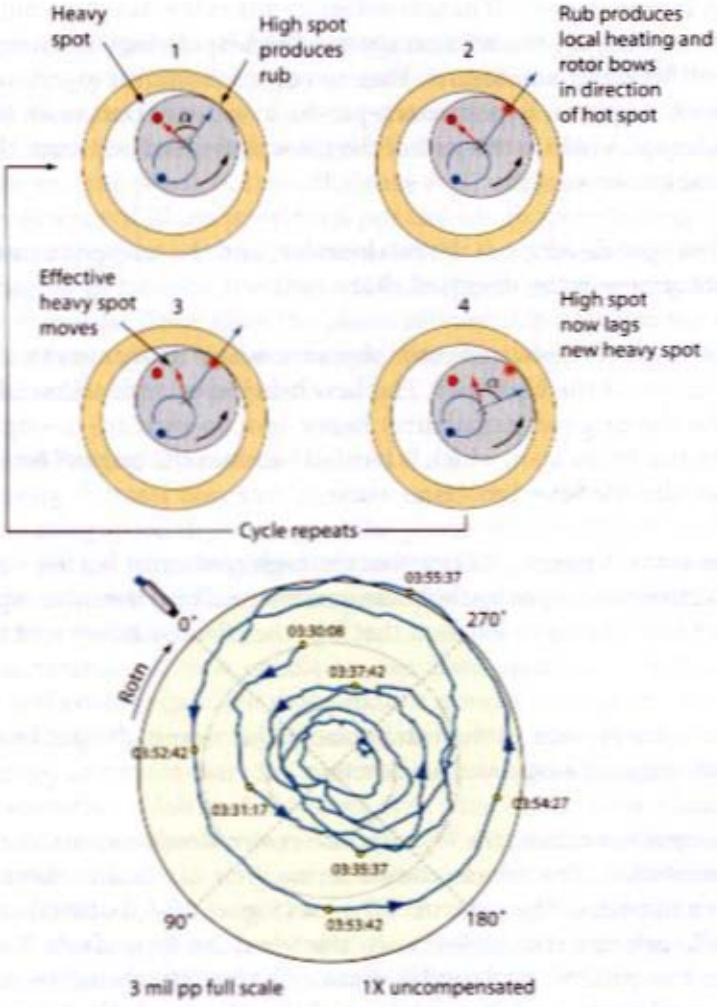


Figure 19-5. Evolution of thermal bow due to rub at constant speed. The four diagrams at the top illustrate the sequence that produces a slowly changing, 1X vector that moves opposite rotation. See the text for details. The 1X APHT plot (bottom) shows the effect on the vibration vector in a high-speed turbine of a thermal bow caused by a mild seal rub. In this machine, the vector completes one revolution of the polar plot in about 5 minutes.

Another source of thermal bow is the differential heating of the rotor due to the difference in fluid-film bearing film thickness that occurs as the journal vibrates. Journals in hydrodynamic bearings typically operate at average eccentricity ratios of 0.6 to 0.8, but dynamic motion produces an orbit around the average position. With large, 1X responses, the journal high spot passes closer to the bearing wall than does the opposite side of the journal (Figure 19-6). In overhung compressors, this type of bow has been observed to produce slow, *forward* motion of the 1X vector [1].

The thermal bow occurs because fluid-film friction is proportional to the velocity gradient in the oil film. As the journal high spot surface nears the bearing wall, the oil film becomes very thin, the velocity gradient between the journal surface and the bearing wall becomes high, and more heat is produced. One-half revolution later, the journal "low spot" passes near the wall, but the rotor centerline is now on the opposite side of the orbit, so the instantaneous eccentricity ratio is lower, and the heat input is smaller.

Thus, the high spot of the journal experiences more heat input from oil film friction than the opposite side. This process produces a temperature gradient between opposite sides of the journal. The differential thermal expansion can produce a temporary bow that has been known to produce high amplitude, cyclic, 1X vibration in machines with overhung rotors or large coupling moments.

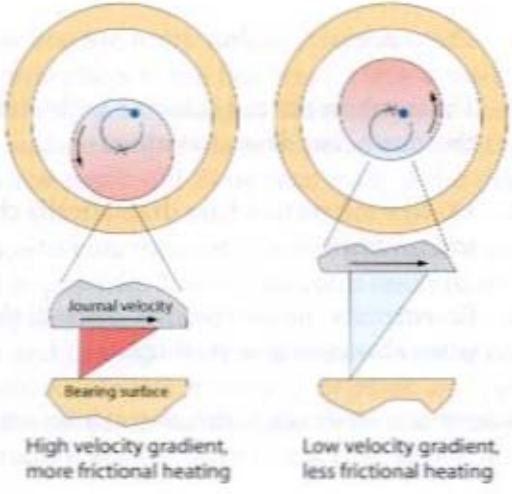


Figure 19-6. Differential heating of rotor due to variable film thickness in a fluid-film bearing. When the 1X response is large and the orbit is offset from the bearing centerline, the high spot side of the journal will pass closer to the bearing surface than the other side. Frictional heating of the oil, which is proportional to the velocity gradient, will heat one side of the journal more than the other, causing a thermal bow of the rotor.

It is possible for a steam turbine rotor to become suddenly and severely bowed during operation. This can happen to HP and IP rotors when a boiler control failure or loss of firing produces low quality (high liquid content) steam, which causes dramatic cooling of the surfaces of both the rotor and casing. Typically, this event will immediately cause a temporary rotor bow, since the quenching effect is uneven from side to side. Heavy rubbing then occurs, and the rotor bow becomes more severe because of the local heating of the rotor surface. It is also quite common for rotors which have been steam quenched to become temperature sensitive afterwards because the thermal properties of the rotor have become anisotropic. Such a severe thermal shock can very easily cause cracking of the rotor surface.

A similar mechanism can occur in LP rotors operating in cold climates. With very cold cooling water, the condensate temperature can become low. If a system upset occurs, very cold saturated steam can move back into the LP turbine casing and thermally shock the rotor. This scenario can produce a sudden rotor bow or shaft surface cracking.

Diagnosing Rotor Bow

Permanent rotor bow is a malfunction that produces 1X vibration behavior that is very similar to unbalance. Unlike unbalance, however, bow produces a significant, forward 1X response *at slow roll.* At high speed, above the resonance, the bow response becomes small, and most of the residual 1X vibration is due to unbalance.

The most obvious symptom of permanent rotor bow is a 1X vibration that occurs on startup. Bow is probable on a machine if

1. The machine was shut down hot and not put on turning gear.

2. The machine did not exhibit 1X vibration at slow roll during the shutdown, but does at startup.

3. 1X slow roll vectors have dramatically changed compared to historical values.

4. Eccentricity measurements exceed the maximum allowable when attempting to start up.

Be aware that a shaft crack can also exhibit symptoms 2, 3, and 4. See Chapter 23.

When trying to diagnose rotor bow, examine *uncompensated* direct and 1X slow roll orbits. The 1X slow roll data from several measurement planes can be correlated to infer the possible shape of the bow. A good way to do this is to plot a set of 1X orbits at the same scale and lay them on a table with spacing proportional to the axial measurement locations.

Several other malfunctions can look like rotor bow. Coupling problems can produce 1X vibration at slow roll speed; a locked gear coupling will produce an eccentric motion of the shaft that can look exactly like bow.

Mechanical and electrical runout (glitch) of the probe viewing surface results from an improperly treated or handled probe viewing surface. Beware if the rotor was lifted using chains, or even straps, that contacted the probe viewing area. The load of the rotor on the straps can modify the mechanical and electrical properties of the surface, producing a 1X runout. Also, a single scratch on the rotor can produce a 1X component and harmonics in the vibration signal. Examine a direct, uncompensated orbit or timebase plot.

A nonconcentric journal and shaft will produce a measurable eccentricity that is unrelated to bow. A shrunk-on part, which is vulnerable to concentricity or cocking problems, can produce an apparent bow in eccentricity or radial vibration measurements.

A shaft crack will almost always produce a bowed rotor. As a shaft crack propagates across the rotor, the rotor bending stiffness becomes lower in the vicinity of the crack. Thus, the rotor is less able to resist the dynamic forces that try to deform the rotor, and the rotor will usually bow as the crack develops. If the size of the crack changes in response to repeated startups, the 1X slow roll vector may change in amplitude and phase. See Chapter 23.

Removing Rotor Bow

For permanent rotor bow, prevention is the best cure. Once a machine is shut down, a hot rotor should be put on turning gear before it has a chance to stop. As the rotor is turning, the surface temperatures will even out, and the top-to-bottom thermal gradient in the rotor will disappear, even if the gradient remains in the casing.

However, the powerful surface-to-core thermal gradient will remain, and the rotor should be cooled as slowly as possible. Turning the rotor tends to even out any plastic deformation that may be taking place.

If a turbine stops while hot and cannot be turned, rotor-to-casing contact may occur at the top of the rotor. *A common mistake is to panic and turn the rotor by force*, which can result in severe internal damage to the machine. A better solution is to wait until the rotor cools enough to turn, then carefully rotate it 180° to even out the bow.

Because the symptoms of rotor bow are so similar to unbalance, there is a temptation to balance a bowed rotor in an attempt to cure the symptoms. *This is not a recommended practice*, because, if the bow goes away, which it often does, the balance weight will become an unbalance weight.

Completely removing a permanent rotor bow requires that the rotor be subjected to a series of steps that produce equal and opposite plastic deformation, a difficult thing to accomplish. In most cases, bow is relatively mild, and it can be reduced by operating the machine at a slow speed. This will *roll out* the rotor bow until the eccentricity readings drop to a low enough level to allow the machine to be started up. Once above the resonance, the bow will often balance itself out.

More difficult bows can be removed by starting the machine, running the speed up until the vibration reaches a predetermined limit, and heat soaking for a period of time. The machine is then slowed down and allowed to cool while on turning gear. This cycle may need to be repeated several times. The object is to get the rotor hot enough so that, when cooled, large surface-to-core thermal gradient stresses will appear and cause enough plastic deformation to even out the bow.

After a hard rub, the rotor may be not only bowed, but also have surface microcracking in the vicinity of the rub. If untreated, these cracks can lead to fatigue cracks in the rotor. Because of this, the affected area should be carefully treated to remove these cracks.

Rotors with permanent bows have been remachined over the outer surface and in the journal area. Because of changed clearances, it may also be necessary to replace seals. Bearing clearances will be affected, and the rotor may still retain a measureable eccentricity.

Extreme bows may require special heat treatment, removal of the entire rotor for heat treatment in a vertical furnace, or scrapping. *Do not casually use a torch to remove a bow!* The residual stress field that remains after local heat application may cause unpredictable thermal characteristics. However, carefully controlled local heat treatment, with attention to resulting material properties, has been used successfully to remove a bow. Heat is applied on the high side of the bow in order to generate local compressive stresses that exceed yield strength, causing the rotor to become shorter in that area.

Summary

There are three general types of rotor bow: elastic, temporary, and permanent. Elastic bows are caused by static radial loads such as gravity or process loads. Temporary bows can be caused by uneven heating of the rotor surface,

which causes differential thermal expansion. Permanent bows are caused by plastic deformation of rotor material when stress exceeds the yield strength.

Permanent rotor bows are often caused by failing to put a hot rotor on turning gear before it stops. Thermal stratification in the machine casing causes a top-to-bottom thermal gradient that warps the rotor upward. If the rotor contacts the stator, high stresses can appear. High tensile stresses also appear on the surface due to the surface-to-core thermal gradient produced when the rotor surface cools below the core temperature. When these thermal stresses are added to the stress associated with elastic gravity bow, they can exceed the yield strength of the material.

During startup, thermal gradients generate stresses with the opposite sign, compressive on the outer surface and tension in the center or at the surface of the bore hole. Startups must be conducted very carefully to avoid thermal shock of the rotor, bowing, and cracking.

Temporary thermal bow can be produced during operation by localized heat sources or cool spots in the rotor, such as shorted turns in generators, or broken rotor bars and shorted rotor laminations in induction motors.

Temporary thermal bow can also be caused by light, once-per-turn rubs on seals and by differential heating effects in the thin film region in fluid-film bearings.

Rotor bow produces a 1X vibration that exists even at low speeds. At speeds well above resonance, the bow will often balance itself out, and the 1X vibration due to bow will be greatly reduced, leaving conventional unbalance vibration.

Modulated 1X vibration can be produced by rub-induced thermal bow. The contact point will tend to move around the rotor as the bow changes the effective heavy spot location and the 1X vibration vector.

Rotor bow is recognized primarily by the 1X vibration that exists at slow roll speed. However, many other malfunctions can also produce low-speed 1X vibration, such as coupling problems, surface problems, concentricity problems, and a shaft crack.

Mild permanent rotor bow can usually be rolled out by operating the machine at low speeds for a length of time.

References

1. de Jongh, Frits M., and van der Hoeven, Pieter, "Application of a Heat Barrier Sleeve to Prevent Synchronous Rotor Instability," *Proceedings of the Twenty-Seventh Turbomachinery Symposium*, Texas A&M University, College Station (September 1998): pp. 17-26.

Chapter 20

High Radial Loads and Misalignment

THE MOST COMMON COMPLAINT BY OPERATORS of rotating machinery is high vibration. The *second* most common complaint is high bearing temperature. High bearing temperature is almost always associated with a high dynamic or static radial load; such a load can be caused by process conditions, various mechanical factors, coupling malfunctions, or misalignment. Radial loads and misalignment have similar symptoms; in fact, misalignment usually produces high radial loads in one or more bearings. Thus, the diagnosis of both of these malfunctions uses closely related techniques. For that reason, we will study both malfunctions in this chapter.

We will start with a discussion of the sources of static radial loads in rotating machinery, only one of which is misalignment. We will then concentrate on misalignment and discuss the definition, causes, and symptoms.

Static Radial Loads

A *static radial load* is a force that acts on the rotor in the radial direction and does not change magnitude or direction with time. This is in contrast with a dynamic radial load, such as unbalance, that changes in magnitude or direction (rotates). Static radial loads are sometimes called *radial sideloads* or *preloads*. The term preload is also used to describe bearing geometry (having nothing to do with a force), so, to avoid confusion, we will reserve the term preload for bearing applications.

Static radial loads can be classified into two general categories, *soft* and *hard*. *Soft* radial loads do not change dramatically with changes in rotor position. Examples include gravity, fluid loads, and partial arc steam admission. Soft radial loads are usually accounted for in the design of machinery and are not

likely to cause damage. Gear mesh forces are also allowed for in machine designs but have the characteristics of hard radial loads. *Hard* radial loads involve rotor interaction with some mechanical boundary. Loads due to locked couplings, misalignment, and rub are examples of hard radial loads, and these loads appear directly or indirectly as a result of a machine malfunction.

The force of gravity is the simplest example of a soft static radial load, and it exists on all horizontal machines. (In vertical machines, this force appears as an axial load.) If the bearings have been properly designed, gravity is not likely to cause any problems in a machine.

Fluid radial loads appear in pumps because of asymmetries in the design of the volute. This type of load is also called a *hydraulic preload* or *hydraulic sideload*. Single volute pumps generate significant loads as a byproduct of the pressure distribution that exists in the pump, while double volute pumps minimize the sideload due to the symmetry of the volute design. The loads are also minimized when the pump operates at its best efficiency point (BEP). When operated off the BEP, these loads can become much larger.

Inlet guide vane (IGV or wicket) problems in a hydro turbine can cause an asymmetric pressure distribution and a static radial load.

Partial arc steam admission in turbines creates a relatively high pressure region on one side of the rotor. The resulting pressure distribution around the rotor produces a static radial load. If the steam admission occurs on the bottom side of the rotor, the load can be strong enough to overcome gravity and lift the rotor. If the rotor moves to a low eccentricity ratio operating point in a fluid-film bearing, this situation can trigger fluid-induced instability.

All of the types of loads presented so far are examples of soft radial loads. Hard radial loads, because of the physical constraint they impose, are much more likely to cause problems in machinery.

Gear mesh forces are more like hard radial loads than soft loads. At the gear tooth contact point, the orientation of the surface is not on a radius of the gear, but is inclined at an angle, the pressure angle. The gear mesh force can be resolved into a *tangential force* and a radial *separation force*. The tangential force is directly related to the torque delivery of the gears. The torque delivered is the product of the tangential force and the *pitch radius* of the gear. The separation force acts in a direction toward the shaft, attempting to separate the gears; for a typical 20° pressure angle, it is about 37% of the tangential force. In addition to these two forces, single helical gears also generate a large axial force.

Gear mesh forces can be quite large. For example, in a high-speed, single-stage, increasing gear drive, transmitting 27 500 hp with an 8640 rpm, 330 mm (13 in) diameter pinion, the tangential force is 140 kN (31 500 lb), and the sepa-

ration force is 52 kN (11 700 lb). The vector sum of these two forces is a large static load that is many times the weight of the pinion and gear.

Hard radial loads can be caused by coupling problems. Properly operating gear couplings have very low axial stiffness and tolerate moderate misalignment. However, these couplings must be properly lubricated; if the lubricant either fails or is inadequate, the coupling can lock and transmit high radial loads in addition to dynamic loads.

Hard radial loads often appear as a result of misalignment. Table 20-1 lists many sources of misalignment loads, all of which can be quite damaging. Because of its importance, misalignment will be dealt with in more detail below.

Rub occurs when the rotor contacts a stationary part of the machine. The contact can be intermittent and generate a dynamic load and response, or, if it is severe enough, it can be continuous and produce a high static radial load.

Table 20-1. Sources of static radial loads.

Source	**Classification**
Gravity	Soft
Fluid loads	Soft
Pump off operating point	
Wicket problems in hydro machines	
Partial arc steam admission	Soft
Gear mesh forces	Soft/Hard
Tangential forces	
Separation forces	
Locked gear couplings	Hard
Misalignment	Hard
Cocked bearings	
Distorted diaphragms	
Locked floating seals	
Thermal warping of casing	
Piping strain	
Foundation problems	
External misalignment	
Rub	Hard

When any rub occurs, the spring stiffness of the rotor system greatly increases and the contact area acts as a hard constraint on the rotor motion. Rub is often caused by misalignment.

Soft radial loads and gear mesh forces are a normal part of machine operation, and, if the machine is properly designed and aligned, they should not produce a problem. However, hard radial loads are not normal and can easily exceed design limits, causing internal rubs, coupling wear, or wear and failure of bearings.

While coupling problems can produce excessive radial loads, the most common source is misalignment. For that reason, the primary emphasis for the rest of this chapter will be on the causes and diagnosis of high radial loads caused by misalignment.

What Is Misalignment?

Before discussing misalignment, we should define alignment. When a machine is operating at thermal equilibrium, and the rotor is subjected to normal operating loads, perfect *internal* alignment exists when, at each axial location, 1) the average shaft centerline position is in the center of each interstage diaphragm or seal, and 2) the average shaft centerline of the rotor operates at the design position in the bearing. Two machines are in perfect *external* alignment if, when operating at thermal equilibrium, the shaft centerlines are collinear at the coupling (Figure 20-1, top) and the shafts operate in their design axial positions within each machine.

In practice, some degree of internal and external misalignment always exists. *When the external misalignment between two machines exceeds the allowable tolerances of the coupling, seals, bearings, etc., the machines are misaligned.*

There are three basic types of external misalignment. *Parallel misalignment* occurs when the shaft centerlines of two machines are parallel, but offset (Figure 20-1, middle). *Angular misalignment* occurs when the shaft centerlines are not parallel (Figure 20-1, bottom). The most common situation is a combination of parallel and angular misalignment. *Axial misalignment* occurs when the axial position of the two, coupled shafts exceeds coupling tolerances. (Axial misalignment can also be a problem in motors and generators when the allowable magnetic centering is exceeded.)

The external misalignment tolerance for two machines will depend on the type of coupling that is used. Rigid couplings have a very low tolerance for misalignment, disk pack and diaphragm couplings, which are usually used in pairs, have more tolerance, and gear and elastomer couplings have the highest tolerance for misalignment.

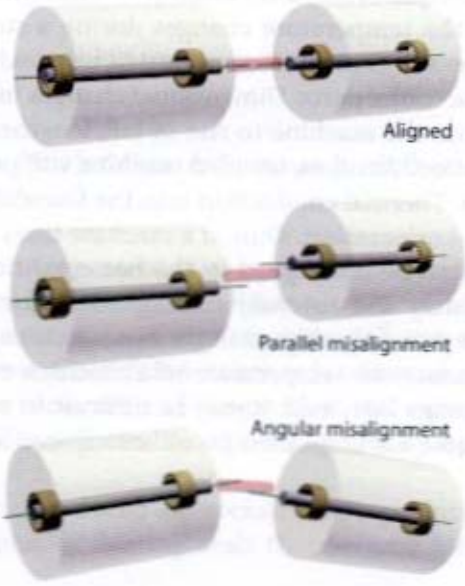


Figure 20-1. Types of external misalignment. Two machines are in perfect external alignment (top) when their shaft centerlines are collinear at the coupling within an allowable tolerance zone (red). Parallel misalignment (middle) occurs when the shaft centerlines are offset, but parallel. Angular misalignment (bottom) occurs when the shafts are not parallel. Misalignment usually involves a combination of parallel and angular misalignment. In the figure, the shafts are shown centered in the bearings. In reality, shafts in properly aligned machines with fluid-film bearings would operate at the design eccentricity ratio.

Temperature Changes and Alignment

While a machine is operating, the temperature can vary significantly from one part of the machine to another and can vary with load. The resulting thermal growth causes changes in the linear dimensions of the machine compared to the "cold" or off condition.

For the purposes of this discussion, "hot" will indicate a machine that is running and in thermal equilibrium, and "cold" will indicate a machine that is stopped and in thermal equilibrium. A machine that handles cold fluids will still be called hot when it is running and in thermal equilibrium. Note that a large machine, like some steam turbine generators, may take one to two days to reach thermal equilibrium after startup or shutdown.

As the temperature changes during a startup, the linear dimensions of a machine will change in complicated ways, with the hotter parts growing more than the cooler parts. Dimensional changes in the machine supports and casing will cause the machine to rise or fall, lengthen or shorten, and change angular orientation. Another, coupled machine will probably exhibit a different growth pattern. Thermal conduction into the foundation will also affect machine position and orientation. Thus, if a machine train is aligned in the cold condition, it is likely to be misaligned in the hot condition. For this reason, machines are deliberately and carefully misaligned in the cold condition so that they will become correctly aligned in the hot condition (Figure 20-2).

Because the temperature of a machine can vary with load, alignment can also change with load. It may be difficult to set a cold alignment that produces acceptable hot alignment for all anticipated load conditions.

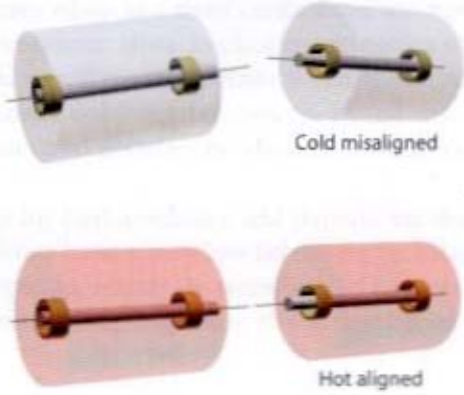


Figure 20-2. Cold alignment compensates for thermal growth. Machines are deliberately and carefully misaligned in the cold condition (top) so that thermal growth will produce correct alignment in the hot condition.

Causes of Misalignment

Assuming that a machine train has been properly aligned to begin with, misalignment symptoms can be caused by anything that produces a change in position of internal parts relative to the shaft, or a change in the position of one machine relative to another. There are several general causes:

1. Internal parts may have been installed incorrectly or may have become damaged.

2. Thermal growth of machine components may not be as expected.

3. Piping strain or a soft foot condition can warp the machine casing.

4. Grouting or foundation deterioration may cause a shift in machine position.

When misalignment symptoms are present, it is important to first eliminate all other possible problems before performing an alignment. Often, when the problem is corrected, the alignment state of the machine will revert to an acceptable level. A true misalignment condition exists only when all of these other factors are accounted for or corrected and the problem still exists. Of all the items in the list, only thermal growth can be compensated for by a physical realignment of the machines.

Internal misalignment can be produced by cocked bearings, incorrectly positioned or distorted diaphragms, or floating seals that become locked off center.

Piping strain, soft foot, and grouting and foundation problems are not the same as misalignment; they are separate malfunctions that produce misalignment symptoms.

Piping strain can misalign a machine by warping the machine casing. The warped casing misaligns the bearing supports, effectively moving the bearings relative to the shaft. Pipe strain can result from loose or locked piping hangers; bent, broken, or missing piping supports; or poor piping fit. Always check the piping system of a machine with misalignment symptoms for signs of pipe hanger or pipe support problems. Poor piping fit can put tremendous loads on the machine casing; *piping should never be forced to mate with the machine.* Instead, the piping and pipe support system must be designed to mate perfect-

ly at the attachment flanges. This mating must be correct in position (3 dimensions) and angle (3 more dimensions).

Soft foot occurs when one or more machine feet are not coplanar with the soleplate (baseplate) or foundation. When one foot (the soft foot) is not properly supported, tightening it down will warp the machine casing, changing the relative position of machine components. Soft foot can be caused by inadequate shimming or by an excessive number of shims; *more than 4 shims under a foot can produce a springy support.* Soft foot can also be caused by a warped, bowed, or improperly installed soleplate; improper machining of feet or the soleplate; a foot not parallel to the soleplate; or a warped or bowed machine casing.

Foundation problems, such as cracked grouting, a loose soleplate, and loose anchor bolts, can cause a shift in machine position over time. Oil-soaked concrete can lead to deterioration of the concrete foundation and loss of support strength. Grouting serves to provide a high stiffness interface designed to distribute loads between the soleplate and the surface of the underlying foundation. Grout is very strong under compressive loads and weak in tension. Because of this, grout should always be kept in compression. *Cracked grout indicates that it has been subjected to tension and that the foundation system of the machine has failed.*

Table 20-2. Symptoms of high radial load and misalignment.

Symptom	High Radial Load	Misalignment
High bearing temp in overloaded bearings	x	x
Low bearing temp in underloaded bearings		x
Bearing wear, premature bearing failure	x	x
Premature coupling failure		x
Low rotor vibration	x	x
High casing vibration	x	x
Axial vibration from disk/diaphragm couplings		x
High eccentricity ratio	x	x
Low eccentricity ratio		x
Abnormal average shaft centerline position	x	x
Fluid-induced instability		x
Flattened or banana-shaped orbits	x	x

Symptoms of High Radial Load and Misalignment

Above- or below-normal radial loads and misalignment produce characteristic symptoms (Table 20-2); *for misalignment, some symptoms occur in pairs.* For example, a bearing that is overloaded due to misalignment will typically have a neighbor that is underloaded. The overloaded bearing will have a high bearing temperature, while the adjacent bearing will have an abnormally low temperature. The average eccentricity ratio can be high in one bearing and low in an adjacent bearing, and the shaft centerline position may be in one quadrant in one bearing and in the opposite quadrant in an adjacent bearing. Often the paired characteristic will occur across a coupling. These symptoms will be discussed in the following sections.

Bearing Temperature

Misalignment will often unload one bearing at the expense of another (Figure 20-3), leading to abnormal bearing temperatures. *High fluid-film bearing temperature is often the first warning of a possible misalignment condition.* Normally loaded fluid-film bearings have a babbitt temperature of 70 °C to 80 °C (160 °F to 180 °F). Overloading a fluid-film bearing will reduce the minimum clearance, producing higher shear forces in the oil and higher oil temperature in

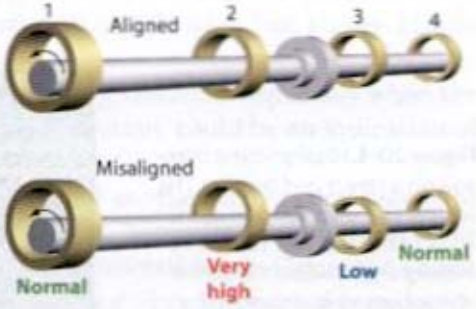


Figure 20-3. Average shaft position in an aligned machine versus a misaligned machine. The coupled shafts are vertically loaded by gravity. Misalignment shifts the load among the bearings. In this example, bearing 2 experiences a very high load and will run abnormally hot. Bearing 3 is essentially unloaded and will run cooler than normal. The two outboard bearings are approximately normally loaded. The shaft centerline position in bearing 3 is in the upper left quadrant; this is an abnormal position for a gravity-loaded shaft that is rotating *X* to *Y* (CCW) in a fluid-film bearing.

the load zone of the bearing. This heat will transfer into the babbitt. Babbitt will be significantly weakened and start to creep at 125 °C (260 °F) and melt at 240 °C (460 °F), leading to bearing failure. Extreme misalignment can result in metal-to-metal contact (wiping), and melting of the babbitt (Figure 20-4).

Note that *oil drain temperature is not a very useful indicator of the temperature in the load zone of the bearing.* It is limited because the oil exiting the bearing is a mixture of oil that has passed through the load zone of the bearing and oil that has bypassed it. Oil drain temperature is best used for plant heat load calculations or oil temperature regulation; it only provides a vague picture of the machine condition.

Resistance temperature detectors (RTDs) or thermocouples can be imbedded in the bearing to provide better warning. Ideally, the RTDs should be installed at several circumferential positions in the bearing backing (not in the babbitt) for two reasons. First, in some machines, the direction of the radial load on a bearing can vary with operating conditions, and it can be difficult to predict where the maximum load occurs in a bearing. Second, if a machine becomes misaligned, load shifting can produce radial loads that act in unpredictable and unexpected directions.

While a high bearing temperature indicates overloading, an abnormally low bearing temperature indicates a below-normal load. Given the load shifting that takes place with misalignment, bearings with above- and below-normal loads will typically be adjacent. For this reason, bearing temperatures along the machine train should be monitored and compared.

Extreme misalignment can lead to polishing or wiping of babbitt material. When a machine is disassembled, bearings should be carefully examined for evidence of any damage (Figure 20-4).

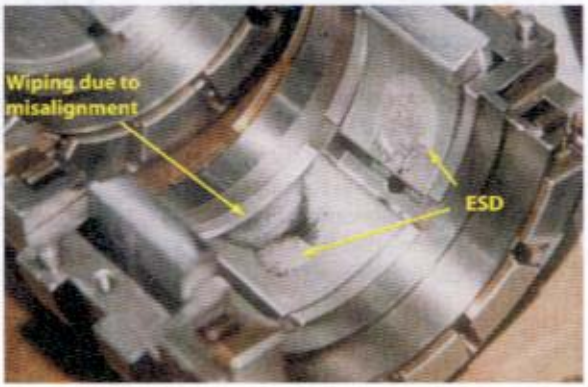


Figure 20-4. Misalignment damage on a tilting pad bearing. The wiping and discoloration at the edge of the pad was caused by misalignment, while the center of the pads were damaged by electrostatic discharge (ESD).

Rolling element bearing life is inversely proportional to the cube of the applied radial load. High radial loads will cause premature failure of rolling element bearings. Chronic premature failure of one or more rolling element bearings in a machine could be evidence of a misalignment problem.

Vibration Changes

Rotor vibration amplitude is related to the spring stiffness in a fluid-film bearing, which is a function of the eccentricity ratio. A rotor operating in a lightly loaded bearing will operate at a lower than normal eccentricity ratio. The reduced rotor support stiffness can lower the resonance frequency and cause a change in vibration.

Assuming that the vibration forcing function originates in the rotor (for example, unbalance), the amount of casing vibration will depend partly on how much rotor vibration is transmitted through the bearings. (Casing vibration will also depend upon how well the machine is mounted to the foundation.) When a rotor is at a high eccentricity ratio, the very high fluid-film stiffness of the bearing more effectively couples the rotor to the casing. Thus, in a misaligned machine, the machine may experience higher than normal casing vibration. Shaft relative vibration, because of the increased constraint on the rotor, may decrease as more of the vibration energy is transmitted to the casing.

If a particular bearing is unloaded, the rotor-to-case coupling will decrease (transmissibility will decrease) at that location, and the casing vibration there may decrease. Under this circumstance, shaft relative vibration may increase as the stiffness of the rotor support decreases.

The foregoing discussion applies when a rotor is operating at a speed where spring stiffness dominates the response of the system. If a rotor operates above a nearby resonance, a reduction in spring stiffness can shift the resonance to a lower speed and cause a *decrease* in shaft relative vibration (see Figure 11-6 in Chapter 11).

Thus, either an increase or decrease in casing vibration, especially when the opposite change occurs in the shaft relative vibration, could be an indication of misalignment.

Casing vibration can change if the machine support structure weakens or loosens, or if the machine develops a soft foot. Sometimes, tightening loose foundation bolts will return casing vibration to normal levels. Figure 9-4 shows an example of increasing shaft relative *and* casing vibration caused by a deteriorating foundation.

Parallel misalignment at the coupling offsets the shafts and makes them act like a crank. This will usually produce 1X and 2X shaft relative vibration components that exist over the entire speed range of the machine. The vibration may

transmit to the casing, but only shaft relative measurements will reveal the cranking action at slow roll speeds. The 2X component occurs because adjacent bearings alternately experience increased loading during each revolution of the shaft.

A skewed coupling bore or a bent shaft (an angular misalignment) can produce 1X or 2X vibration. An out-of-round coupling can produce a runout indication that looks like shaft displacement, but is not.

Disk and diaphragm couplings can produce an axial "pumping" action (axial vibration) when shafts are misaligned. This axial forcing can excite axial resonance frequencies of the rotor system. It is also possible for the axial vibration to couple into lateral (radial) vibration. Properly functioning gear couplings are much more axially compliant and less likely to produce axial vibration from misalignment.

Stresses and Wear

The increased load and changes in position that are associated with misalignment can affect critical components of the machine. Excessive loads can produce high stresses, leading to low- or high-cycle fatigue of shafts, couplings, and bearings. A misaligned rotor may come into contact with seals, causing wear or damage to the seals or shaft and a loss of efficiency.

Misalignment can cause wear, damage, or coupling lockup of flexible couplings. It can also cause fatigue failure of both flexible and rigid couplings.

The high bearing loads associated with misalignment can cause overheating, wear, and fatigue of babbitt in fluid-film bearings. Misalignment can also drastically reduce a rolling element bearing's useful life, which is a strong function of radial load. The L_{10} life (the time during which 10% of similar bearings will fail) for a point contact ball bearing is inversely proportional to the cube of the applied load.

Abnormal Average Shaft Centerline Position

In machines without gearing loads, and where the primary radial load is gravity, normally loaded, plain cylindrical, hydrodynamic bearings will usually have a rotor position angle of between 30° and 45° and an eccentricity ratio of 0.6 to 0.8. During a startup or shutdown, the shaft centerline will move in a typical way due to the effect of the fluid support wedge (Figure 20-5, green).

Remember that the *attitude angle* is determined by the shaft position relative to the applied radial load (Chapter 6). In a horizontal machine with a dominant gravity load, we expect the attitude angle to be equal to the position angle. If a significant radial load appears that acts in another direction, the position angle will be different. Gear mesh forces can easily overcome gravity and move

gears to positions that are very different from what would be expected for a horizontal machine.

Load shifting among bearings will cause shaft centerline behavior to depart from normal. The radial load due to misalignment is likely to be in an unexpected direction, and the direction and amount of the misalignment load can change as the machines heat up. Heavily loaded bearings will have operating eccentricity ratios that are higher than normal, and lightly loaded bearings lower than normal. If the misalignment becomes severe enough, shaft operating positions may move to unusual locations, such as near the top of a bearing (Figure 20-5, red). In a horizontal machine, this is a sign that the radial load is no longer dominated by gravity.

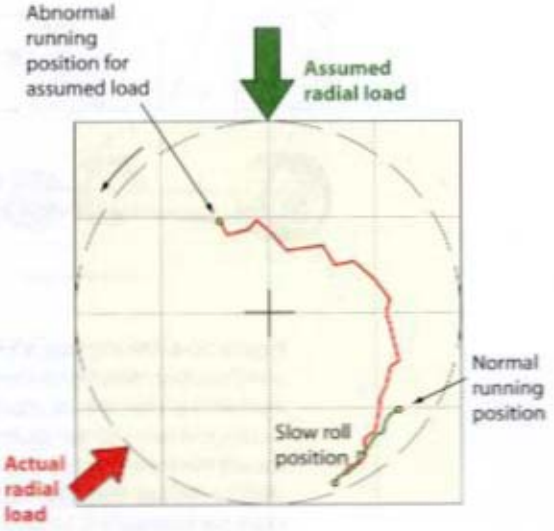


Figure 20-5. Shaft centerline behavior for a plain, cylindrical, fluid-film bearing. A typical startup or shutdown centerline plot (green) is shown for a horizontal machine with gravity as the dominant load. When the load changes (red), the startup and shutdown centerline response, for this case, changes significantly.

While a high radial load can be detected with shaft position data from a single bearing, misalignment is often revealed by comparing position data from adjacent bearings. Differences in operating position can be most apparent across a coupling, where the rotor may operate in different or opposite quadrants of the bearings (Figure 20-6). Note that the misalignment effectively moves the bearings relative to the shaft; the combination of bearing position shift and changing load in the bearings causes the measured shaft position to change.

Average shaft centerline plots should be examined at every axial position and compared for signs of abnormality. They are most useful when clearance circles are included on the plot; operation in an abnormal quadrant can be more easily detected. The plots should also be compared to archived data.

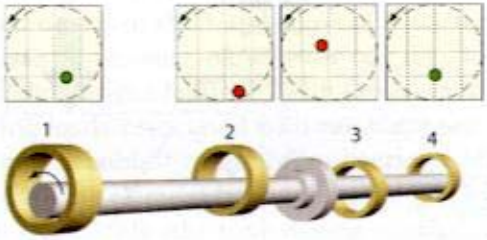


Figure 20-6. Misalignment and operating position. Misalignment of the machines moves the locations of the bearings relative to the shaft. This causes the measured position of the shaft to deviate from normal. In this example, bearing 2 is heavily loaded, and the shaft operates at a high eccentricity ratio. Across the coupling, bearing 3 is lightly loaded, and the shaft operates at a low eccentricity ratio in the opposite quadrant.

Most couplings are designed to accommodate the residual misalignment that exists in all machine trains. A properly operating coupling isolates the shafts from the loads that would occur due to this residual misalignment.

If, for any reason, a gear coupling becomes locked in position, unable to freely move, then large loads can appear in the bearings of the coupled machines. Often, this will cause sudden changes in the average shaft centerline position in these bearings (Figure 20-7). This situation can also produce large changes in 1X vibration behavior.

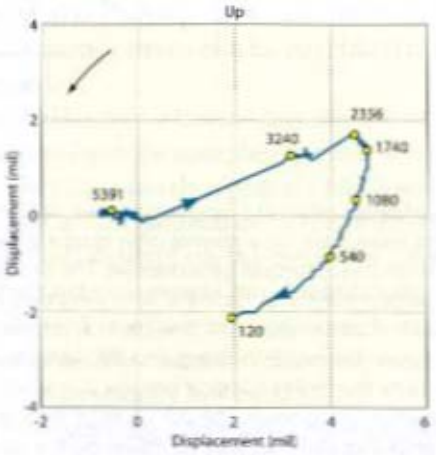


Figure 20-7. Coupling lockup and shaft centerline behavior. During this shutdown, the gear coupling between the pipeline compressor and its gas turbine driver was locked up from operating speed down to about 2300 rpm. The average shaft centerline plot at the inboard, low-pressure, gas turbine bearing shows abnormal response until the coupling unlocks. The shaft behaves normally for the remainder of the shutdown.

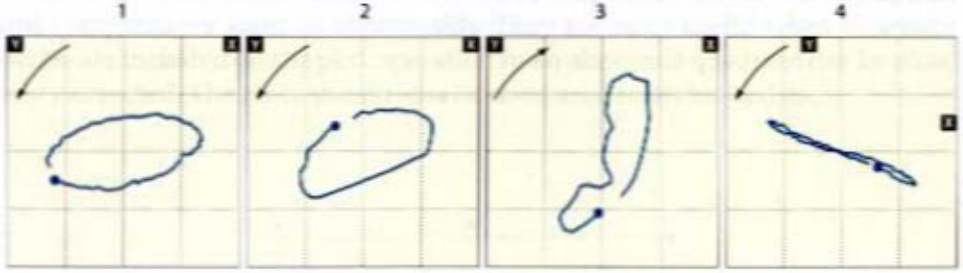


Figure 20-8. Unfiltered orbits showing progressive degrees of misalignment symptoms. Case 1 is a normal orbit from a generator bearing on a small steam turbine generator set. The orbit is mildly elliptical and predominantly 1X. Case 2 is from a Frame 6 gas turbine bearing. The orbit shows evidence of constraint along the lower edge. Case 3 is from the exciter bearing on a 500 MW steam turbine generator set. Note the highly elliptical banana shape, which produces a 2X vibration component. The curvature of the right side of the orbit suggests that the shaft may be following the geometry of the bearing. Case 4 is from an HP steam turbine bearing. The orbit is highly flattened, suggesting a high, misalignment-induced, radial load. (Note that line orbits can occur for other reasons.)

Orbit Shape

Orbits are very helpful for diagnosis of high loads and misalignment. The orbits in Figure 20-8 illustrate increasing degrees of misalignment-induced bearing loading in four different machines. Normally loaded rotors operating in fluid-film bearings tend to produce unfiltered (direct) orbits that are elliptical in shape and predominantly 1X (Figure 20-8, case 1). The ellipticity of such orbits can fall into a wide range and still be considered normal.

The shape of a direct orbit is sensitive to the amount of radial load that acts on the rotor. As the radial load increases, the orbit becomes more flattened, and part of the orbit path may follow the curvature of the bearing (Figure 20-8, case 2). (Note that orbits in elliptical and lemon bore bearings tend to be aligned with the bearing geometry and more elliptical than orbits in plain, cylindrical bearings.)

With increasing radial load, the orbit may become banana shaped, partially follow the curvature of the bearing, and contain a 2X vibration component (Figure 20-8, case 3). 2X components can be amplified if the rotor operates near half of a resonance speed.

In cases of high radial load or extreme misalignment, the rotor may become so constrained that it may become completely flattened (Figure 20-8, case 4) or, if unbalance is small, shrink to nearly a point. This is why gear shaft orbits tend to be very small. Assuming that unbalance is the primary source of rotor vibration, the orbit behavior will depend on the degree of radial load, the amount of unbalance, and the shaft attitude angle and eccentricity ratio.

A rotor that is unloaded in a bearing will operate at a low eccentricity ratio and have an orbit that is nearly circular. When elliptical orbits are the norm, a circular orbit suggests an unusually low radial load.

Orbits should be compared for every axial position in the machine train during steady state and transient operation. If startup or shutdown data is available, these orbits should be examined over the entire speed range of the machine for evidence of high radial loads. The orbits (dynamic position information) should be correlated with average shaft centerline plots (average position information).

Rub

Rub may occur on seals if the rotor is misaligned. The rub can occur during startup, shutdown, or steady state operation. Symptoms may include changes in 1X vibration behavior through resonances and changes in steady state vibration behavior. See Chapter 21.

A rub on interstage seals may open up clearances, resulting in higher leakage flows and a loss of efficiency. Any machine that shows a loss of efficiency

over time should be carefully evaluated for the root cause, and misalignment should be considered as a possibility.

Fluid-Induced Instability

If a plain, cylindrical, fluid-film bearing becomes underloaded because of misalignment, the journal may operate near the center of the bearing at an abnormally low eccentricity ratio. This can cause the bearing to become fully (360°) lubricated, making the rotor system vulnerable to fluid-induced instability.

Fluid-induced instability manifests itself as a predominantly forward, subsynchronous vibration, usually in the frequency range from 0.3X to a little below 0.5X, although it can occur outside this range (Figure 20-9). See Chapter 22.

There have been several cases of machines that have run for years without any problems, were overhauled, and encountered fluid-induced instability when started up. Misalignment during reassembly was the culprit. After the machines were realigned, the fluid-induced instability disappeared. Correct alignment properly loaded the bearings, resulting in normal, partially lubricated operation.

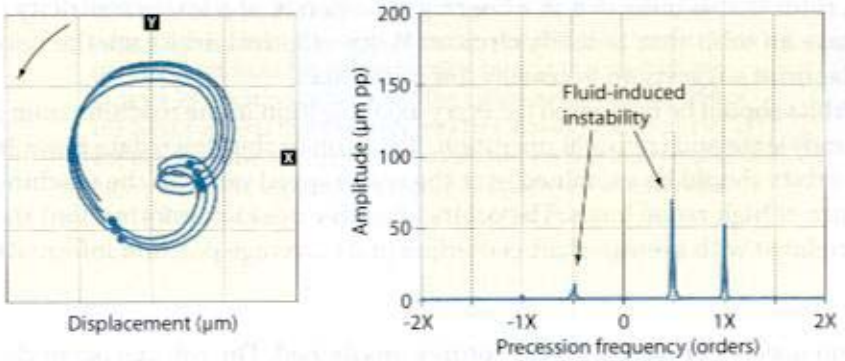


Figure 20-9. Fluid-induced instability orbit and full spectrum. The orbit contains a mixture of 1X and 0.48X instability frequencies. The full spectrum is plotted to the same scale.

Summary

A *static radial load* is a force that acts on the rotor in the radial direction and does not change magnitude or direction with time.

Static radial loads can be classified into two general categories, *soft* and *hard*. *Soft* radial loads do not dramatically change in magnitude with changes in rotor position. *Hard* radial loads involve rotor interaction with some mechanical boundary. Excluding gear mesh forces, hard radial loads usually appear as a result of a machine malfunction. Misalignment is only one potential source of high radial loads.

Internal misalignment occurs when the centerlines of internal parts of a machine are not collinear within an acceptable tolerance. External misalignment occurs when the centerlines of the shafts of two coupled machines are not collinear within an acceptable tolerance at the coupling.

Parallel misalignment exists when centerlines are parallel but offset from each other, while angular misalignment exists when the centerlines are not parallel. Usually, misalignment involves a combination of both types. Axial misalignment occurs when the axial position of two, coupled machines exceeds coupling tolerances.

There are several general causes of misalignment: internal parts may have been installed incorrectly or may have become damaged, a machine train may not have been properly aligned to begin with, thermal growth of machine components may not be as expected, piping strain or a soft foot condition may have warped the machine casing, or grouting or foundation deterioration may have caused a shift in machine position.

There are many symptoms of high radial loads and misalignment: high bearing temperatures in overloaded bearings, low bearing temperatures in underloaded bearings, bearing wear, premature bearing failure, fluid-induced instability, changes in rotor vibration or casing vibration, axial vibration, unusual orbit shapes, premature coupling failure, and abnormal shaft centerline position. Misalignment symptoms often occur in pairs.

Chapter 21

Rub and Looseness

RUB AND LOOSENESS ARE TWO DIFFERENT PHENOMENA that, because of their similar rotor dynamic behavior, produce remarkably similar symptoms. Both malfunctions are capable of producing changes in 1X vibration, as well as generating super- and subsynchronous vibration. The two phenomena are, in some respects, mirror images of each other. For that reason, we will discuss both types of malfunctions in this chapter.

We will start by defining rub and looseness; we will then discuss the physical sequence of a partial rub, the most common manifestation of rub and looseness, and briefly discuss the more unusual full annular rub, an extension of partial radial rub.

Rub and looseness affect the forces applied to and the Dynamic Stiffness of the rotor system; thus, the effects can be complex. We will discuss the forces that act on the rotor, how rub and looseness change the Dynamic Stiffness of the system, how the rotor system dynamically responds to these changes, and the symptoms. We will also discuss other malfunctions that can produce similar symptoms.

Rub and Looseness

Rub is an undesired contact between a rotating and stationary part. Normally, bearings keep the rotating part of a machine from contacting the stationary part. When machine parts move to a position where clearances are reduced, or vibrate so that clearances are taken up, contact can occur at places other than the bearings; the moving and stationary parts will "rub." During the contact, rub produces an additional constraint on rotor motion.

On the other hand, looseness is a condition where a normally present con-

straint on the rotor motion does not act during part of the vibration cycle. Worn or oversize fluid-film bearings or a loose bearing support are examples of looseness. While the root cause is different, the dynamic response due to looseness often leads to rub; for that reason, looseness will be discussed as a type of rub.

Rub causes direct damage to the contacting parts (Figure 21-1). The damage can range from mild, for light rub, to complete destruction of the machine.

Rub is always a secondary effect that is caused by some other malfunction that produces a combination of average and dynamic shaft centerline position that uses up the available clearance between the rotor and the stator. Extreme average shaft centerline position can be caused by excessive radial loads, looseness, and external or internal misalignment caused by a warped casing, piping strain, foundation problems, uneven thermal growth, a locked gear coupling, or an out-of-position internal part. Extreme dynamic shaft position can be caused by high vibration, due to excessive unbalance, rotor bow, or instability. Even when vibration is not extreme, it produces dynamic motion about the average shaft centerline position. Rub will occur whenever the instantaneous position of the rotor uses up the available clearance (Figure 21-2).

Rub contact can occur in the radial direction or in the axial direction, or in combination (Figure 21-3). Axial rub can result from a mismatch in the thermal growth rates between the rotor and casing. During a cold startup, a steam turbine rotor expands faster than the more massive casing. For this reason, large steam turbines are usually heat soaked several times during the startup process to allow the casing time to catch up to the thermal growth of the rotor. A related problem can occur when a slideway on the machine casing hangs up and prevents free movement of the casing during startup or shutdown. The constrained machine casing can deform, displace internal parts, and cause an internal rub. Surge in compressors or in the compressor stage of gas turbines can cause highly damaging axial rubs on blades and impellers; thrust bearing failure can produce catastrophic axial rub.

Partial Radial Rub

When the actual rub/stator contact occurs over a small fraction of the vibration cycle, it is called *partial rub.* When it occurs over the entire vibration cycle, maintaining continuous contact, it is called *full annular rub.* Partial rub is the most common manifestation of rub. Partial radial rub is the most common form of rub; it can be divided into two general categories, normal-tight and normal-loose.

In *normal-tight* rub, the machine rotor normally operates in an *unconstrained* condition. When the rotor contacts a bearing wall, seal, or some other part, it becomes constrained, or *tight* (Figure 21-4). By restricting free motion of



Figure 21-1. Seal damage due to a rub. A rub destroyed the labyrinth seal on the discharge end of this compressor.

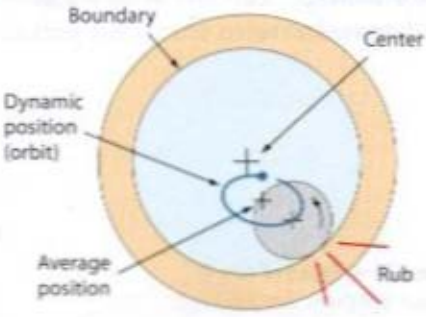


Figure 21-2. Limit of shaft motion in a clearance. The dynamic motion of the shaft centerline (orbit) adds to the average shaft centerline position. When the total displacement equals the allowable clearance, a rub will occur.

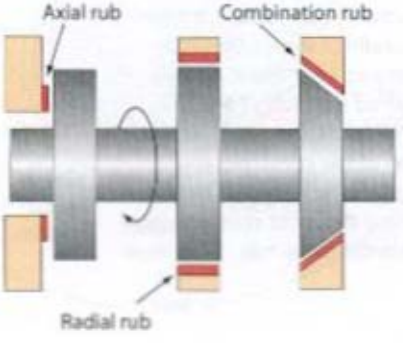


Figure 21-3. Directions of rub contact. Rub can occur in the radial direction, in the axial direction, or in a combination of both.

the rotor shaft, the rotor system spring stiffness increases. Normal-tight is the most common form of partial radial rub.

Normal-loose rub is the manifestation of looseness in a machine. In this situation, the rotor operates normally in a *constrained* condition. At some point, usually because of high vibration, the rotor moves clear of the constraint and becomes *loose*. A worn bearing or loose support structure can allow normal-loose behavior. If a rotor normally operates close to a bearing wall (is constrained by the bearing, perhaps operating near the bottom), and the rotor vibration reaches a level such that it dynamically moves the rotor well away from the bearing constraint, the spring stiffness of the rotor system will decrease, producing a change in vibration response. Alternatively, a loose bearing support structure can allow the rotor and bearing to lift or move away from contact with the support, producing a sudden reduction in support spring stiffness. Normal-loose rub occurs when the rotor recontacts the bearing or support structure at some point in the vibration cycle. This produces behavior which is very similar to normal-tight rub, but with some important differences in behavior that we will discuss below.

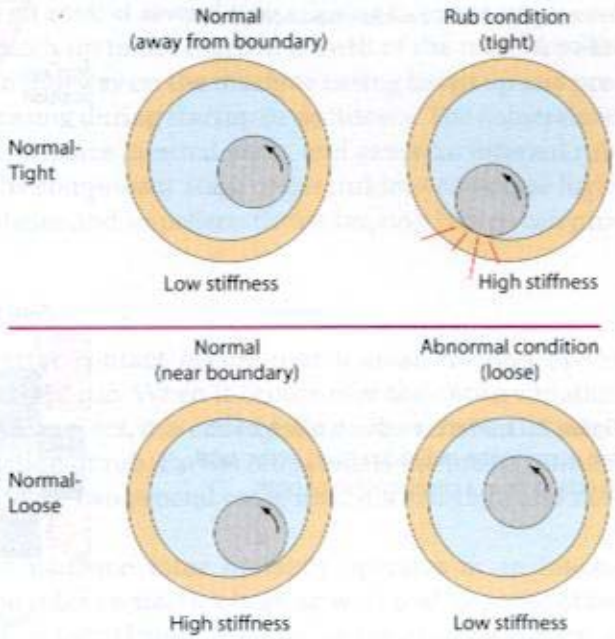


Figure 21-4. Types of rub. In *normal-tight* rub, the machine rotor normally operates in an *unconstrained* condition (left). When the rotor contacts a bearing wall, seal, or some other part, it becomes constrained, or *tight* (right). *Normal-loose* (bottom) implies a condition where the rotor operates normally in a *constrained* condition (left). At some point, the rotor moves clear of the constraint and becomes *loose* (right). Normal-tight is the most common form of partial radial rub.

The mechanism of partial rub usually involves a temporary, sliding rotor contact with the stationary part. During some part of its vibration cycle, the rotating shaft approaches the stationary part. While the stationary part has zero velocity, the shaft surface velocity is nonzero. The shaft contacts the stationary part and maintains contact for some period of time determined by the dynamics of the situation (Figure 21-5).

During the period of contact, a force acts perpendicular to the surface (called a *normal* force), pressing the two objects together. At the same time, the shaft spins relative to the stationary part, so a tangential friction force develops that is proportional to the normal force and the coefficient of friction at the interface. This force acts in the direction opposite to the shaft surface velocity. Usually, this is also opposite to the direction of precession of the shaft centerline. Thus, the friction force acts to accelerate the rotor in the reverse direction, producing reverse components in the full spectrum.

At some point, the rotor breaks contact with the stationary part and moves away to complete the vibration cycle. During the next vibration cycle, the rotor repeats the process. The period of time during which the rotor maintains con-

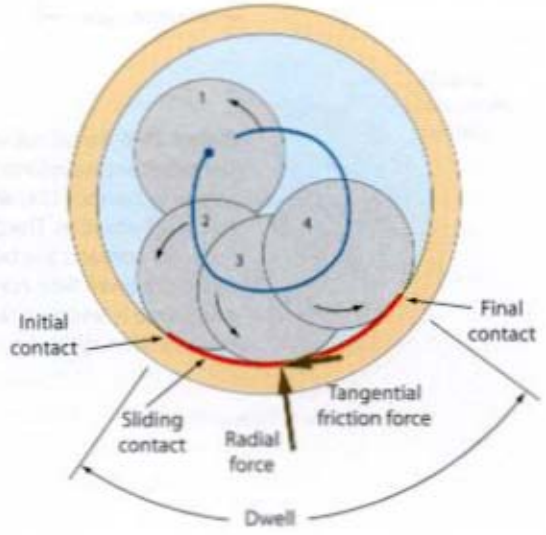


Figure 21-5. The mechanism of partial rub.

1. The rotor moves in its orbit, free of contact, for a period of time determined by the dynamics of the situation.

2 - 4. The rotor contacts and slides, generating radial and tangential forces (shown for position 3). The tangential force acts in a direction opposite to rotation.

4. The rotor eventually breaks contact and continues in its orbit until contact recurs.

tact with the stationary part is called the *dwell time* (Figure 21-6). Partial rub, by definition, has a dwell time that is less than the period of vibration.

Most of the time, partial rub contact occurs once per revolution of the rotor (1X rub). Less often, rub contact will occur once per several revolutions of the rotor, producing subsynchronous vibration.

Rub can occur with lubricated or unlubricated contact, but most often occurs with unlubricated contact. Because of this, local friction forces can be quite high and generate sufficient local heating to cause rotor bowing, severe wear, local melting or welding of the contacting surfaces, or plastic deformation of the shaft.

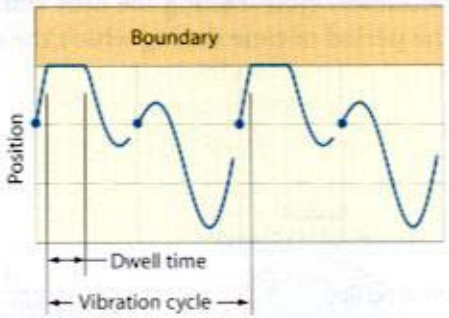


Figure 21-6. Partial rub and dwell time. The timebase waveform contains a mixture of unbalance (1X) and rub-generated (1/2X) vibration. The period of time the rotor contacts the boundary is called the *dwell time*. For partial rub, the dwell time is less than the period of one vibration cycle.

Full Annular Rub

When the dwell time is 100% of the vibration cycle, the result is a *full annular rub*. Imagine a rotor that is precessing in a 1X, circular orbit that is almost perfectly concentric with a seal (Figure 21-7). If the rotor orbit is large enough, it will come into contact with the clearance. Full annular rub occurs when that contact is maintained for a complete vibration cycle.

Full annular rub can be either forward or reverse. If the rotor is precessing in the forward direction (most common), the relative surface velocity of the shaft at the contact point can be quite high. If the contact surface is lubricated and the contact force is small, so that the friction force is small, a forward, full annular rub will occur. If the surface is poorly lubricated or the contact force high, the friction force will try to drive the rotor in the reverse direction. In both cases, the rotor will lock into a rotor system natural frequency that has been modified by the contact stiffness.

In both cases, there will almost always be sliding between the rotor and the stator. For a reverse rub, if there is no sliding contact (that is, the rotor and stator have zero relative surface velocity), the rotor system will behave like a planetary gear system. This situation is unlikely to occur outside the laboratory. Reverse full annular rub, while rare, can generate high vibration and rapidly destroy the machine.

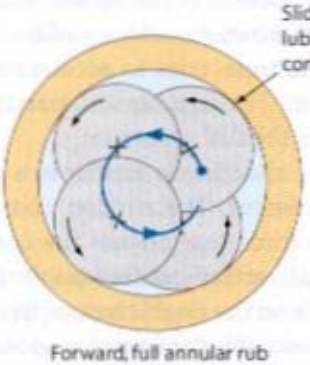


Figure 21-7. Full annular rub. The rotor maintains continuous contact with the boundary. For forward, full annular rub (shown), which involves sliding, lubricated contact, the rotor precession direction is forward.

Rub-Induced Forces and Spring Stiffness Changes

Remember that vibration is the ratio of the input force to the Dynamic Stiffness of the rotor system (Chapter 11):

$$\text{Vibration} = \frac{\text{Force}}{\text{Dynamic Stiffness}} \tag{21-1}$$

Rub produces simultaneous, nonlinear changes in both the force and the Dynamic Stiffness, so the rotor dynamics of rub can become quite complicated. In this section, we will look at how the forces and stiffness are changed; in the next section, we will discuss how these changes affect vibration.

When a rub occurs, contact forces suddenly appear and disappear. As the rotor contacts the stationary part, the stator pushes on the rotor while the rotor pushes in an equal and opposite manner on the stator. This contact force can be separated into *radial* and *tangential* (frictional) components.

When contact occurs, the radial force acts in the direction of the rotor center to strongly accelerate the rotor away from the contact point. The radial force changes during the dwell time of the contact period and is proportional to the instantaneous radial acceleration, $\boldsymbol{a}$, of the rotor ($\boldsymbol{F} = M\boldsymbol{a}$). For a short time, the rotor shaft may deflect in response to the inertia of the rotor while the average radial velocity of the rotor centerline decreases. After the radial velocity reaches zero, the rotor shaft rebounds and the rub contact is broken.

During the period of contact, the tangential friction force appears, which is proportional to the instantaneous magnitude of the radial force and the coefficient of friction at the sliding interface. The tangential friction force acts opposite to the surface velocity of the shaft. It produces a torque on the rotor and, at the same time, tries to accelerate the rotor centerline in the reverse precession direction. For this reason, partial radial rub usually produces reverse components in the full spectrum.

This situation is similar to a wheel spinning on ice that suddenly encounters dry pavement. The sudden contact between the tire and the road produces a friction force that pushes the car in a direction opposite to the surface velocity of the tire at the contact point. The amount of force depends on the weight of the car and the coefficient of friction between the road and tire.

A side effect of the tangential friction force is that it acts as an agent to transfer the energy of rotation to radial vibration. Thus, the magnitude of unfiltered vibration is likely to change with rub onset, as is the amplitude and phase of filtered vibration. Direct (unfiltered) orbits can reveal changes in rotor trajectory due to rub.

For partial radial rubs, a shallow (low angle of incidence) approach to the contact zone will produce a gentle, wiping contact, which does not greatly change the forces in the system. This kind of rub usually produces only a modified 1X response.

However, if the approach to the contact zone is steep, the rub contact can also be sudden and relatively violent. In this case, the radial and tangential frictional forces due to rub are *impulsive* by nature. They appear suddenly, build to high levels, and disappear suddenly. The effect is similar to hitting the rotor shaft with a hammer; an impulse force produces a *impulse response* that includes many of the free radial vibration modes in the rotor. When the rotor rebounds from the contact, it will ring in free vibration at one or more natural frequencies.

The tangential force impulse, combined with the radius of the shaft, also produces a torque impulse that can excite the torsional natural frequencies of the system.

Rotor system spring stiffness consists of the series and parallel combination of all of the stiffness elements that exist between the effective rotor mass and solid ground. This includes the stiffness of the rotor shaft, the bearing stiffness, bearing support stiffness, and foundation stiffness. The shaft spring acts in series with the springs in the bearings, supports, and the foundation. For any series combination of springs, the stiffness of the combination is always weaker than the weakest spring in the series.

Flexible rotor shafts are often one of the weakest springs in the system. They behave like beams; the stiffness of the shaft is related to the cross-section geometry and the length of the shaft between supports. For a simply supported shaft with a constant cross section, a concentrated midspan load, and no coupling effects at the ends, the spring stiffness is

$$K_{shaft} = 48\frac{EI}{l^3} \tag{21-2}$$

where E is the modulus of elasticity (Young's Modulus) of the shaft material, I is the section moment of inertia, and l is the unsupported length of the shaft. As the equation shows, the spring stiffness of the shaft is very sensitive to its unsupported length.

When a rotor shaft contacts the stator surface (a normal-tight rub), the contact point is usually somewhere well away from the bearings. It provides an additional constraint that effectively decreases the length of the shaft and increases

the stiffness of the rotor system. This sudden increase in stiffness lasts for the dwell time of the contact and disappears when the rotor breaks contact with the stationary part. If this cycle repeats periodically, then the step increase in stiffness also repeats. This cycle produces a time-averaged, effective stiffness that is a function of the two values of stiffness and the dwell time. The average stiffness due to normal-tight rub will be *higher* than for the no rub case (Figure 21-8); normal-loose behavior *lowers* the average stiffness.

The stiffening due to temporary contact can be illustrated with a simple pendulum where the mass is supported by a string (Figure 21-9). The pendulum is in free vibration at a natural frequency that is related to the length of the string. When the string is not contacting the boundary (left), the frequency of vibration is low. When the string contacts the boundary (right), the pendulum is shorter, and the natural frequency is higher while the contact is maintained (the dwell time). The pendulum actually switches between low and high natural frequencies (long and short string). When viewed over the entire vibration cycle, the pendulum has an average natural frequency between these two values.

Rub and Steady State 1X Vibration

The modification of the spring stiffness part of the Dynamic Stiffness causes changes in both steady state and transient vibration behavior. Any change in the spring stiffness, K, affects the natural frequency, ω_n, or balance resonance speed, Ω_{res}, of the rotor system:

$$\omega_n = \Omega_{res} = \sqrt{\frac{K}{M}} \tag{21-3}$$

where M is the rotor mass (Chapters 10 and 11). Increasing K increases the balance resonance speed, while decreasing K decreases it.

A change in resonance speed can have a direct affect on the steady state vibration of the system. Figure 21-10 shows how steady state, 1X vibration can change due to normal-tight (left) and normal-loose (right) conditions. If a machine is operating at a speed slightly below a balance resonance (Ω_1), a normal-tight rub will shift the natural frequency upward and move the resonance peak away from the operating speed. This will decrease the vibration amplitude and phase lag. Conversely, if the rotor was originally operating above a resonance (Ω_2), the normal-tight rub will shift the natural frequency upward and closer to the operating speed. This could cause an increase in vibration amplitude and a decrease in phase lag.

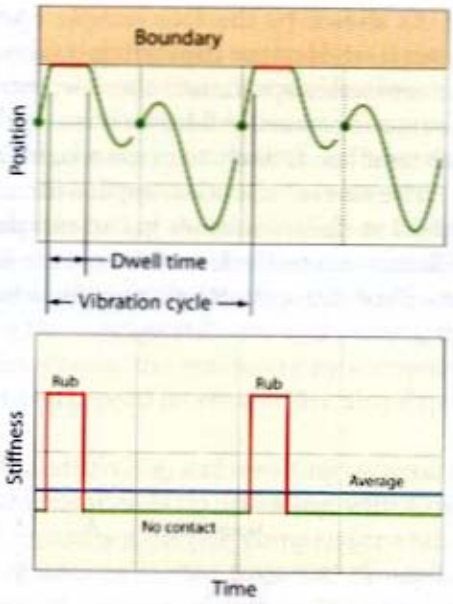


Figure 21-8. Stiffness change during normal-tight rub. Shaft spring stiffness plot (bottom) and timebase plot (top) show correlation of stiffness versus shaft position. When there is no contact (green), the stiffness is lowest. During contact (red), the stiffness is much higher. The average stiffness (blue) depends on the rub stiffness and the dwell time

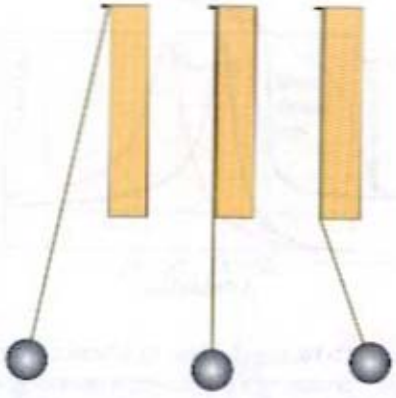


Figure 21-9. The stiffening effect in a pendulum. A simple pendulum consists of a mass supported by a string. The pendulum is in free vibration at a natural frequency that is related to the length of the string. When the string is not contacting the boundary (left), the frequency of vibration is low. When the string contacts the boundary (right), the pendulum is shorter, and the natural frequency is higher while the contact is maintained (the dwell time). The pendulum actually switches between low and high natural frequencies (long and short string). When viewed over the entire vibration cycle, the pendulum has an average natural frequency between these two values.

As shown by the four sample operating speeds, vibration amplitude can increase or decrease depending on the relationship of operating speed, the no-rub resonance speed, and the new, rub-modified resonance speed. Note that if the rotor operates well below a resonance, then the higher stiffness due to normal-tight rub is likely to cause a significant reduction in vibration response.

The reverse situation applies for normal-loose behavior. The average rotor system stiffness becomes lower, and the rotor response changes from the high stiffness case to the low stiffness case in the figure. The changes in rotor behavior will be the opposite of those for normal-tight rub.

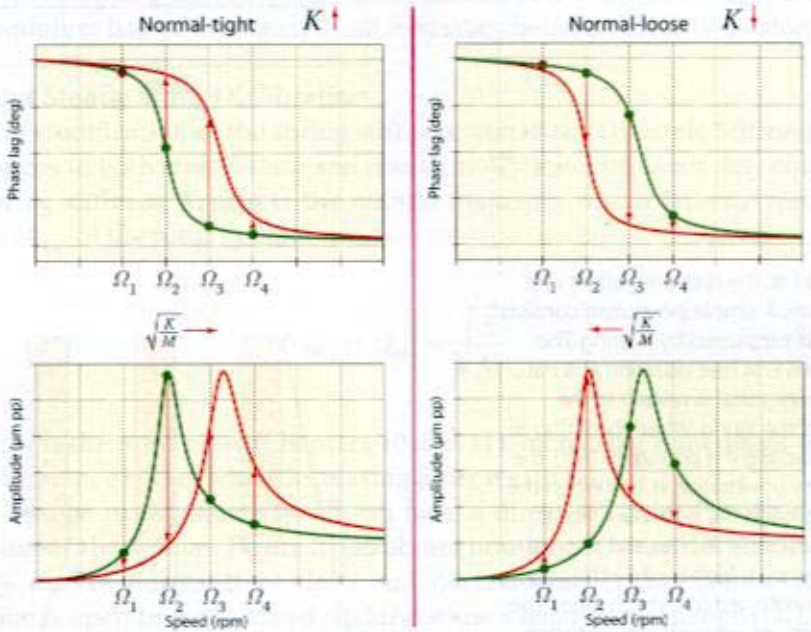


Figure 21-10. Steady state, 1X vibration changes due to normal-tight (left) and normal-loose (right) conditions. If a machine is operating at Ω_1 below a balance resonance, and a normal-tight rub develops, the natural frequency increases, moving the resonance peak away from the operating speed, and the vibration amplitude and phase lag decrease. Conversely, if the rotor is operating well above a resonance (Ω_4), and a normal-tight rub develops, the resonance will be shifted closer to the operating speed, the vibration amplitude will increase, but the phase lag will decrease. These results are reversed for normal-loose behavior.

Rub and 1X Vibration During Resonance

It is possible for a rotor to go into normal-tight rub due to the high vibration associated with a balance resonance. When contact occurs, the sudden increase in stiffness will shift the resonance speed upward and modify the rotor response.

The response is similar to a nonlinear mechanical system with stiffness that increases with displacement. The response also depends on whether the rotor is approaching the resonance during startup or during shutdown (Figure 21-11). During startup (red), the phase lag will be delayed since the resonance peak is shifted to a higher speed. If vibration amplitude increases further, the dwell time of the rub is likely to increase, increasing the average stiffness and moving the resonance even higher. The rotor, in effect, chases the resonance by increasing the average spring stiffness of the system as its speed increases. (See also Figure 18-2.)

Under some circumstances, this gradual stiffening, and resulting increase of the resonance speed, can continue until the rotor ends up operating, with a rub, on or very close to the highly modified resonance. In one compressor with a heavy rub on a locked floating seal, the resonance shifted from the nameplate value of 5200 rpm to somewhere near the operating speed of 10 480 rpm (see Chapter 24).

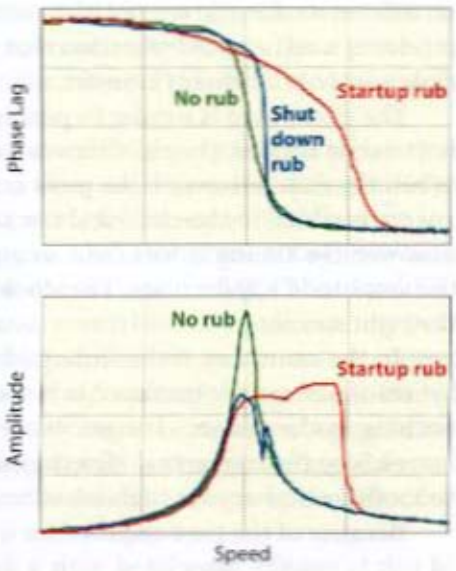


Figure 21-11. Experimental data showing modification of the balance resonance speed by normal-tight rub. During startup (red), the phase lag will be delayed because the resonance peak is shifted to a higher speed. If vibration amplitude increases further, the dwell time of the rub is likely to increase, increasing the average stiffness and moving the resonance even more to the right. During shutdown (blue), the system will behave normally until just above the resonance, when the rotor initiates rub contact. The sudden increase in stiffness increases the resonance frequency, possibly above the current rotor speed. The rotor encounters the resonance peak sooner than normal, or may even find itself operating slightly below the rub-modified resonance. When this occurs, the phase lag will decrease very suddenly, and the rotor will pass through the resonance more quickly than normal.

During shutdown (blue), the situation is different. The system will behave normally until vibration increases, at just above the resonance, and the rotor initiates rub contact. The sudden increase in stiffness immediately moves the resonance to a higher speed, which may even be slightly above the current rotor speed. The rotor encounters the resonance peak sooner than expected, or may find itself operating slightly below the rub-modified resonance. When this kind of jump occurs, the phase lag will decrease very suddenly, and the rotor will pass through the resonance more quickly than normal.

Subsynchronous Vibration

Partial radial rub usually occurs once per turn of the rotor, but under some circumstances, it can produce a self-excited, large amplitude, subsynchronous vibration at subharmonic (½X, ⅓X, ¼X, etc.) or, more rarely, simple integer ratio (⅔X, ¾X, etc.) frequencies. This self-excited vibration is always associated with a modified natural frequency of the rotor system.

Impulse forces produce a wide spectrum of energy, which is available to excite system natural frequencies. If the impulse *does not* recur, then the resulting free vibration will die away. However, if the rotor system has a natural frequency at a subharmonic of the rotor operating speed, then, after one cycle of this subsynchronous free vibration, the rub impulse *will* recur (Figure 21-8). 1X vibration can also play a role by increasing the lateral velocity of the rotor at the moment of contact, thereby increasing the contact force.

During the dwell time of the contact, the tangential friction force acts to transfer some kinetic energy of rotation to radial vibration energy. This process produces a self-excited vibration that is larger in amplitude than would be possible without the energy transfer.

The effect here is similar to pushing a child on a swing. When the push (the impulse) is applied, the child moves away, acting as a pendulum in free vibration. When the child returns, if the push is applied *with the correct timing*, then more energy is added to the child and the amplitude of free vibration will increase. If, however, the timing is incorrect, energy may be removed from the system, and the amplitude will decrease. The whole process depends on adding energy at just the right moment.

In the same way, if the timing of repeated rub contacts is correct, the subsynchronous rotor vibration can be excited, amplified, and sustained as long as nothing in the system changes. Wear of the contacting surface is a factor that can change the contact relationship, as can local heating and rotor bow, leading to modification or eventual cessation of the rub.

Because of the time required for a subsynchronous vibration cycle, this kind of rub is usually associated with a *light* radial rub with a short dwell time. A

heavy rub (with a typically long dwell time) so strongly constrains shaft motion that there is insufficient time for a complete cycle of free vibration.

Even though we are discussing a subsynchronous, self-excited vibration, the 1X unbalance response also plays a role; the 1X response, when added to the free subsynchronous vibration, affects the timing of the contact between the rotor and the stator.

Subharmonic (½X,⅓X, ... 1/nX) vibration produces a timing relationship where every rub contact occurs at the same location on the rotor surface and the contact occurs once every n revolutions. If the contact friction causes local heating and bowing of the rotor, then the effective heavy spot and unbalance response vector may change, causing modification or cessation of the rub.

However, vibration at simple integer ratios (⅔X, ¾X, ... k/nX) produces *k* rub contacts for every n revolutions; the contacts occur at *k different* locations on the rotor surface, with a contact recurring at the same place on the surface once every n revolutions. For this type of vibration to maintain itself, the 1X unbalance response of the rotor must be relatively small; a large 1X vibration component would cause the contact timing to be significantly different for the *k* successive vibration cycles. Thus, only shafts with a small unbalance response have the potential to exhibit nonsubharmonic integer ratio vibration, and this kind of subsynchronous rub is less common.

An important factor in the timing of subsynchronous rub is the modification of the rotor natural frequency caused by the rub itself. For normal-tight rub, the constraint active during the dwell time of the rub causes an increase in the average spring stiffness of the rotor and its natural frequency, which we will call the *modified natural frequency*. For subsynchronous vibration to take place, the modified natural frequency must be a subharmonic (½X, ⅓X, etc.) or simple integer ratio (⅔X, ¾X, etc.) of running speed.

For a normal-tight rub, the modified natural frequency will be *higher* than the no-rub natural frequency. For a normal-loose condition, the spring stiffness is modified downward, and the modified natural frequency will be *lower*.

Because of the critical impulse timing needed to sustain subsynchronous vibration, it can appear only if the machine obeys certain rules. There are different, but similar rules for normal-tight and normal-loose conditions. In both cases, for subharmonic vibration to be sustained, *the rotor must rotate at a speed that is an integer multiple of the modified natural frequency*. For example, for ½X rub (either normal-tight or normal-loose) to occur, the rotor must operate at 2 times the *modified* natural frequency of the rotor. This seems obvious when you think about it, but this statement is the key to understanding why the rules that are presented below work.

The first rule applies to *normal-tight* rub. It states that, for 1/nX vibration to appear, the rotor speed, Ω, must be greater than n times the no-rub natural frequency:

Normal-tight subharmonic $\Omega > n\omega_n$ (21-4)

where n is an integer (2, 3, 4, ...), and ω_n is the no-rub (unmodified) natural frequency of the rotor given by Equation 21-3.

For example, for normal-tight rub, ½X vibration can appear only if the running speed of the rotor is greater than two times the *unmodified* natural frequency of the rotor. Similarly, for ⅓X vibration to appear, the rotor must operate at a speed greater than three times the unmodified natural frequency.

Why is this true? Figure 21-12, top shows the case for normal-tight rub. The green frequency line corresponds to the unmodified natural frequency, ω_n. Two possible rotor speeds are shown, Ω_1 (which is twice ω_n) and Ω_2. Any rub will stiffen the system and move ω_n to a higher frequency, the modified natural frequency (red). If this happens, then Ω_1 is now less than twice the modified natural frequency, and ½X vibration cannot occur. The rotor speed, Ω_2, which is greater than twice ω_n, is now twice the modified natural frequency, and ½X vibration is possible.

The second rule applies to a *normal-loose* condition. 1/nX normal-loose subharmonic vibration requires that

Normal-loose subharmonic $\Omega < n\omega_n$ (21-5)

For example, for ½X vibration to appear, the running speed of the rotor must be less than twice the *unmodified* natural frequency of the rotor. Similarly, for ⅓X vibration to appear, the rotor must operate at a speed less than three times the unmodified natural frequency.

The normal-loose condition (Figure 21-12, bottom) modifies ω_n by moving it to a *lower* frequency (red). If the rotor operates at exactly twice the unmodified natural frequency (Ω_1), then any shift of ω_n to a lower frequency will leave the rotor operating at more than twice the modified natural frequency, and ½X vibration will be impossible. For ½X vibration to occur, the rotor speed must be *less* than twice the unmodified natural frequency (Ω_2). Then, when the natural frequency shifts down to exactly one half rotor speed, ½X vibration can take place.

The examples in the figure are for ½X vibration. The same relationships apply to rub vibration at ⅓X, ¼X, etc.

These two rules can be helpful for diagnosing whether a subsynchronous vibration is due to a normal-tight or normal-loose condition. If startup or shutdown data is available, and if the balance resonance associated with the subharmonic vibration can be identified, then, for 1/nX vibration,

1. If the rotor speed is more than n times the balance resonance frequency, or if the subharmonic frequency is higher than the balance resonance frequency, then the rub is probably normal-tight.

2. If the rotor speed is less than n times the balance resonance frequency, or if the subharmonic frequency is less than the balance resonance frequency, then it is probably normal-loose.

In both cases, the self-excited, subharmonic vibration is occuring at the modified natural frequency of the balance resonance.

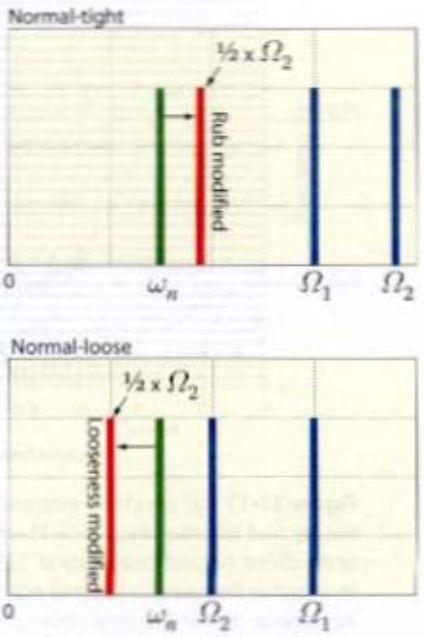


Figure 21-12. Rotor speed requirements for 1/2X vibration due to rub. For normal-tight rub, top, the green frequency line corresponds to the unmodified natural frequency, ω_n, which is exactly one half Ω_1. The rub will move ω_n to a higher frequency, the *modified* natural frequency (red). Ω_1 is now less than twice the modified natural frequency, and 1/2X vibration cannot occur. Ω_2 is greater than twice ω_n, and it is possible to increase the natural frequency up to a value that is exactly one-half Ω_2, making 1/2X vibration possible. For the normal-loose condition, bottom, ω_n will move to a *lower* frequency (red). Ω_1, which is twice the unmodified natural frequency, is more than twice the modified natural frequency, and 1/2X vibration will be impossible. When the natural frequency is shifted down so that it is exactly one-half Ω_2, 1/2X vibration can take place. The behavior for other subsynchronous frequencies is similar.

Subsynchronous vibration, when it appears, will often track running speed for a while during a startup or shutdown. For the frequency to satisfy the rules when the rotor speed is changing, some mechanism must exist that modifies the natural frequency appropriately. The rotor system adjusts the dwell time of the rub to control the modified spring stiffness.

This adjustment can only be done over a limited speed range. When the rotor system is no longer able to adjust the stiffness to satisfy the rule, then the subsynchronous vibration will either stop, or it will shift to another frequency that does satisfy the rule. Rub during the startup of a gas turbine was observed to begin with ½X, track running speed for a while, then shift to and track at ⅓X, and continue shifting to and tracking at ¼X and ⅕X.

A full spectrum cascade plot (Figure 21-13) can show these subsynchronous components. As the rotative speed increases, the vibration may suddenly drop to the next lower ratio when the condition for that ratio is satisfied.

Occasionally, subsynchronous vibration has been observed at simple integer

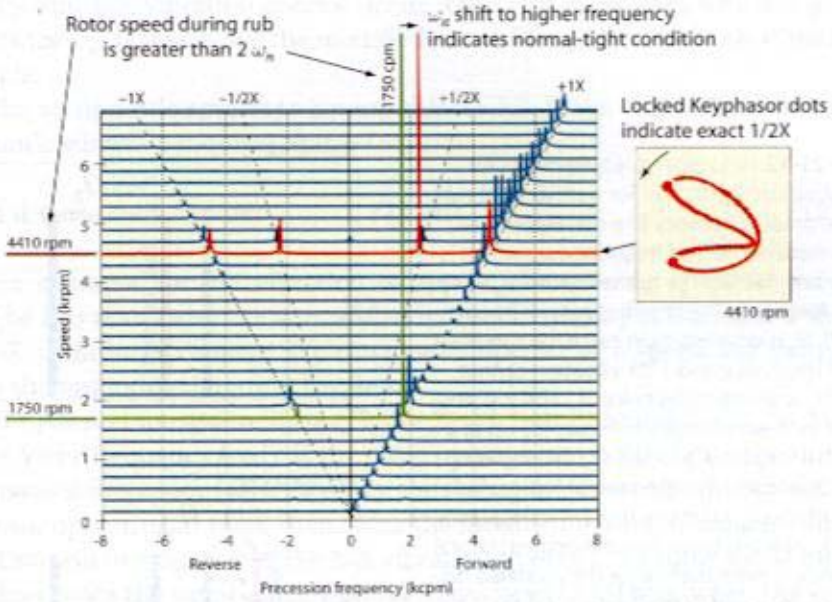


Figure 21-13. Full spectrum cascade of normal-tight rub. The 1/2X rub occurs during the second balance resonance. The rotor speed of 4410 rpm is more than twice the unmodified natural frequency of 1750 cpm, satisfying Equation 21-4. The shift of ω'_n to a higher frequency indicates normal-tight rub is taking place. The locked Keyphasor dots in the orbit show that the frequency of vibration is exactly 1/2X.

ratios such as ⅔X, ¾X, etc. The basic rules above still apply to this situation. For example, for normal-tight rub to produce vibration at ⅔X, the rotor speed must be greater than 3/2 times the unmodified natural frequency of the system; a normal-loose response would require a rotor speed less than 3/2 times the unmodified natural frequency.

Symptoms of Rub

Rub produces changes in both the forces and the Dynamic Stiffness of the rotor system. Because rub involves rotor interaction with a hard constraint, rub also introduces nonlinearities in the rotor system. The result of these effects is a complex rotor dynamic response that produces a wide variety of symptoms.

Like most malfunctions, diagnosis of rub involves correlation of different types of data. It is important to look at steady state and transient data, including direct orbit and timebase plots; full spectrum, including full spectrum cascade plots; 1X Bode and polar plots; and average shaft centerline plots. The symptoms of rub are summarized in Table 21-1.

Table 21-1. Symptoms of rub. Not all symptoms will be present at the same time.

Changes in 1X vibration
Abnormal orbit shape
Subsynchronous or subharmonic vibration
Reverse precession components
Harmonics in spectrum
Thermal bow
Changes in average shaft centerline position
Wear, damage, loss of efficiency

Changes in 1X vibration. During steady state operation, rub will produce changes in 1X vibration because of rotational energy transfer to radial vibration energy and because of changes in stiffness. A light rub is more likely to increase 1X vibration amplitude, while heavy rub can severely constrain the rotor and reduce 1X vibration. Heavy rub can also result in more energy transfer to the machine casing, causing an increase in 1X casing vibration.

If the rotor system is operating near a balance resonance, the 1X vibration amplitude can increase or decrease, depending on whether the machine is operating above or below the resonance (Figure 21-10). For normal-tight rub, the resonance is moved to a higher speed because of the rub-induced stiffness increase.

During startup and shutdown, rub-induced changes of the balance resonance speed can produce changes in observed behavior through the resonance (Figure 21-11). For this reason, it is always good to have reference startup and shutdown Bode and polar plots available.

Abnormal orbit shape. Direct (unfiltered) orbits should be examined and correlated with any other unusual activity that may be taking place in the machine. Changes in direct orbit shape should be noted, although the farther away the measurement point is from the rub, the less distinct the changes will be. Very importantly, only direct orbits with Keyphasor dots can be used to verify that subsynchronous vibration is an integer ratio of running speed. On an orbit, the number of displayed Keyphasor dots yields the denominator of a frequency ratio. For example, two Keyphasor dots could indicate ½X, ³⁄₂X, ⁵⁄₂X and so on. If the frequency really is locked to an integer ratio, then the Keyphasor dots will remain locked in place through subsequent vibration cycles (Figure 21-13). If they move along the path of the orbit, then the frequency is not an integer ratio.

Vibration that consists of a mixture of 1X and rub-induced ½X can produce orbits with complicated shapes (see the orbit in Figure 21-14). This orbit shows the path of the shaft centerline for eight shaft revolutions. The two stationary sets of Keyphasor dots show the vibration to be exactly ½X.

Subsynchronous/subharmonic vibration. Subsynchronous rub will follow the rules in Equations 21-4 and 21-5. Because of stiffening, normal-tight rub will produce vibration at a frequency *above* the unmodified natural frequency, while normal-loose will produce vibration at a frequency *below* the unmodified natural frequency. Full spectrum cascade plots can be used to identify which way the frequency shift is occurring, aiding diagnosis (Figure 21-13).

Note that it is not possible to verify that a vibration frequency is an exact integer ratio by using spectrum data. There is always some uncertainty in the

displayed frequency on a spectrum plot because of the limited resolution of the spectrum. A direct orbit with Keyphasor dot display should be used to verify the integer relationship.

Harmonics in the spectrum. As the rotor meets and rebounds from the contact surface, the changes in stiffness can produce sudden changes in the rotor trajectory and sharp corners on orbits and timebase plots. Because of this, harmonics of the fundamental vibration frequency can appear. If the fundamental frequency is 1X, then 2X, 3X, or higher harmonics may be visible in vibration data. If the predominant rub-induced vibration frequency is ½X, it is possible to see 1X (as a mixture of normal 1X rotor response and the harmonic of ½X), 3/2X, 4/2X, 5/2X, etc. (Figure 21-14).

Reverse precession components. Because the rub usually involves tangential friction forces acting against the direction of rotation, full spectrum plots will often show significant reverse precession components at the subsynchronous frequency (Figures 21-13 and 21-14).

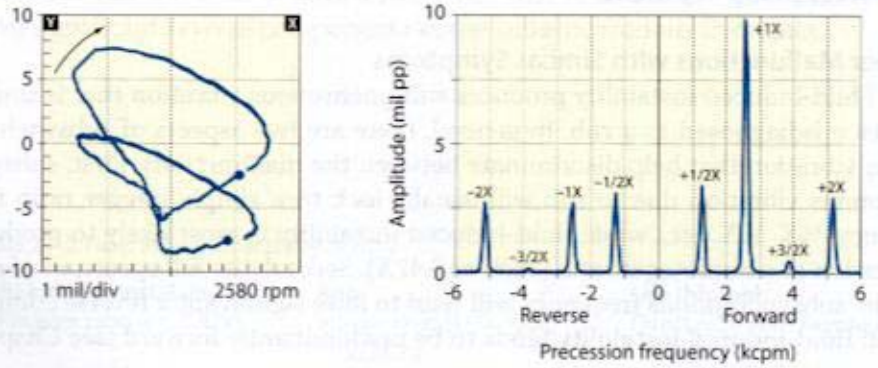


Figure 21-14. Orbit and full spectrum from a steam turbine with a 1/2X rub. The full spectrum shows forward and reverse 1/2X fundamental components and harmonics, including 3/2X and 2X. The nearly equal amplitudes of the 1/2X full spectrum lines indicate that the response at that frequency is nearly a line orbit.

Thermal bow. The frictional forces that act during radial rub produce local heating of the surface. If, at a steady operating speed, a rub occurs repeatedly in the same place on the rotor, the frictional heating of the surface and associated thermal expansion in that area will cause the rotor to bow in the direction of the rub contact. This bow effectively changes the unbalance magnitude and direction, which changes 1X rotor response. Under special circumstances, the local heating due to a light, 1X rub can produce a continuously changing 1X response vector (see Chapter 19).

Changes in average shaft centerline position. Rub will often change the trajectory (orbit) of the rotor, which can change the average shaft centerline position, too. Sudden changes in average shaft centerline position during startup, shutdown, or steady state operation can be symptomatic of a rub. Correlation of shaft centerline position changes with other data can help confirm a diagnosis.

Wear, damage, and loss of efficiency. Rub can cause extreme wear of contacting parts (Figure 21-1). Seals are especially vulnerable, and, because machine efficiency often depends on tight clearances, seal wear will usually result in degraded operating efficiency.

Machines suspected of rubbing should be carefully inspected for evidence of seal damage. Also look for discoloration of parts due to high temperature, scratched or smeared babbitt, and damaged seals, turbine blades, compressor blades, or pump impellers.

Other Malfunctions with Similar Symptoms

Fluid-induced instability produces subsynchronous vibration that is sometimes misdiagnosed as a rub. In general, there are two aspects of subsynchronous vibration that help discriminate between the malfunctions. First, subsynchronous vibration due to rub will usually lock to a simple integer ratio frequency, ½X, ⅓X, etc., while fluid-induced instability is most likely to produce vibration at a noninteger ratio (such as 0.47X). Second, the full spectrum of rub at the subsynchronous frequency will tend to have significant a reverse component; fluid-induced instability tends to be predominantly forward (see Chapter 22).

The diagnosis is further complicated because the large amplitude, subsynchronous vibration due to a fluid-induced instability can create a rub. This causes the rotor system to lock to an integer ratio frequency and the root cause fluid-induced instability to manifest itself as a rub. Both of these malfunctions may be active at the same time. They have a similar rotor dynamic property: the sub-

synchronous vibration is associated with a natural frequency of the rotor system.

Direct orbits (with Keyphasor dots) and the full spectrum are the best tools for discriminating between fluid-induced instability and rub. Vibration that occurs at an integer ratio will produce an orbit on which the Keyphasor dots are locked (do not change position with time). Fluid-induced instability is unlikely to occur at an exact integer ratio; thus, the Keyphasor dots will move around the orbit. A spectrum does not usually have the resolution to make this determination with certainty.

Full spectrum does give us powerful information about the direction of precession of frequency components (see Chapter 8). The subsynchronous vibration due to fluid-induced instability is predominantly forward, while rub may contain a significant reverse component. If the only diagnostic tool available is half spectrum, it is nearly impossible to tell the difference between these two malfunctions (Figure 21-15, top). On an oscilloscope, the fluid-induced instability orbit is not steady because the components are not harmonically related; the orbit shrinks and grows (breathes) as the components go in and out of phase, an internal loop rolls around the orbit, and the Keyphasor dots move along the orbit path. Some of this activity can be seen in the instability orbit (Figure 21-15, bottom right). Compare this to the rub orbits on the left, which are relatively steady.

To summarize (see Table 21-2), the subsynchronous vibration due to a partial radial rub will tend to have a frequency that is an integer ratio, and it may have significant reverse components at the subsynchronous frequency.

Table 21-2. Fluid-induced instability or rub?

Fluid-induced instability	**Rub**	**Use this tool**
Noninteger ratio (e.g. 0.47X)	Integer ratio (1/2X, 1/3X, 2/3X, ...)	Direct orbit with Keyphasor dots
Circular orbit, forward precession at instability frequency	Significant reverse components	Full spectrum, Direct orbit

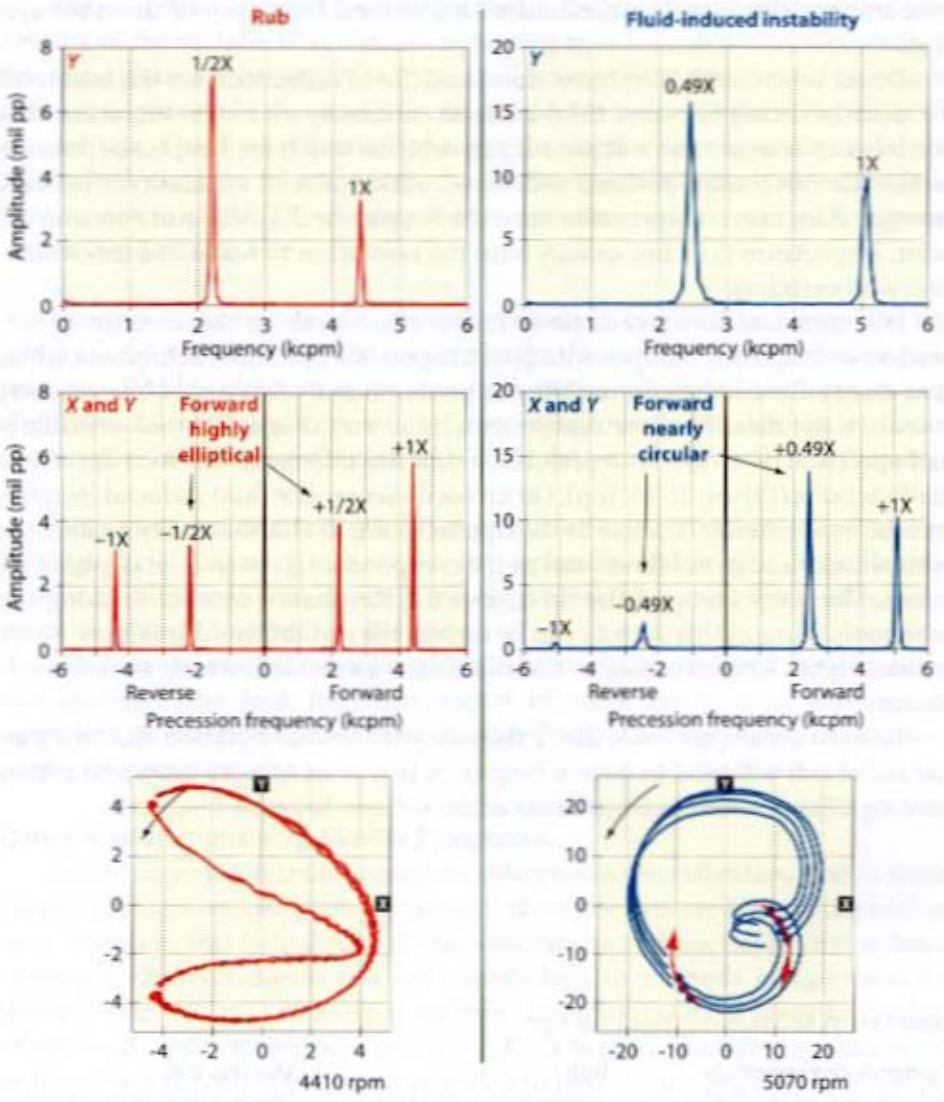


Figure 21-15. Comparison of rub (red) and a fluid-induced instability (blue) data. Note the similarity of the two half spectrum plots (top). The full spectrum plots clearly reveal the difference in the subsynchronous vibration: the subsynchronous component of the rub orbit is extremely elliptical; that of the fluid-induced instability orbit is forward and nearly circular. In the rub orbit, the Keyphasor dots are locked, indicating exact 1/2X vibration; in the fluid-induced instability orbit, the Keyphasor dots move around the orbit path indicated by the red arrows.

Scratches seen by eddy current probes will also produce harmonics in the spectrum. If the shaft has multiple scratches, a full spectrum display may show a complicated set of forward and reverse lines at harmonics of running speed. On a spectrum cascade plot, the amplitude of these spectrum lines will show little change with speed, from slow roll speed to operating speed. Harmonics due to rub, when they appear at all, will be visible only for the speed range that the rub is active.

A locked gear coupling can produce a sudden change in 1X vibration response together with a change in average shaft centerline position. A locked coupling may produce 2X vibration as well. Coupling problems will not produce subsynchronous vibration.

Summary

Rub is an undesired contact between a rotating and a stationary part, while looseness is a significant decrease in stiffness of the rotor constraint during part of the vibration cycle. *Rub is always a secondary malfunction*: it is caused by some other malfunction that reduces the available clearance in the rotor system. The reduced clearance can be caused by rotor deflection toward the stator, by out-of-position stationary parts, or by high vibration.

Rub contact can occur in the radial direction or in the axial direction, or in a combination of both.

The actual rub/stator contact can occur over a small fraction of the vibration cycle, a *partial rub*, or it can occur (rarely) over the entire vibration cycle, a *full annular rub*. Full annular rub can be *forward*, involving lubricated contact, or *reverse*, usually involving poorly lubricated or unlubricated contact.

Partial rub is the most common manifestation of rub. The period of time during which the rotor maintains contact with the stationary part is called the *dwell time*.

There are two general categories of partial rub, normal-tight and normal-loose. In *normal-tight* rub, the machine rotor normally operates in an *unconstrained* condition. When the rotor contacts a bearing wall, seal, or some other part, it becomes constrained, or *tight*. *Normal-loose* implies a condition during which the rotor operates normally in a *constrained* condition. At some point, the rotor moves clear of the constraint and becomes *loose*.

Rub contact produces both radial and tangential friction force components. The friction force acts opposite to the direction of rotation and tries to push the rotor in the reverse direction; it also produces a torque that can excite torsional vibration.

Normal-tight rub increases the stiffness of the rotor system, and a normal-loose condition decreases it. The stiffness change, and resulting natural frequency change, modifies the vibration response of the rotor system.

Heavy rub generally produces only modified 1X vibration. Light rub can produce subsynchronous vibration that is an integer ratio of running speed, usually ½X, ⅓X, etc., but occasionally ⅔X, ¾X, etc. For subsynchronous, $\frac{1}{n}$X vibration to appear with normal-tight rub, the rotor must operate at a speed more than n times the unmodified natural frequency; for a normal-loose condition, the rotor must operate at less than n times the unmodified natural frequency.

Rub can occur due to high vibration during a balance resonance. When this happens, the resonance frequency increases as the system becomes stiffer.

Rub symptoms can include changes in 1X vibration; abnormal orbit shape; subsynchronous and subharmonic vibration; reverse precession frequency components; harmonics in spectra; thermal bow; changes in average shaft centerline position; and wear, damage, and loss of efficiency.

Chapter 22

Fluid-Induced Instability

FLUID-INDUCED INSTABILITY IS A CONDITION that is caused by rotor interaction with a surrounding fluid. It can produce large amplitude, usually subsynchronous, self-excited vibration of a rotor, leading to rotor-to-stator rubs on seals, bearings, impellers, or other rotor parts. The vibration can also produce significant alternating stresses in the rotor, leading to fatigue failure. Because it is so potentially damaging, fluid-induced instability should be eliminated and avoided.

The term "instability" is somewhat of a misnomer. As we discussed in Chapter 14, when a rotor operates in fluid-induced instability, it is actually operating in a stable limit cycle of high vibration. But the rotor is unstable in the sense that it is operating outside desired operational limits.

In Chapter 14, we presented two mathematical approaches to stability analysis based on Dynamic Stiffness and eigenvalues. In this chapter, we will be primarily concerned with the detection of fluid-induced instability as a rotor system malfunction. First, though, we will revisit the Dynamic Stiffness view of instability and combine it with a discussion of the physical origin of fluid-induced instability. We will follow that with a discussion of the two primary modes of instability, whirl and whip, and show how they are actually the same phenomenon, caused by different sources of dominant stiffness in the rotor system.

After discussing the symptoms of fluid-induced instability, we will list other malfunctions that have similar symptoms and the characteristics and techniques used to make a correct diagnosis. Finally, we will show various strategies for eliminating and preventing fluid-induced instability.

The Cause of Fluid-Induced Instability

Rotating machines exist to convert the energy of rotation into useful work. This often involves rotor interaction with a working fluid, either gas or liquid. Also, most bearings in large machines are hydrodynamic, fluid-film bearings. The rotational interaction with surrounding working or bearing fluid always ends up swirling the fluid to some extent. This swirling is the primary cause of fluid-induced instability.

When a fluid, either liquid or gas, is contained in the annular region (gap) between two, concentric cylinders that are rotating relative to each other, the fluid will be set into motion. This situation exists in fully (360°) lubricated fluid-film bearings, in seals, around impellers in pumps, or when any part of a rotor is completely surrounded by fluid trapped between the rotor and the stator. It also exists to a lesser extent in partially lubricated bearings. In this chapter, we will talk primarily about cylindrical, fluid-film bearings. However, everything written here about these bearings also applies to seals, pump impellers, and any other region in a machine where a liquid or gas is contained in a small clearance between two parts that rotate at different angular velocities.

The fluid next to the rotor will have the same velocity as the surface of the rotor, while the fluid next to the bearing will have its velocity, usually zero. The fluid in the gap will have some average angular velocity, $\lambda\Omega$, where λ is the Fluid Circumferential Average Velocity Ratio, and Ω is the rotor speed. For a plain, cylindrical, fully lubricated bearing, λ is typically a little under ½, around 0.48 or so. But the value of λ is influenced by the geometry of the bearing, the rate of end leakage out of the bearing, the eccentricity ratio of the rotor in the bearing, and the presence of any pre- or antiswirling that may exist in the fluid. (See Chapter 10 for a full discussion of the meaning of λ.)

As we saw in Chapter 10, when a rotor is displaced from the center of a bearing, the converging fluid forms a pressure wedge. The pressure profile creates a force that can be separated into two components, a radial spring force, $-K\mathbf{r}$, that points back toward the center of the bearing, and a tangential force, $jD\lambda\Omega\mathbf{r}$, that acts at 90°, in the direction of rotation.

The strength of the tangential force depends on the rotative speed, Ω, and on the fluid circulation around the rotor, λ. Since properly loaded hydrodynamic bearings are normally only partially lubricated, λ is usually relatively small. Thus, a *properly loaded* bearing is unlikely to be a source of very large tangential forces unless it becomes flooded. Note that, because of misalignment, a fluid-film bearing can become lightly loaded and transition to fully lubricated operation, increasing λ and generating high tangential forces.

To see how the tangential force affects rotor stability, imagine a rotor rotating in the center of a fluid-film bearing. If the rotor is displaced from the center

and released, the spring force will push the rotor back toward the center. But, at the same time, the tangential force will push the rotor in the direction of rotation, 90° from the spring force. The rotor will begin to move at some angle determined by the ratio of the tangential force and the spring force. Once the rotor starts moving in response to these forces, the fluid damping will produce a force that is opposite to the direction of instantaneous rotor centerline motion. This force acts to slow down the motion, while the inertia of the rotor tries to keep the rotor moving in the same direction. All of these forces act together in a complicated way. The result is that the stable, rotating rotor cannot return directly to the center of the bearing; it follows a decaying spiral path (Figure 22-1a). The precession frequency of this spiral path is equal to the *damped natural frequency* of the rotor system.

As rotor speed increases, the tangential force becomes stronger, while the spring force and damping forces do not, because they are relatively independent of rotor speed. At some speed (the Threshold of Instability), the tangential force becomes so strong that the rotor will not return to its original position (Figure 22-1b). If the rotor speed remains constant, the rotor will continue in a constant amplitude orbit, or perhaps an increasing spiral orbit (Figure 22-1c). When this happens, the rotor is experiencing fluid-induced instability.

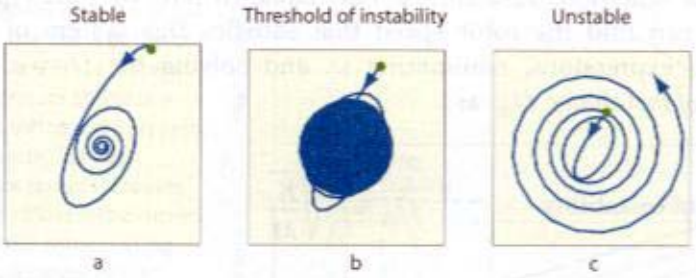


Figure 22-1. Path of the rotor centerline for three conditions of stability. In all three cases, the rotor is released after being displaced from its equilibrium position to the green point. In (a), the system is stable, but the displaced rotor cannot return directly because of the tangential force. The rotor returns to equilibrium by following the decaying spiral path. In (b), the rotor is operating at the Threshold of Instability, and, after an initial transient, the rotor moves in a circular orbit of constant diameter. In (c), the system is unstable; the orbit diameter increases.

These various forces can be viewed as stiffness elements and combined into the nonsynchronous Dynamic Stiffness, $\boldsymbol{K}_N$,

$$\boldsymbol{K}_N = K - M\omega^2 + jD(\omega - \lambda\Omega) \tag{22-1}$$

Dynamic Stiffness controls the response when the rotor is subjected to a static or dynamic force. The onset of fluid-induced instability occurs when the Dynamic Stiffness become zero.

Dynamic Stiffness is a complex quantity: it contains both direct and quadrature parts. For the Dynamic Stiffness to equal zero, both the direct and quadrature parts must be zero, *simultaneously*. Thus, the Direct Dynamic Stiffness is zero,

$$K - M\omega^2 = 0 \tag{22-2}$$

and the Quadrature Dynamic Stiffness is zero:

$$jD(\omega - \lambda\Omega) = 0 \tag{22-3}$$

For this last expression to be true, the term in parentheses must be to equal zero:

$$\omega - \lambda\Omega = 0 \tag{22-4}$$

Because Equations 22-2 and 22-4 are equal to zero, they are equal to each other. We can find the rotor speed that satisfies this system of equations. Combining expressions, eliminating ω, and solving for Ω, we define the *Threshold of Instability*, Ω_{th}, as

Threshold of Instability

$$\Omega_{th} = \frac{1}{\lambda}\sqrt{\frac{K}{M}} \tag{22-5}$$

This is the rotor speed at or above which the rotor system will be unstable. *This expression is a very powerful diagnostic tool, and it is the key to understanding how to prevent and cure fluid-induced instability problems in rotor systems.*

As the system becomes unstable, it begins to vibrate at a frequency equal to the undamped natural frequency of the system. Equations 22-2 and 22-4 tell us that, at the Threshold of Instability, the vibration frequency is

$$\omega = \lambda\Omega = \sqrt{\frac{K}{M}} \tag{22-6}$$

Because λ is usually less than 0.5, the vibration is subsynchronous. More complex rotor systems can experience the onset of subsynchronous instability vibration at frequencies different from $\lambda\Omega$.

How does the increase in tangential stiffness with rotor speed cause the complete loss of Dynamic Stiffness? Figure 22-2 shows the nonsynchronous Direct Dynamic Stiffness (top) and Quadrature Dynamic Stiffness (bottom) for our simple rotor model (see Chapter 11). The horizontal axis of these plots represents the frequency, ω, of some outside disturbing force that is independent of rotor speed (a nonsynchronous perturbation). The Direct Dynamic Stiffness is

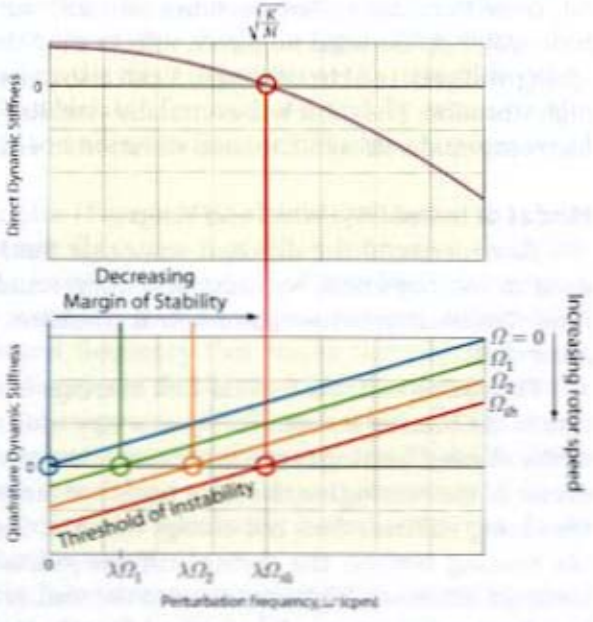


Figure 22-2. Nonsynchronous Direct (top) and Quadrature (bottom) Dynamic Stiffness plots and the Threshold of Instability. When rotor speed is zero (blue), the quadrature stiffness line starts at the origin of the plot. At this rotor speed, when the quadrature stiffness is zero, the direct stiffness is non zero, and some Dynamic Stiffness remains. As rotor speed increases, the quadrature stiffness line moves down, causing the zero crossing point of the quadrature plot to move to the right. As long as the quadrature stiffness and the direct stiffness are not simultaneously zero at any frequency, the rotor system will be stable. When the rotor reaches a speed at which the quadrature stiffness and the direct stiffness are both zero (red), the rotor is at the Threshold of Instability, Ω_{th}.

independent of rotor speed, and the zero crossing of the direct stiffness parabola occurs at the undamped natural frequency of the rotor system,

$$\omega = \omega_n = \sqrt{\frac{K}{M}} \tag{22-7}$$

The Quadrature Dynamic Stiffness changes with rotor speed because of the tangential stiffness term, which is the y intercept of the plot. When the rotor speed is zero, the quadrature stiffness line (blue) passes through the origin of the plot, and the quadrature stiffness is zero. The direct stiffness at this frequency is nonzero, though, and the system is stable. We define the *Margin of Stability* as the frequency difference between the direct and quadrature zero crossings.

As rotor speed increases, the quadrature stiffness line moves down (the tangential stiffness increases), causing the zero crossing point of the quadrature stiffness line to move to the right and the Margin of Stability to decrease. The system remains stable until the quadrature stiffness and the direct stiffness become zero at the same frequency (value of ω). This happens when the rotor speed reaches the Threshold of Instability, Ω_{th} (red). At this point, the Margin of Stability is zero.

Once the rotor system becomes unstable, subsynchronous vibration amplitude will increase until nonlinear effects near the bearing surface increase the spring stiffness. In extreme cases, K can also increase because of a rub due to the high vibration. The rotor will eventually establish itself in a "stable" limit cycle of high amplitude, subsynchronous vibration in either whirl or whip.

Modes of Instability: Whirl and Whip

To understand the different ways that fluid-induced instability manifests itself in rotor systems, we must first understand how the spring stiffness of a fluid-film bearing behaves and how it combines with the stiffness of a flexible rotor shaft.

The spring stiffness in fluid-film bearings is a strong function of rotor position in the bearing and can vary over a very wide range of values. The stiffness of a fully flooded bearing is at a minimum when the rotor is operating in the exact center of the bearing. As the rotor begins to move away from the bearing center, the spring stiffness does not change much at first. However, as the rotor nears the bearing surface, the spring stiffness starts to increase dramatically and becomes extremely high very close to the wall. It is this effect that produces the high load-carrying capacity of a fluid-film (hydrodynamic) bearing.

Up to now, we have treated the spring stiffness, K, as a single spring that acts

in the radial direction. However, the situation is usually more complicated than that. A flexible rotor can be thought of as a mass that is supported by a shaft spring, K_S, which is in turn supported by a bearing spring, K_B (Figure 22-3, left). Thus, K actually consists of two springs in series, with the total stiffness given by:

$$K = \frac{1}{\left(\frac{1}{K_S} + \frac{1}{K_B}\right)} = \frac{K_B}{\left(1 + \frac{K_B}{K_S}\right)} = \frac{K_S}{\left(1 + \frac{K_S}{K_B}\right)} \tag{22-8}$$

For any series combination of springs, the total stiffness is always less than the stiffness of the weakest spring, and the weak spring controls the system stiffness, K.

Typically, the bearing stiffness at the center of the bearing is much lower than the shaft stiffness. In this case, the ratio K_B/K_S is small, and K is a little less than K_B (Equation 22-8, center). Therefore, *at low eccentricity ratios, the bearing stiffness controls the system stiffness.*

On the other hand, the bearing stiffness close to the bearing surface is typically much higher than the rotor shaft stiffness. In this case, the ratio K_S/K_B is small, and K is a little less than K_S (Equation 22-8, right). Thus, *at high eccentricity ratios, the shaft stiffness controls the system stiffness.* The behavior of the system stiffness, K, versus eccentricity ratio is shown in Figure 22-3, right.

Fluid-induced instability causes the rotor to precess at the natural frequency of the system (Equation 22-7), which is a function of K and, by extension, the eccentricity ratio. This is the key to understanding the difference between whirl and whip.

The full spectrum cascade plot (Figure 22-4) shows the startup data of a rotor system where the rotor is initially centered in a fully lubricated, fluid-film bearing. When the rotor speed crosses the Threshold of Instability (2470 rpm), the rotor enters fluid-induced instability and develops a forward, subsynchronous vibration at a system natural frequency that tracks the rotor speed. At about 2900 rpm, the instability disappears due to the high vibration amplitude during the balance resonance, but reappears at 4670 rpm when the 1X amplitude decreases. The subsynchronous vibration frequency continues to track rotor speed for a while and then transitions to a constant frequency of about 3100 cpm. *The instability behavior is called whirl when it tracks rotor speed, and whip when it locks to a particular frequency.*

At the beginning of whirl, the rotor starts precessing at the natural frequency that corresponds to the stiffness near the center of the bearing: the *low-eccentricity natural frequency* of the rotor system. At this point, the bearing stiffness

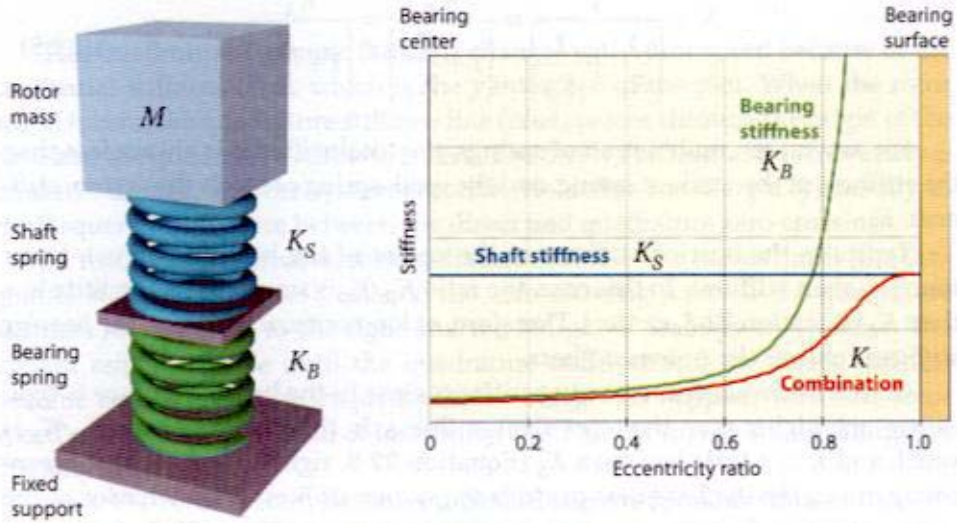


Figure 22-3. Rotor system stiffness versus eccentricity ratio. The system stiffness consists of the series combination of the shaft spring stiffness, K_S, (blue) and the bearing spring stiffness, K_B (green). The combination stiffness (red) is always less than the weakest spring in the system and is a function of eccentricity ratio. At low eccentricity ratios, the bearing stiffness, which is a strong function of eccentricity ratio, controls the combination stiffness; at high eccentricity ratios, the shaft spring, which is constant, controls the combination stiffness.

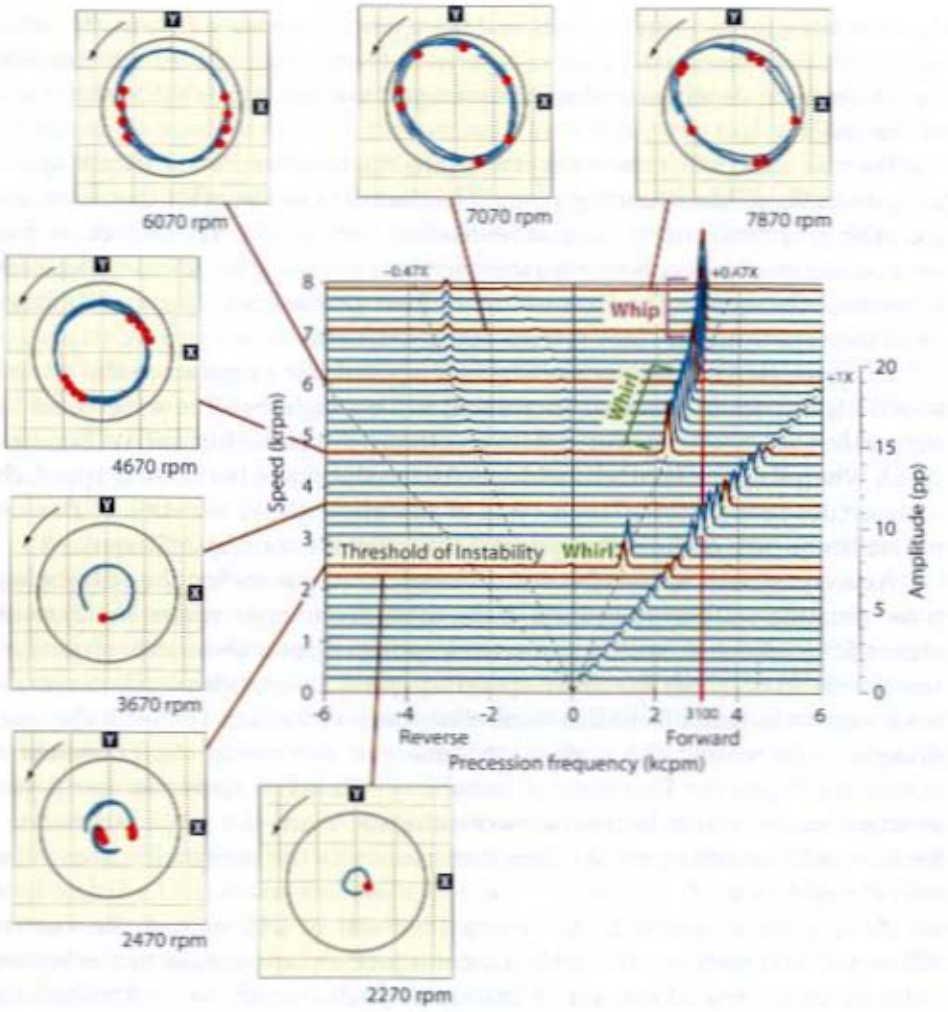


Figure 22-4. Full spectrum cascade plot showing whirl and whip during startup of an experimental machine. The measurement probes were mounted in the fully flooded, fluid-film bearing. The rotor system starts into whirl instability at about 2470 rpm, the Threshold of Instability. The orbits show the dynamic position of the rotor versus the approximate bearing clearance (gray). The Keyphasor dots (red) in the whirl orbits (2470 and 4670 rpm) are shifting slowly in a direction opposite to rotation, indicating that the frequency of vibration is a little less than 1/2X. See the text for details.

controls the rotor system stiffness and the natural frequency. Because the shaft spring stiffness is relatively high compared to the bearing stiffness, the instability vibration mode shape is likely to be a rigid body conical or cylindrical mode of the system.

For this case, when the rotor starts into fluid-induced instability, it spirals away from its stable operating point. The diameter of the orbit increases, and the orbit is approximately circular. While the *average* eccentricity ratio is close to zero, the *dynamic* eccentricity ratio increases as the orbit diameter increases. If the fluid-film bearing had a constant stiffness profile, this orbit would continue to increase until the rotor system destroyed itself.

However, this is not the case. The bearing stiffness increases as the dynamic eccentricity ratio increases, increasing K. This higher stiffness increases the natural frequency of the rotor system and the Threshold of Instability (Equation 22-5). When K moves the threshold speed to a value equal to the rotor speed, the rotor settles into a "stable" limit cycle of subsynchronous vibration at the new natural frequency of the rotor system. This is the situation at 2470 rpm.

At any constant speed above the Threshold of Instability, the subsynchronous vibration will maintain itself. If the orbit diameter increases, the dynamic eccentricity ratio and bearing and system spring stiffness also increase, pushing the Threshold of Instability above operating speed. This momentarily causes the rotor system to begin to return to *normal* stable operation. The orbit diameter decreases, decreasing the system stiffness, and decreasing the Threshold of Instability. When the Threshold of Instability falls below operating speed, subsynchronous vibration begins to increase again. Thus, the rotor system has a built-in, self-regulating mechanism that maintains the subsynchronous vibration at a certain level.

As the rotor speed continues to increase, it will exceed the current Threshold of Instability. The orbit diameter further increases until the bearing stiffness drives the Threshold of Instability high enough to "restabilize" the rotor. This cycle continues, and the subsynchronous vibration frequency (which is the natural frequency of the rotor system) *tracks running speed* as the rotor accelerates. The rotor operates on the continuously changing Threshold of Instability, with increasing orbit diameter, dynamic eccentricity ratio, bearing and system spring stiffness, and natural frequency.

For the simple rotor we have been discussing, the frequency of precession in fluid-induced instability can be found from Equation 22-6. The simple model is a good fit to the rotor in the experiment. The rotor is operating on the Threshold of Instability, and the precession frequency is close to $\lambda\Omega$. If λ is about 0.47, then the subsynchronous precession in whirl will take place at a frequency of about

0.47X. This relationship is not so simple for more complex rotor systems, but it does provide a useful estimate.

At about 2900 rpm, the rotor system encounters a balance resonance associated with a bending mode of the system. The temporarily higher 1X vibration amplitude moves the rotor to a higher dynamic eccentricity ratio, increasing K and moving the Threshold of Instability above running speed. The fluid-induced instability disappears until the 1X vibration amplitude decreases above the resonance. When the dynamic eccentricity ratio decreases again, K decreases, moving the Threshold of Instability back down through running speed, causing the instability to reappear at 4670 rpm.

As speed continues to increase, the dynamic eccentricity ratio approaches its limit as the rotor approaches the bearing surface, and the bearing stiffness becomes so high that it is no longer the weakest spring in the rotor system. The shaft spring stiffness, which is constant, becomes the weakest spring, and it controls the stiffness of the rotor system.

Because the shaft stiffness is now the upper limit of K, the vibration frequency (the natural frequency of the rotor system) asymptotically approaches a constant value, the *high-eccentricity natural frequency*. In this region, the whip region, the frequency of precession of the rotor system is constant. The high-eccentricity natural frequency often correlates well with the "nameplate critical" in rotating machinery, because normally loaded, fluid-film bearings operate at moderately high eccentricity ratios, where the bearing stiffness is significantly higher than the shaft stiffness.

Because $K_B >> K_S$, whip vibration is usually associated with a bending mode of the rotor. Note that the whip frequency of 3100 cpm in Figure 22-4 is associated with the 1X balance resonance that occurred earlier. During whip, vibration at the bearing is limited by the bearing clearance, but the bending mode can produce very high vibration amplitudes at locations between bearings. The resulting rubs and severe stress cycling can be very destructive.

Rotors normally operate at high average eccentricity ratios in partially lubricated bearings. If such a bearing becomes flooded with lubricant, it is possible for fluid-induced instability to suddenly appear as whip, completely skipping whirl. For whirl and its characteristic speed tracking to occur, the rotor natural frequency must be less than its maximum value at high eccentricity. This condition is most easily met when the rotor operates at a low eccentricity ratio near the center of a fluid-film bearing (as can happen due to misalignment). If the rotor encounters fluid-induced instability while operating at a high eccentricity ratio, then the instability will appear as whip.

Symptoms of Fluid-Induced Instability

Subsynchronous vibration. The primary symptom of fluid-induced instability is forward, subsynchronous vibration, typically at less than ½X. The frequency of the subsynchronous vibration in whirl is related to the Fluid Circumferential Average Velocity Ratio, λ, at the location that is the source of the instability. This subsynchronous frequency can range from 0.3X to 0.8X or higher if fluid has been preswirled. However, in whip the frequency of vibration will lock to the frequency of a rotor system bending mode (typically the lowest).

Unlike rub, fluid-induced instability almost never produces a simple integer ratio vibration frequency such as ⅔X, ½X, ⅓X, ¼X, etc. Instead, fluid-induced instability is most likely to produce mostly irrational subsynchronous frequencies. *However, if the lubricant film breaks down between the rotor and the bearing, or if the large amplitude instability vibration causes a rub, then fluid-induced instability can transition to rub and lock to an integer ratio.*

The subsynchronous vibration caused by fluid-induced instability is almost purely forward (Figure 22-4). This is a very useful way to discriminate between rub and fluid-induced instability. Rub tends to produce significant reverse components at the subsynchronous frequency.

During a startup or shutdown, whirl due to fluid-induced instability will track a percentage of running speed, while whip tends to lock to a constant frequency (Figure 22-4). It is possible for whip to suddenly appear without any whirl.

Fluid-induced instability is always associated with a natural frequency of the rotor system (usually the lowest mode). The whip frequency in Figure 22-4 is associated with a visible 1X balance resonance frequency (seen almost directly below the whip frequency). The 1X balance resonance vibration of a mode will not be visible, though, if the mode is supercritically damped (overdamped), as can happen with rigid body modes. For example, the whirl onset in Figure 22-4 is associated with an overdamped rigid-body mode, which does not produce a visible 1X resonance.

Orbit. If the vibration at the measurement plane is dominated by fluid-induced instability, then the direct orbit will be predominantly forward and circular (Figure 22-4). If the orbit is filtered to the instability frequency, it will always be approximately circular and forward.

The behavior of the Keyphasor dots will depend on the relationship of the subsynchronous frequency to running speed. The number of visible Keyphasor dots is related to the denominator of the integer ratio that is closest to the actual vibration frequency. Their direction of motion tells whether the ratio is greater or less than the integer ratio. For example, given a subsynchronous

vibration frequency near ½X, the denominator indicates that two Keyphasor dots will be visible. If the Keyphasor dots move in a direction *opposite* to rotation, then the ratio is slightly *below* ½X (this is the case in Figure 22-5). If the Keyphasor dots move in the *same direction* as rotation, then the ratio is slightly *above* ½X. Vibration near ⅓X will produce a set of three Keyphasor dots in the orbit that behave in a similar way. See Chapter 5.

When the subsynchronous vibration is not near an integer ratio, the Keyphasor dots will tend to form a jumbled pattern consisting of a great many dots. This is more typical for whip orbits. Whip orbits, because of the lower subsynchronous frequency ratios at which they usually occur, are more likely to show complicated Keyphasor dot behavior than whirl orbits (Figure 22-4, whip orbits).

If the vibration at the measurement plane contains a mixture of 1X and subsynchronous vibration, then the orbit will be more complex in shape. The subsynchronous vibration will cause the orbit to continuously change shape, but the motion of the Keyphasor dots (for frequencies close to an integer ratio) will still tend to migrate in a circle (Figure 22-5).

Average shaft centerline position. It can be useful to correlate the onset of subsynchronous vibration with movement of the shaft centerline toward the center of the bearing and the resulting modification of the natural frequency. When fluid-induced instability occurs and the orbit grows larger in whirl or whip, the average shaft centerline position may move closer to the bearing center. However, if the source of the instability is in a seal, which is normally concentric with the shaft, then the shaft centerline position may not change significantly.

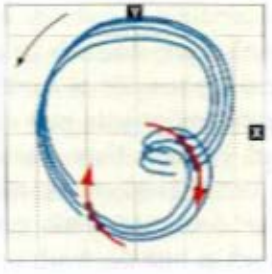


Figure 22-5. Whirl orbit showing a mixture of 1X and 0.48X vibration. In this case, because the frequency of vibration is slightly less than 1/2X, the Keyphasor dots move slowly opposite to the direction of rotation, in a small, circular path. The orbit continuously changes shape as the frequency components change their relative phase.

Other Malfunctions That Can Produce Similar Symptoms

Rotating stall and aerodynamic instabilities in compressors. Rotating stall is a phenomenon that can occur in both axial and centrifugal compressors when they are operated near the surge point. In rotating stall, partial flow separation creates one or more regions of impaired flow that rotate around the rotor disk or diffuser at a subsynchronous frequency. The vibration produced by rotating stall is, like fluid-induced instability, forward at the subsynchronous frequency, and it tends to track running speed. At times, rotating stall can look so much like fluid-induced instability that it can be very difficult to make an accurate diagnosis.

There are some clues that can be helpful. Rotating stall can occur at subsynchronous frequencies that are both lower and higher than the typical range for fluid-induced instability. Rotating stall in an impeller or axial blade section typically occurs at a frequency from 0.60X to 0.80X. Diffuser stall typically occurs in the range 0.10X to 0.30X.

Rotating stall, while *aerodynamically* self-excited, is not usually associated with a rotor system natural frequency. Thus, it is likely to occur at frequencies unrelated to observed balance resonance frequencies. However, the frequency of fluid-induced instability, which *is* associated with a system natural frequency, may also not correlate to a visible balance resonance if the rotor system is overdamped in that mode.

The subsynchronous spectral line from rotating stall can be sharp, or it can be broad and unsteady. Fluid-induced instability tends to produce a smooth, sharp, well-defined spectral line.

Rotating stall most often occurs when compressors are operated near the surge point. If rotating stall is suspected, keep the speed constant and change the machine operating condition to a point farther away from the surge point. If the problem is rotating stall, it may disappear. (An additional complication can result if something is obstructing the flow path in a compressor. Then it is possible to trigger rotating stall when the compressor is operating far from the surge point.)

Aerodynamic instability in compressors can produce a broad band of frequencies that are available to excite a rotor natural frequency. This type of excitation usually manifests itself with both forward and reverse subsynchronous frequency components in a full spectrum. The frequency lines are broad and noisy and, because they excite natural frequencies, the subsynchronous vibration frequency is often correlated with a rotor balance resonance. This type of excitation does not track running speed. It can sometimes look like fluid-induced whip, but the whip subsynchronous vibration is usually predominantly forward with a sharp spectral line.

Surge is an aerodynamic instability in compressors that involves large scale axial flow oscillations. Surge typically occurs at very low frequency, such as 0.10X. Even though surge is an axial flow phenomenon, it usually produces detectable lateral vibration as well as significant axial vibration of the rotor.

Rub. Rub can also produce subsynchronous vibration. However, there are significant differences between the subsynchronous vibration due to rub and that due to fluid-induced instability.

Fluid-induced instability tends to produce a predominantly forward, subsynchronous precession of the rotor. However, the subsynchronous vibration due to rub usually has significant reverse components, and it will lock to a frequency that is an integer ratio, such as ½X or ⅓X, or, more rarely, ⅔X, ¾X etc. Although fluid-induced instability occurs over a range that includes many of these frequencies, it is improbable that it will occur exactly at one of these integer ratios. It is very difficult to tell the difference between fluid instability and rub if the only tool available for diagnosis is a half spectrum, which has limited resolution and lacks rotor precession information.

A direct orbit with Keyphasor dots is the best tool for determining whether or not the frequency is an integer ratio (Figure 22-6). If the Keyphasor dots steadily change position in the orbit from one vibration cycle to the next, then the vibration is not an integer ratio, and the malfunction is not likely to be rub. In the figure, the Keyphasor dots steadily shift position in the fluid-induced instability orbit, and the orbit shape changes over time because of the changing phase of the frequency components. On an oscilloscope, the Keyphasor dots would steadily change position, following a nearly circular track. Part of that track can be seen in the figure.

Partial radial rub, when it manifests itself as steady subsynchronous vibration, tends to produce vibration at a subharmonic frequency. On a direct orbit, the Keyphasor dots will remain in approximately the same position for many cycles. Note that, in the figure, the Keyphasor dots on the rub orbit remain more tightly locked in position, and the orbit maintains the same shape and position over time.

Subsynchronous rub usually causes reverse precession components in the rotor response, whereas fluid-induced instability tends to produce a forward, nearly circular orbit when filtered to the instability frequency. The full spectrum can be very helpful for determining if the subsynchronous frequency is predominantly forward or has significant reverse components (Figure 22-6).

When a rotor is precessing in fluid-induced instability whip, the high amplitude vibration may cause occasional breakdown of the fluid film in the bearing. When this happens, metal to metal contact can occur, producing a rub, and the

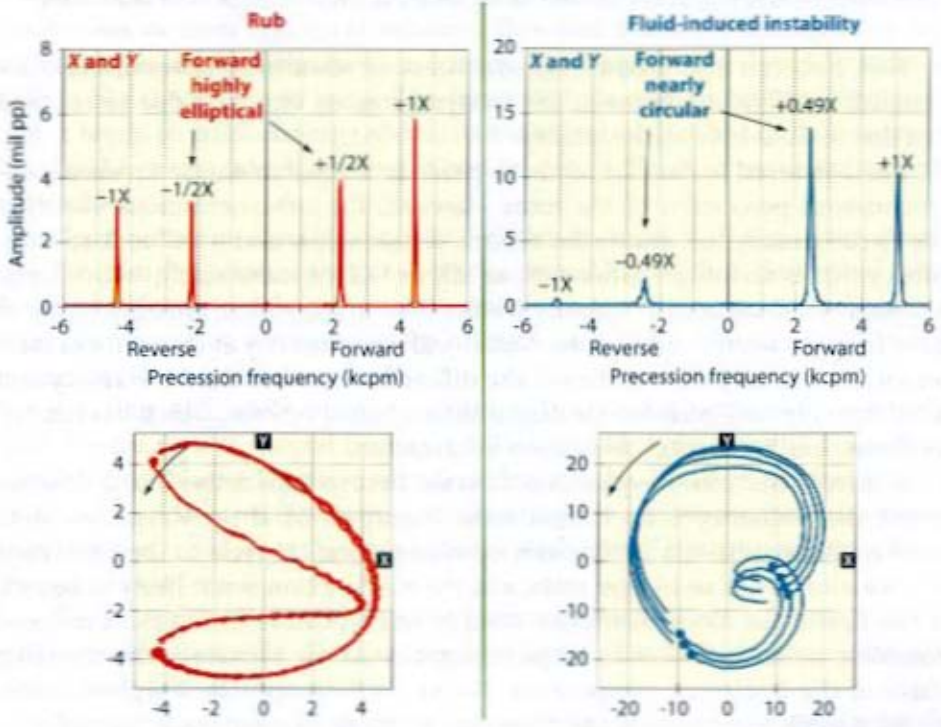


Figure 22-6. Comparison of rub and fluid-induced instability orbits and full spectra. Both orbits contain a mixture of 1X and subsynchronous vibration. This is frequently called "rabbit ears." The fluid-induced instability orbit (blue) is predominantly forward at the subsynchronous frequency. Also, the Keyphasor dots move, indicating that the orbit is not exactly 1/2X. The rub orbit (red) is locked to exactly 1/2X, and the subsynchronous frequency contains a significant reverse precession component in the full spectrum.

vibration frequency may lock to an integer ratio. During startup, Keyphasor dots may temporarily lock, then unlock, as the rotor passes through narrow speed ranges that are centered on integer ratios of the whip frequency to rotor speed. This phenomenon is essentially a type of lubricated rub, but the root cause is fluid-induced instability.

To summarize (Table 22-1), the subsynchronous vibration due to a partial radial rub will tend to have a frequency that is an integer ratio, and it will have significant reverse components at a, typically, subharmonic frequency.

Table 22-1. Partial radial rub or fluid-induced instability? This table compares the symptoms of fluid-induced instability and rub, and shows the best diagnostic plot to discriminate between the two conditions.

Fluid-induced instability	**Rub**	**Use this tool**
Noninteger ratio (e.g. 0.47X)	Integer ratio (1/2X, 1/3X, 2/3X, ...)	Direct orbit with Keyphasor dots
Circular orbit, forward precession at instability frequency	Significant reverse components	Full spectrum, Direct orbit

Locating the Source of Instability

When a machine is diagnosed as having a fluid-induced instability problem, the next step is to locate the source of the problem. Fluid-induced instability can originate in fluid-film bearings and seals, around pump impellers, or in any part of the machine where fluid is contained between two concentric cylinders that are rotating relative to each other. Machines that are equipped with stabilizing bearing designs (such as tilting pad bearings or externally pressurized, or hydrostatic bearings) can still experience fluid-induced instability in oil control rings, seals, or some other part of the machine.

The subsynchronous vibration due to fluid-induced instability originates at the source of the instability and propagates through the machine. A phase delay will occur that depends on the distance of the measurement plane from the source, the dynamic behavior of the rotor system, and, of course, the relative angular orientation of the measurement transducers.

While in fluid-induced instability, the rotor is operating on a rotor system natural frequency. Thus, the amount of the phase lag in the rotor will also depend on the mode shape of the natural frequency associated with the instability. Rigid body modes are likely to have small relative phase lags (perhaps only a few degrees), while bending modes can have substantial relative phase lags of many tens of degrees.

For any particular set of measurements taken at different distances from the source and adjusted for transducer orientation, the plane showing the largest relative phase *lead* is likely to be closest to the source of the instability.

Eliminating Fluid-Induced Instability

To ensure stability, all we need to do is ensure that Ω_{th} stays above the highest anticipated operating speed of the rotor, where

$$\Omega_{th} = \frac{1}{\lambda}\sqrt{\frac{K}{M}} \tag{22-9}$$

We can increase Ω_{th} by adjusting the parameters in this equation; we can decrease the fluid circulation, λ, increase the spring stiffness, K, or decrease the rotor mass, M. In reality, system damping can also play a role, but it has a relatively small effect, and our simple model neglects it. (See Chapter 29 for a case history where this effect may have been important.) We cannot easily change the rotor mass, M, but we *can* increase the spring stiffness, K, or reduce the fluid circulation in the clearance, λ. The following discussion will address the practical aspects of changing λ and K in rotor systems.

Reduction of fluid circulation (λ). The fluid circulation around the rotor is fundamentally responsible for the instability. The Fluid Circumferential Average Velocity Ratio, λ, is a measure of the strength of this circulation. Note that, in Equation 22-9, λ is outside the radical. Therefore, *anything that acts to disrupt fluid flow around the clearance will reduce λ and be most effective in increasing the Threshold of Instability.*

Reduction of λ is commonly done by using bearing geometries that depart from simple cylindrical shapes or introduce discontinuities to the lubricant flow (Figure 22-7). Tilting pad bearings are an example of this; because the pads are not continuous, fluid flow is disrupted in the bearing and stability is enhanced.

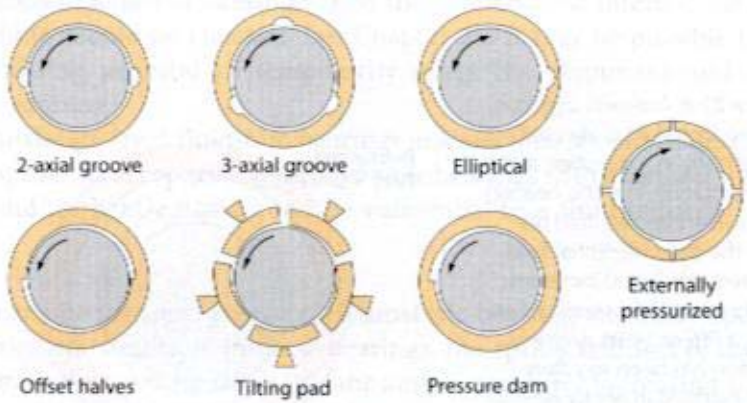


Figure 22-7. Bearing geometries that impair fluid circulation. These are several different types of bearings that have been developed over the years to break up circumferential fluid flow (reduce λ) and stabilize the bearing. The externally pressurized bearing can also enhance bearing stiffness.

Antiswirl injection involves injecting working fluid tangentially into the bearing or seal in a direction opposite to rotation (Figure 22-8). The injected fluid acts to reduce the average fluid angular velocity and, consequently, reduce λ. This technique has been proven to be very effective in both bearings and seals.

Externally pressurized bearings with antiswirl. New, externally pressurized (hydrostatic) bearing designs are emerging that can also incorporate antiswirl injection into their design. These externally pressurized bearings are fully lubricated by design, and they can produce high inherent spring stiffness, K_B. Additionally, the antiswirl injection reduces λ, making these bearings highly stable. The principle can be extended to pressurized seal design, which, by acting

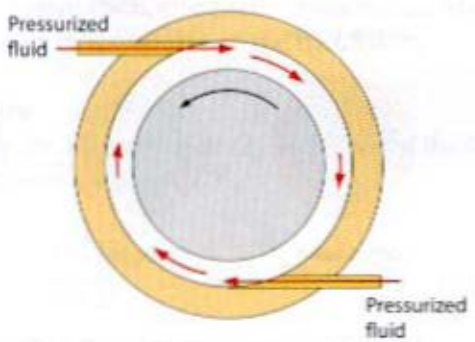


Figure 22-8. Antiswirl injection. The working fluid is injected tangentially, in a direction opposite to rotation, into the bearing or seal. The injected fluid disrupts the circumferential fluid flow, reduces λ, and increases the Threshold of Instability. For the last fifteen years, antiswirl injection has been regularly demonstrated in Bently Nevada lectures on rotating machinery behavior.

to stiffen the rotor system near midspan, can eliminate whip. Chapter 23 explains in detail externally pressurized bearings and machinery diagnostics.

Proper loading of hydrodynamic bearings. Fluid-induced instability often originates in hydrodynamic bearings that are operated below design load. Misalignment can shift the load from one bearing to one or more other bearings in the machine. The lightly loaded bearing will tend to position the rotor closer to the center of the bearing. This will weaken the bearing spring stiffness, K_B, and reduce the Threshold of Instability. At the same time, if the bearing becomes flooded and fully lubricated, then λ can increase dramatically and further reduce the Threshold of Instability. Either one of these changes may trigger fluid-induced instability.

Proper loading of the bearing will force the rotor to operate at a normal eccentricity ratio, which will keep K_B high and λ low. If a machine that once ran acceptably now exhibits fluid-induced instability, check the shaft centerline plot to see where the rotor is operating in the bearing clearance. If the rotor is found to be operating at a low eccentricity ratio in a particular bearing and high eccentricity ratios in adjacent bearings, then the external and internal alignment of the machine should be checked. See Chapter 20. It may be possible to cool or heat a bearing pedestal to temporarily affect the alignment and keep the machine running.

At the design level, fluid-film bearings in a machine should be designed with an appropriate load capacity. Hydrodynamic bearings with excessive load capacity can end up lightly loaded and be vulnerable to a fluid-induced instability problem.

Adjustment of supply pressure. Hydrostatic bearings normally operate in a fully lubricated condition. In these bearings, the spring stiffness of the bearing is strongly influenced by the lubricant supply pressure. Increasing the supply pressure will increase the bearing stiffness and may eliminate the instability.

Some seals act like hydrostatic bearings. The rotor in the seal area is normally completely surrounded by the working fluid of the seal, and the rotor is designed to operate in the center of the seal. Thus, increasing the seal fluid supply pressure may increase the stiffness of the seal and, if the seal is the source of the instability, eliminate the instability. Unfortunately, this will usually require a modification of the seal system.

Hydrodynamic bearings, on the other hand, normally operate in a partially lubricated condition at a relatively high eccentricity ratio. *Increasing the lubricant supply pressure may actually flood the bearing,* causing it to operate in a fully lubricated condition, potentially destabilizing the rotor system. If a hydro-

dynamic bearing is suspected of being the source of the fluid-induced instability, then reducing lubricant supply pressure may eliminate the flooded condition and stop the instability. Obviously, care must be taken to avoid reducing the supply pressure to such a low level that it causes damage to the bearing.

Adjustment of lube oil temperature. Fluid viscosity affects both K_B and the bearing damping, D. For the simple rotor model, the bearing damping has no effect on the Threshold of Instability (Equation 22-9). For more complex rotor systems, the bearing damping can have an effect. Thus, changing the fluid viscosity may have a significant effect on the fluid-induced instability.

The viscosity of lubricating oil is a strong function of temperature. Increasing the oil temperature reduces the viscosity, reducing both K_B and D. Decreasing the oil temperature increases the viscosity, increasing both K_B and D. In complex machines, a favorable change in K_B may be offset by an unfavorable change in D, or vice versa.

Thus, it is difficult to predict ahead of time how changes in oil temperature will affect the Threshold of Instability. In some cases, a change in oil supply temperature of only a few degrees has produced dramatic changes in the fluid-induced instability behavior of the machine. Try cooling down low viscosity fluids or heating up high viscosity fluids, a few degrees at a time.

Summary

Fluid-induced instability is a condition that can produce large amplitude, usually subsynchronous, self-excited vibration of a rotor, leading to rotor-to-stator rubs on seals, bearings, impellers, or other rotor parts.

Fluid-induced instability can occur wherever a fluid is contained between two concentric cylinders that are rotating relative to each other. It can occur in fluid-film bearings and seals and around impellers.

The onset of fluid-induced instability involves a loss of Dynamic Stiffness of the rotor system. Both the Direct and Quadrature Dynamic Stiffness become zero simultaneously.

The speed at which the instability onset occurs is called the *Threshold of Instability*, and is given by

$$\Omega_{th} = \frac{1}{\lambda}\sqrt{\frac{K}{M}}$$

Whirl occurs when the shaft operates near the center of a bearing and is usually associated with a rigid-body mode of the rotor. In whirl, the subsynchronous vibration tracks running speed.

Whip occurs when the shaft operates in the bearing at a high ecentricity ratio and is usually associated with a bending mode of the rotor. In whip, the subsynchronous vibration locks to a single frequency.

Curing fluid-induced instability involves increasing Ω_{th} above the highest anticipated operating speed of the machine. In the most practical sense, this can be done by reducing the fluid circulation, λ, or by increasing the system spring stiffness, K.

Reduce λ by using bearing geometries, such as tilting pad bearings or other bearings of noncontinuous geometry, that break up fluid circulation, or by anti-swirl injection.

Increase K by using externally pressurized (hydrostatic) bearings or by properly loading hydrodynamic bearings at normal eccentricity ratios (which also reduces λ).

Chapter 23

Externally Pressurized Bearings and Machinery Diagnostics

In previous sections of this book, we have discussed machinery diagnostics from the point of view of conventional bearing technology. The technology continues to evolve, and new bearing developments promise to change the behavior of machines and the way we look at machinery data. The externally pressurized bearing is undergoing rapid development, and, because of its external pressure source, it has the capability of producing variable spring stiffness under operator or automatic control. As the externally pressurized bearing enters general use, it promises to change rotor behavior in ways that will affect diagnostic methodology.

In this chapter, we will discuss the machinery diagnostic implications of variable stiffness. We will start by comparing the stiffness behavior of rolling element bearings, conventional hydrodynamic (internally pressurized) bearings, and externally pressurized bearings. Then, we will show how variable stiffness can affect rotor behavior. Finally, we will discuss the implications of variable stiffness for future machinery diagnostics.

Types of Bearings

The bearings used in large rotating machinery can be divided into two general types: rolling element bearings and fluid-film bearings. Fluid-film bearings can be further subdivided into internally and externally pressurized bearings.

Rolling element bearings most often find their application in small, balance-of-plant machinery, very low-speed machinery, or machines where the weight or complexity of a lubricant supply system for a fluid-film bearing cannot be justified. Rolling element bearings do not allow significant rotor motion near the bearing; they have very high spring stiffness and low damping, both of which are

essentially independent of load. Because of their high stiffness, rotor nodal points tend to be located within or very close to the bearings.

To provide more damping to machines equipped with rolling element bearings, squeeze-film dampers are sometimes used. For the same bearing design, a squeeze-film damper will tend to reduce the overall bearing stiffness and allow more shaft motion in the bearing plane.

Fluid-film bearings are used in most large rotating machines, where a thin film of lubricant keeps the journal and bearing separated. The film thickness in these bearings is typically on the order of tens of micrometers (a few mils). The lubricant is most often oil, but it can also be the working fluid of the machine; for example, water or even a gas.

Fluid-film bearings can be *internally* or *externally* pressurized.

Internally Pressurized Fluid-Film Bearings

Internally pressurized bearings support the rotor with the dynamic pressure of the lubricating fluid created inside the bearing by the action of the rotating journal. In these bearings, a continuous flow of oil is supplied to the bearing at low pressure, typically around 1 to 2 bar (15 to 30 psi). The bearing normally supports the journal at a moderate to high average eccentricity ratio, and the journal acts like a pump that produces circumferential flow of the fluid. As the fluid is drawn into the reduced clearance by the journal, the local fluid dynamic pressure increases, some of the fluid escapes axially, and the remaining fluid provides the support for the rotor. When a rotor is normally loaded and operating at design eccentricity ratios, plain cylindrical journal bearings are partially lubricated; that is, the bearing cavity is often not completely flooded with lubricating fluid.

When plain cylindrical journal bearings are lightly loaded, perhaps due to misalignment or radial loads on the rotor that act against gravity, the journal can move to low eccentricity ratios, and the bearing can become fully flooded. When this happens, it can trigger fluid-induced instability. The root cause of this instability is the tangential force produced by the differential pressure associated with the converging and diverging circulating fluid (see Chapter 22).

In an attempt to counteract this effect by disrupting the fluid flow, various bearing geometries have evolved, such as axial groove, elliptical, offset half, and pressure dam bearings. Even with these designs, fluid-induced instability is occasionally a problem.

Tilting pad bearings were developed to provide an inherently more stable, internally pressurized, fluid-film bearing. They have several separate, pivoted pads that support the journal. Each pad rests on a single pivot or rocker that, ideally at least, cannot support a moment. The design ensures that the support

force is directed through the pivot toward the center of the journal. When this condition is satisfied, no tangential forces can develop, and the bearing will be stable. However, fluid-induced instability has been known to occur in tilting pad bearing machines because of circumferential oil flow in the oil control (retainer) rings of these bearings or because of circumferential fluid flow elsewhere in the machine, such as in seals or around turbine or impeller disks. Tilting pad bearings are effectively fully flooded, with a hydrodynamic oil film existing at all times on all of the pads, and the journal normally operates at eccentricity ratios close to zero. Tilting pad bearings are in widespread use on many types of rotating machinery.

Externally Pressurized Fluid-Film Bearings

An externally pressurized fluid-film bearing operates in a fully flooded condition by design. Fluid at relatively high pressure is supplied to a set of inlet orifices that restrict the flow and create a pressure drop, and pockets distribute the fluid pressure over parts of the journal (Figure 23-1). The fluid escapes axially through the journal/bearing radial clearance.

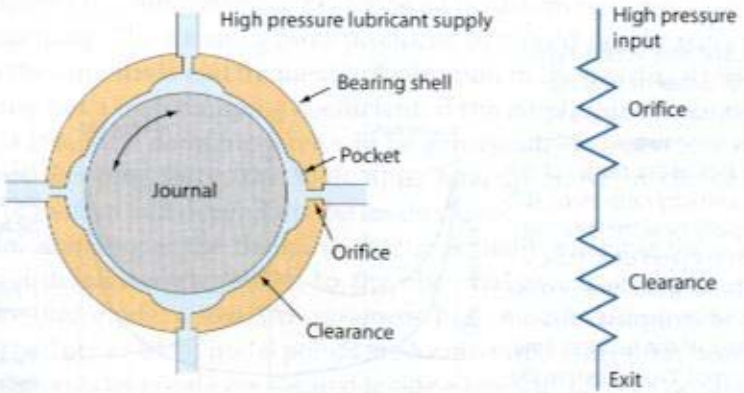


Figure 23-1. Externally pressurized bearing and equivalent resistance network. The high pressure supply feeds multiple inlet ports. The pressure drops across an orifice to an intermediate level. The clearance between the journal and the bearing creates another flow resistance that changes when the journal position changes in the bearing.

The pressures and fluid flow in an externally pressurized bearing are similar to the voltages and electric current in a series resistor circuit. The orifice is equivalent to one resistor; the flow resistance created by the journal/bearing clearance is the second. Thus, the fluid in the pocket is at a lower pressure than the supply pressure, but is higher than the pressure at the exit of the bearing.

When a nonrotating journal is centered in an externally pressurized bearing, the fluid flow is equally divided between the bearing pockets. In this condition, the pressure in each bearing pocket is equal, and no net force is produced on the rotor.

When a nonrotating journal is displaced from the center, the flow becomes more restricted in the narrow clearance and less restricted in the wider clearance. The lower restriction in the larger clearance lowers flow resistance, reducing the pressure in that pocket. Meanwhile, on the opposite side of the bearing, the narrower clearance increases flow resistance, increasing the pocket pressure on that side. The net result is a pressure differential and a restoring force that attempts to push the rotor back toward the center of the bearing. The differential pressure is approximately proportional to the displacement from the center and produces the equivalent of a spring stiffness.

If the journal is rotating, then displacement from the center also creates a hydrodynamic pressure wedge similar to that in an internally pressurized bearing. Thus, the stiffness of an externally pressurized bearing is a due to a combination of static and dynamic pressure effects (Figure 23-2). If the supply pres-

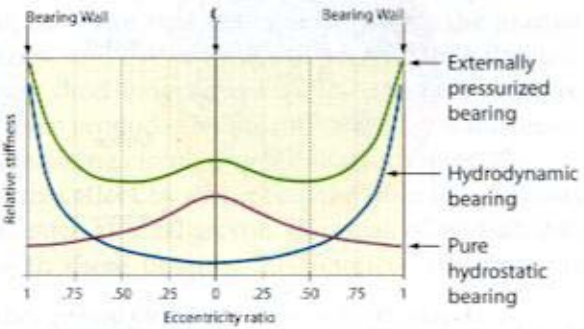


Figure 23-2. Stiffness versus eccentricity ratio for different bearing types. Hydrodynamic bearing stiffness (blue) is minimum when the journal operates at the center of the bearing (eccentricity ratio zero); stiffness rises rapidly near the bearing wall (eccentricity ratio one). Pure hydrostatic bearing stiffness (violet) is maximum when the journal is near the center of the bearing (low eccentricity ratio). Typical externally pressurized bearing stiffness (green) has characteristics of both types: high stiffness at low eccentricity ratios and high stiffness at high eccentricity ratios.

sure is sufficiently high, static pressure effects dominate, hence the name *hydrostatic* bearing.

For the same size bearing, externally pressurized bearings have higher stiffness than internally pressurized bearings. Also, an externally pressurized bearing can be pressurized before startup to lift the rotor, eliminating metal to metal contact and bearing wear, and allowing very low starting torque. The high stiffness of this bearing forces the journal to operate near the center of the bearing clearance most of the time. At such low eccentricity ratios, the attitude and position angles can become unimportant.

The hydrodynamic behavior of the externally pressurized bearing can cause fluid-induced instability problems. The hydrodynamic pressure wedge produces a tangential force component similar to that in an ordinary, plain cylindrical, internally pressurized bearing. In relatively low-speed turbomachinery applications, it is possible to defeat this circumferential flow problem by antiswirl injection. At the time of this writing, considerable research is taking place on how to eliminate circumferential flow and ensure stability in high speed turbomachinery applications.

Damping in externally pressurized bearings is more independent of stiffness than in hydrodynamic bearings. When oil is used as a lubricant, damping can be largely controlled by changing lubricant temperature (viscosity). Spring stiffness is largely independent of damping in these bearings.

Stiffness and Modal Damping in Fluid-Film Bearings

Compared to rolling element bearings, all fluid-film bearings have inherently high damping. The damping force produced by a fluid-film bearing is proportional to the amplitude and frequency of vibration in the bearing. However, even if a bearing has a high damping coefficient, if the amplitude of vibration in the bearing is low, little damping force will be generated. To generate a significant amount of damping force, the shaft must laterally move in the bearing; the amount of motion will depend on the mode shape.

Modal damping is the damping that is actually available for a particular mode, and it is inversely related to the observed Synchronous Amplification Factor *for that mode*: a low SAF produces high modal damping. High modal damping will occur when nodal points are located well away from bearings; conversely, when nodal points are located inside a bearing, little or no shaft motion can take place, resulting in low modal damping.

Rigid body modes (high shaft stiffness relative to bearing stiffness) tend to have nodal points located well away from the bearings, relatively large vibration amplitudes inside the bearings, and high modal damping. For this reason, rigid body modes are often overdamped (supercritically damped); machines will not

exhibit a visible balance resonance associated with overdamped modes during startup or shutdown. Rigid body modes that are underdamped still tend to have relatively high modal damping with low Synchronous Amplification Factors.

Flexible rotor modes (low shaft stiffness relative to bearing stiffness) tend to have nodal points closer to or inside the bearings and lower vibration amplitudes inside the bearings. High bearing stiffness, by constraining rotor motion, tends to move nodal points closer to the bearing. (This is easy to understand if you imagine tightly squeezing the rotor at the bearing. The tighter you squeeze, the smaller the allowable motion at that point.) Nodal points near the bearing reduce shaft vibration in the bearing, decrease interaction with the lubricating fluid, and result in low modal damping. Thus, bearing stiffness, because of its effect on the location of nodal points relative to bearings, has a powerful effect on modal damping. Small changes in nodal point location can have a dramatic effect on modal damping. It is not sufficient to have bearings with high damping; the bearing stiffness must position nodal points in a way that allows enough movement in the bearing to provide adequate *modal* damping. Often the best compromise design occurs when bearing stiffness is approximately equal to the shaft stiffness.

Variable Stiffness in Internally Pressurized Bearings

While internally pressurized bearings are passive devices, they have spring stiffness that is a strong function of eccentricity ratio (Figure 23-2). Spring stiffness is minimum near the center of the bearing (eccentricity ratio of zero) and increases dramatically near the bearing wall (eccentricity ratio of one). In horizontal machines with plain cylindrical bearings, operation near the center of the bearing can only occur when a bearing is very lightly loaded, either as a result of misalignment or a radial load that cancels the gravity load of the rotor; for example, partial arc steam admission in a steam turbine.

Rotor speed and radial load both affect the average shaft centerline position of the journal within the bearing. The dependence of stiffness on eccentricity ratio means that it also depends on speed and load. As rotor speed increases, the dynamic pressure in the fluid film also increases, causing an increase in stiffness and a change in rotor position. This is easily visible on an average shaft centerline plot as a rotor moves off the bearing during startup. At any given rotor speed, changes in the magnitude of the radial load will also produce changes in the rotor eccentricity ratio and changes in stiffness.

Stiffness changes will change the rotor dynamic behavior. The undamped natural frequency, ω_n, of a rotor system can be expressed in its simplest form as

$$\omega_n = \sqrt{\frac{K}{M}} \tag{23-1}$$

where K is the modal stiffness of the rotor system, and M is the modal mass. In this equation, K includes the modal effects of the series and parallel combination of all the stiffnesses in the rotor system: the shaft stiffness, bearing stiffness, bearing support stiffness, and foundation stiffness. If we assume that the support and foundation stiffnesses are essentially infinite (rigid), K can be expressed as the series combination of the bearing stiffness, K_B, and the modal shaft stiffness, K_S:

$$K = \frac{1}{\left(\frac{1}{K_S} + \frac{1}{K_B}\right)} = \frac{K_B}{\left(1 + \frac{K_B}{K_S}\right)} = \frac{K_S}{\left(1 + \frac{K_S}{K_B}\right)} \tag{23-2}$$

When the bearing stiffness is relatively low, as is usually the case for the lowest rigid body modes, it will control the stiffness of the combination, and K can be expressed by the middle equation in Equations 23-2. In this case, because K is a function of eccentricity ratio in the bearing, ω_n will also be a function of eccentricity ratio.

This effect can often be observed in the lowest, rigid body modes on a critical speed map, a plot of rotor system natural frequencies versus bearing stiffness (Figure 23-3). On these plots, the natural frequencies of these modes often show a strong dependence on bearing stiffness, while the natural frequencies of the higher, bending modes, where stiffness is controlled by the shaft stiffness, are

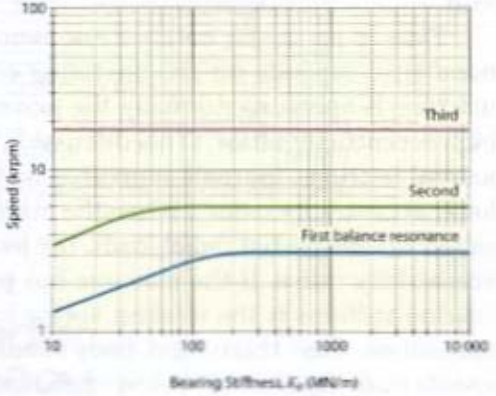


Figure 23-3. Typical critical speed map. The map shows how rotor system balance resonance speeds (vertical axis) change with different values of bearing stiffness (horizontal axis). Several modes are shown. The lowest modes tend to be more sensitive to changes in bearing stiffness than higher modes.

relatively insensitive to changes in bearing stiffness. This last situation can be represented by the rightmost equation in Equations 23-2.

These stiffness effects are responsible for the observed behavior of whirl and whip during fluid-induced instability (Figure 23-4). During whirl, the frequency of the instability vibration tracks rotor speed because, as the subsynchronous orbit diameter increases, the dynamic eccentricity ratio increases, which increases the bearing stiffness and the natural frequency. In the speed range where whirl is taking place, the bearing stiffness is significantly less than the shaft stiffness, the middle equation of Equations 23-2 applies, and changes in eccentricity ratio and bearing stiffness cause a change in rotor system natural frequency. When the eccentricity ratio becomes high enough, the bearing stiffness becomes significantly larger than the shaft stiffness, and the system transitions to whip. In whip, the natural frequency is constant because the shaft stiffness is significantly less than the bearing stiffness, and the shaft stiffness controls the combination stiffness, the right equation of Equations 23-2, and cannot be changed.

At very high eccentricity ratios, where the bearing stiffness is higher than the shaft stiffness, the natural frequency of that mode is relatively high compared to the natural frequency at lower eccentricity ratios; this natural frequency is called the *high-eccentricity natural frequency*. The high-eccentricity natural frequency is approximately the same as the balance resonance frequency documented by the manufacturer, the *nameplate critical*. Most often, in large horizontal machines, the high-eccentricity natural frequency is observed in conjunction with shaft bending modes.

If, for some reason, the lightly loaded rotor operates at an abnormally low eccentricity ratio, the bearing stiffness and natural frequency will also be relatively low, the *low eccentricity natural frequency*. It is only likely to be encountered during abnormal conditions, where the bearing operates below design load.

Thus, a particular balance resonance frequency may exist in a *frequency band* that depends on the operating conditions of the machine. When the machine is operating normally, the journals operate, by design, at moderate to high eccentricity ratios. (This discussion pertains primarily to plain cylindrical journal bearings. Journals supported by tilting pad bearings tend to operate at lower eccentricity ratios.) When the machine is significantly misaligned or subjected to unexpected radial loads, the journals operate at very low or very high eccentricity ratios. If the machine has primarily rigid body modes, where the bearing stiffness is the weakest spring in the system during normal operating conditions, then these rigid body natural frequencies may actually occur at speeds from significantly below to above their nameplate criticals. Systems with

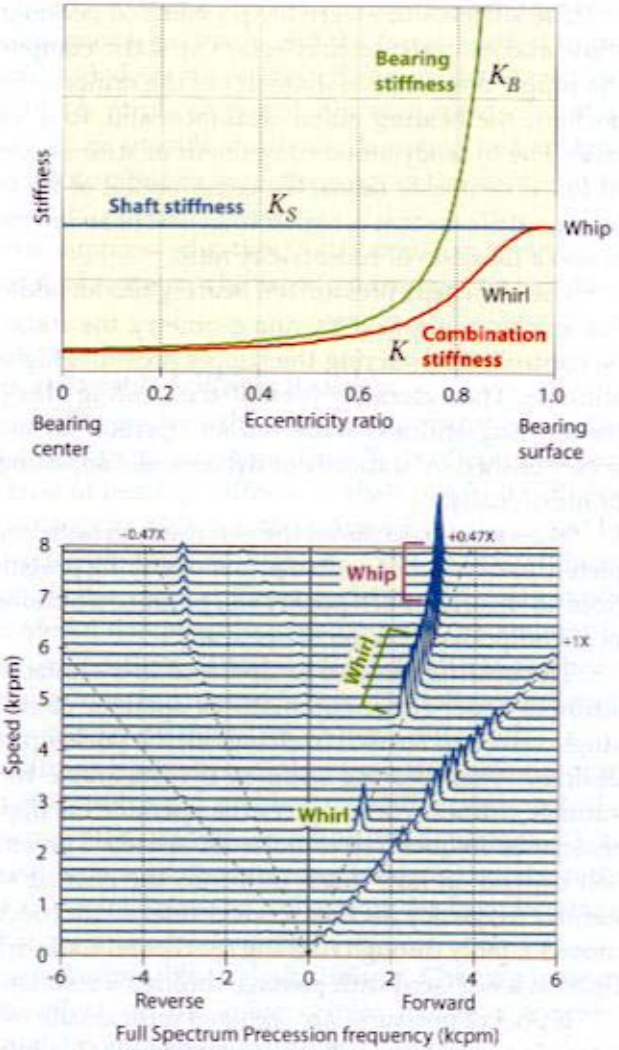


Figure 23-4. Bearing stiffness, shaft stiffness, and whirl and whip. The upper plot shows stiffness as a function of eccentricity ratio. The combination stiffness (red) is always less than its weakest component, bearing stiffness (green) or shaft stiffness (blue). As journal eccentricity ratio increases during whirl, the overall stiffness increases until it becomes approximately equal to the shaft stiffness. At this point, the whirl transitions to whip and locks to a single natural frequency controlled by shaft stiffness (lower plot).

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flexible rotors, where shaft stiffnesses control resonance frequencies, will show a much smaller change.

Variable Stiffness in Externally Pressurized Bearings

The stiffness of externally pressurized bearings includes a combination of static and hydrodynamic effects. The static component of stiffness depends on the supply pressure; the diameter of the orifices; the number, area, and shape of pockets; the bearing radial clearance; and, to a lesser extent, the eccentricity ratio. The hydrodynamic component of stiffness depends on eccentricity ratio. At low eccentricity ratios, the combination of these effects produces an overall bearing stiffness that is higher than that of an internally pressurized bearing and is also a function of eccentricity ratio.

The externally pressurized bearing has an additional, important advantage. For any given physical bearing geometry, the static component of stiffness can be controlled by varying the supply pressure; higher pressure produces higher stiffness. The externally pressurized bearing also provides the opportunity to put bearing stiffness under either operator or automatic control. Control is accomplished by statically or dynamically adjusting pocket pressures in groups or individually.

In its most basic form, the externally pressurized bearing operates in a completely passive mode, with a preset operating pressure applied to all the inlet orifices. In this case, each pocket will produce a uniform, constant hydrostatic stiffness component around the bearing.

The bearing can also be operated in a semiactive mode to provide variable stiffness under either automatic or operator control. In this operating mode, a single valve can be used to change all the pocket pressures simultaneously when desired. When bearing stiffness is comparable to or less than shaft stiffness, variable stiffness control gives the operator the opportunity to shift the balance resonance frequencies to different speeds. The amount of the frequency shift will depend on how much influence the bearing stiffness has over the system natural frequency for that mode. Under some conditions, the resonance can be moved rapidly through running speed, and some or most of the vibration amplification associated with passage through a resonance can be avoided.

If pocket pressures are adjusted individually by separate control valves, it is possible to produce a change in average shaft centerline position. For example, given a radial load which moves the journal away from the center of the bearing, if the pressure is increased in pockets opposite the load and decreased in pockets on the same side of the load, the net force will push the average shaft centerline of the journal back to the bearing center. This can be done by sensing the position of the journal, comparing it to a predefined set point, and using an

automatic control system to make the adjustments. Because the average of the individual pocket pressures does not change, the overall bearing stiffness remains approximately the same.

Finally, the bearing can be operated in a fully active mode, where rotor position information is provided to an automatic feedback control system that continuously adjusts the individual pocket pressures and the forces applied to the journal. The type of control used determines the effects of these pressure changes. The use of proportional control can provide the equivalent of a rotating stiffness, derivative control can provide synthetic damping, and integral control can be used to adjust the dc offset to move the rotor to the desired average set point.

The fully active bearing will suppress vibration (for example, in the event of sudden blade loss), and the control signal can be used to provide input to a diagnostic system.

Rotor Dynamic Implications of Variable Stiffness Bearings

Changing bearing stiffness changes how the rotor system responds to the static and dynamic forces that exist in the machine. The effects of variable bearing stiffness depend on the ratio of bearing stiffness to shaft bending stiffness. In the following discussion, we assume a shaft with stiffness K_S is supported by two variable stiffness bearings with total stiffness K_B. The modal stiffness depends on the combination of the shaft stiffness and the deflection mode shape of a particular mode. Modal damping depends on both the amount of damping in the bearings and the locations of nodal points relative to the bearings. These effects can be divided into three general cases. The following discussion is qualitative in nature; particular machines may behave differently.

$K_B << K_S$. The bearing stiffness is significantly less than the shaft stiffness. In this situation, the rotor is effectively rigid, and bearing stiffness controls the rotor system natural frequency (the middle equation of Equations 23-2). Changing bearing stiffness will have a relatively strong influence on the rotor system natural frequency. Nodal point locations will remain far from bearings, producing relatively small changes in modal damping.

$K_B \approx K_S$. Bearing stiffness is comparable to shaft stiffness. Changes in bearing stiffness will have some effect on the rotor system natural frequency. Stiffness changes will move nodal points and affect modal damping.

$K_B >> K_S$. The bearing stiffness is significantly greater than the shaft stiffness. In this situation, the rotor is flexible, and shaft stiffness controls the rotor system natural frequency, which will be relatively insensitive to changes in bearing stiffness. With stiff bearings, rotor nodal points will be close to the bearings, and stiffness changes will have a large influence on nodal point locations rela-

tive to the bearing. As the bearing stiffens, nodal points will approach and enter the bearings, shaft vibration in the bearings will decrease, modal damping will drop, and amplification factors at resonance will increase. Similarly, reducing bearing stiffness will increase shaft vibration in the bearings, increase modal damping, and reduce amplification factors at resonance. Thus, for this case, variable stiffness will have more of an effect on modal damping than on natural frequency. Note that the American Petroleum Institute recommends that $K_B \leq 4K_S$ to avoid high amplification factors at resonance [1].

Figure 23-5 shows, when $K_B \approx K_S$, how the balance resonance frequency can be changed by changing bearing stiffness. The left Bode plot displays the vibration response of the simple rotor model of Chapter 10 for three ratios of K_B/K_S. This model is not sophisticated enough to show the effects of changes in nodal point location. The value of K was calculated using Equations 23-2, and the values of mass and damping were unchanged. The resonance frequencies increase with increasing stiffness ratio. The Synchronous Amplification Factor is affected somewhat by the increase in stiffness, even though the modal damping was not changed.

The right Bode plot displays the vibration behavior for the same three stiffness ratios, using a finite element rotor model. This model is sophisticated enough to show nodal point location effects on modal damping. The change in amplification factor is much more dramatic because the stiffness changes produced changes in the nodal point location and, hence, in the modal damping. The change in modal damping is also revealed by the changes in the slope of the phase data.

For most machines, variable stiffness bearings will allow some adjustment of natural frequencies and mode shapes while the machine is running. The stiffness adjustment will also affect modal damping. Since spring stiffness and modal damping are a part of Dynamic Stiffness, changes in bearing stiffness will cause changes in vibration amplitude and phase (see Figure 23-5). Shifting the resonance away from running speed may cause dramatic changes in amplitude and phase.

At any particular speed, a change in bearing stiffness will change the influence vectors of the machine, which are closely related to the Dynamic Stiffness (Chapter 16). This has important implications, discussed below, for the repeatability needed for balancing.

Changes in bearing stiffness will also have an effect on the unbalance response of the rotor system. As we discussed in Chapters 16 and 18, 1X vibration depends on the amount, angular location, and axial distribution of unbalance along the rotor and *how well that distribution fits the rotor vibration mode*

shape. If that mode shape is changed by adjustments in bearing stiffness, the unchanged unbalance distribution may fit the new mode shape better or worse than the original mode shape, increasing or decreasing the unbalance response of the rotor. Thus, when bearing stiffness is changed, 1X vibration will change for two reasons: the unbalance force distribution fit to new the mode shape and the new Dynamic Stiffness.

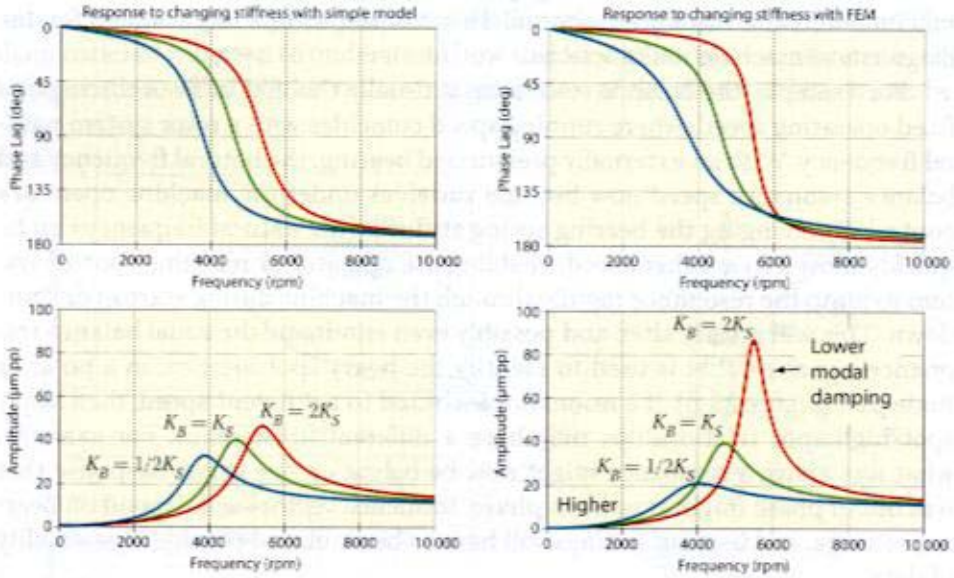


Figure 23-5. Variable stiffness effects on natural frequency and modal damping. The left Bode plot displays the vibration response of the simple rotor model of Chapter 10 for three ratios of K_B/K_S. The value of K is calculated using Equations 29-2, and the values of mass and damping are held constant. The resonance frequencies increase with increasing stiffness ratio. The Synchronous Amplification Factor is affected by the stiffness increase, even though the modal damping is not changed (the phase slope remains constant). The right Bode plot displays the vibration behavior of a more sophisticated finite element rotor model for the same three stiffness ratios. The model includes the effects of nodal point location changes. The change in amplification factor is much more dramatic because the stiffness changes produce changes in the nodal point location and hence in the modal damping.

Diagnostic Implications of Variable Stiffness Bearings

Changes in static spring stiffness are often the primary indicator of various malfunctions: rub increases stiffness, a shaft crack decreases stiffness, and misalignment can do either. Spring stiffness is a component of Dynamic Stiffness. We use the rotor behavior resulting from forces interacting with Dynamic Stiffness to interpret the state of a machine; for example, we deduce the heavy spot location for a particular mode by examining the Bode or polar plot data from a startup or shutdown. This vector data depends on the Dynamic Stiffness of the machine, which, in healthy machines with conventional bearings, remains approximately constant over speed and time.

However, variable stiffness bearings allow changes in spring stiffness that affect Dynamic Stiffness. Thus, changing bearing stiffness during a startup or after a machine reaches running speed alters the Dynamic Stiffness of the machine and the machine response. This has important implications for the diagnosis of machine malfunctions.

For example, the balance resonance is usually thought of as occurring at a fixed operating speed, where running speed coincides with a rotor system natural frequency. With an externally pressurized bearing, the natural frequency and balance resonance speed now become variables under the machine operator's control. By changing the bearing spring stiffness, the natural frequency can be quickly moved to another speed, enabling the operator or machine control system to jump the resonance rapidly through the machine during startup or shutdown. This will greatly alter, and possibly even eliminate, the usual balance resonance signature that is used to identify the heavy spot location in a polar or Bode plot (Figure 23-6). If a resonance is shifted to a different speed, then heavy spot/high spot relationships may have a different appearance. For example, what was above a resonance might now be below, or vice versa. Response that was out of phase might now be in phase. Influence vectors will depend on bearing settings, and bearing settings will have to be similar to provide repeatability of data.

Changes in bearing stiffness can also change the rotor mode shape. A mode associated with low bearing stiffness, for example, a rigid body mode, could be modified by higher bearing stiffness to a bending mode. This change in mode shape could change the match to the unbalance distribution, producing a change in unbalance response. It is possible that the existing unbalance distribution would become a better match to the new mode shape and that the rotor would have to be balanced specifically at particular bearing settings.

Some malfunctions manifest themselves as a self-excited vibration at a system natural frequency. Because of the new, variable nature of the balance resonance, this natural frequency will exist somewhere in a frequency band, which

will depend on the range of bearing settings and their effect on rotor modal stiffness.

Some malfunctions produce vibration at a system natural frequency. Subsynchronous rub vibration and fluid-induced instability whip are examples. It is a common diagnostic practice to look at a cascade plot to see if a malfunction vibration frequency is near a balance resonance frequency. With rub, the vibration frequency is often shifted above the original balance resonance speed because of the stiffening of the rotor due to the rub contact. With variable stiffness bearings, these relationships may be shifted farther than expected, and in a different direction. For example, the frequency of a resonance encountered with a stiff bearing setting may be higher than the frequency of a subsynchronous rub vibration with a softer bearing setting.

Under some circumstances, the bearing will allow the operator to move the natural frequency to a place where the malfunction vibration cannot occur. The diagnostician will need to understand how this kind of variable-parameter bearing operation will affect his or her interpretation of the data and how it can be

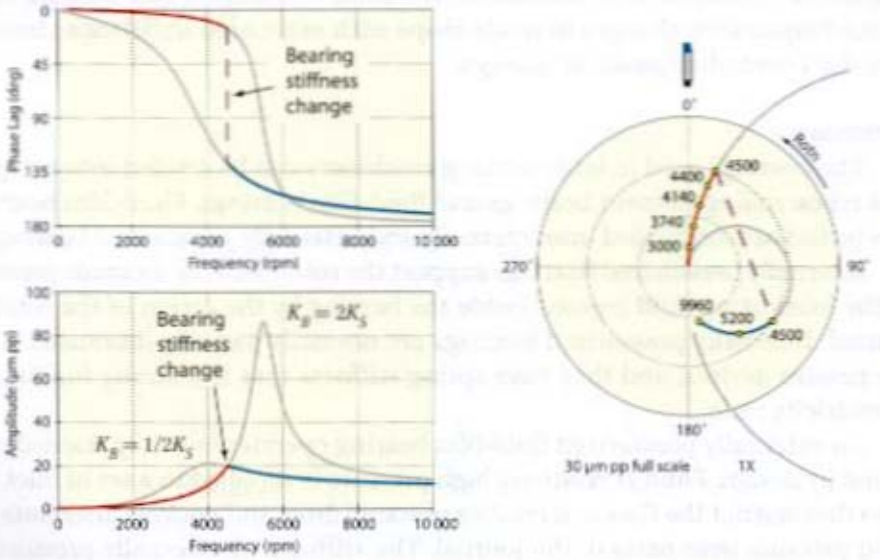


Figure 23-6. Effect of changing bearing stiffness during startup or shutdown. Finite element rotor model data shows the effect of suddenly changing bearing stiffness. The machine is started up with relatively high bearing stiffness (red). The bearing stiffness is suddenly lowered when the machine reaches 4500 rpm, changing the machine response (blue). A sudden phase change (dashed) is associated with the resonance moving through the machine, and the polar plot looks significantly different from the circular shape that would be expected with a normal bearing.

used to suppress unwanted vibration. For example, for ½X rub to occur, a machine must operate at a speed that is more than twice the natural frequency for the no rub condition; an externally pressurized bearing can be used to move the natural frequency associated with the ½X vibration to a higher frequency where this condition is no longer satisfied, eliminating the subsynchronous vibration.

Externally pressurized bearings, because of their high stiffness, will force the journal to operate near the center of the bearing clearance by default. Thus, average shaft centerline plots will show smaller changes during a normal start-up or shutdown. Variable stiffness, when used, will change the average shaft centerline position somewhat. Externally pressurized bearings can also be used to control the average shaft centerline position without significantly changing bearing stiffness. In either case, the appearance of average shaft centerline plots will be different from what would be expected with a conventional bearing.

All of the possible changes caused by variable stiffness will need to be kept in mind by the machinery diagnostician when examining data: Shifts in natural frequencies and resonances, including changes in amplification factors; misalignments between subsynchronous vibration frequencies and balance resonance frequencies; changes in mode shape with associated unbalance changes; and shaft centerline position changes.

Summary

The bearings used in large rotating machinery can be divided into two general types: rolling element bearings and fluid-film bearings. Fluid-film bearings can be further subdivided into internally and externally pressurized bearings.

Internally pressurized bearings support the rotor with the dynamic pressure of the lubricating fluid created inside the bearing by the action of the rotating journal. Internally pressurized bearings are normally partially lubricated. They are passive devices, and they have spring stiffness that is a strong function of eccentricity ratio.

An externally pressurized fluid-film bearing operates in a fully flooded condition by design. Fluid at relatively high pressure is supplied to a set of inlet orifices that restrict the flow and create a pressure drop, and pockets distribute the fluid pressure over parts of the journal. The stiffness of externally pressurized bearings includes a combination of static and hydrodynamic effects. The static component of stiffness can be controlled by varying the supply pressure; higher pressure produces higher stiffness.

The externally pressurized bearing also provides the opportunity to put bearing stiffness under either operator or automatic control. Control is accomplished by statically or dynamically adjusting pocket pressures in groups or indi-

vidually. Changing bearing stiffness changes how the rotor system responds to the static and dynamic forces that exist in the machine. The effects of variable bearing stiffness depend on the ratio of bearing stiffness to shaft bending stiffness.

Variable stiffness can produce shifts in natural frequencies and resonances; changes in amplification factors; misalignments between subsynchronous vibration frequencies and balance resonance frequencies; changes in mode shape with associated unbalance changes; and minor shaft centerline position changes.

References

1. American Petroleum Institute, *Tutorial on the API Standard Paragraphs Covering Rotor Dynamics and Balancing: An Introduction to Lateral Critical and Train Torsional Analysis and Rotor Balancing*, API Publication 684 (Washington, D.C.: American Petroleum Institute, 1996).

Chapter 24

Shaft Cracks

A SHAFT CRACK IS A SLOWLY GROWING FRACTURE of the rotor. If undetected in an operating machine, a crack (also called a fatigue crack) will grow until the remaining, reduced cross section of the rotor is unable to withstand the dynamic loads that are applied to it. When this happens, the rotor will fail in a fast brittle fracture mode. The sudden failure will release the large amount of energy that is stored in the rotating system, and the rotor will fly apart. Shaft fractures have caused machine parts to penetrate the machine casing and even penetrate building walls. Damage due to this kind of failure is catastrophic and can cause serious injury or death to anyone unfortunate enough to be standing near the machine at the moment of failure. Obviously, shaft crack detection is a very serious matter, and machines that are suspected of having a crack must be treated with the utmost respect.

In this chapter, we will start with a description of how shaft cracks start, propagate, and affect the shaft stiffness and rotor dynamic response of the machine. We will discuss these effects and develop two important rules for shaft crack detection. After discussing design and operation considerations for minimizing the risk of crack development, we will provide monitoring recommendations that can help in the early detection of cracks.

Crack Initiation, Growth, and Fracture

Rotors are subject to periodically changing, or cyclic, stresses as they rotate. If a rotor orbits about the center of the rotor system in pure 1X, circular precession, no location in the rotor will see a change in stress. However, if the rotor is deflected in bending from the center of the rotor system (the rotor sags, for example), then the rotor will see a 1X variation in stress, and the maximum

stress will occur at the outer surface. If the 1X orbit is elliptical (as it typically is), then the rotor sees 2X stress cycling, whether or not the rotor is deflected (Figure 24-1). Thus, even under normal, 1X operation, real rotors live in a complicated stress environment that contains a mixture of 1X and 2X stress cycling. Any sub- or supersynchronous vibration that may be present will produce an additional complicated pattern of cyclic stresses in the rotor.

Cracks are initiated in the shaft in regions of high local stress. Shafts are subjected to large-scale stresses due to static and dynamic bending and torsional twisting, static radial loads, constrained thermal bows, thermal shock, and residual stresses from heat treatment, welding, or machining operations. These larger-scale stresses can be concentrated by geometric factors such as step changes in shaft diameter, shrink fits, keyways, drilled holes, threads, or other discontinuities. Further concentration can occur at the microstructure level, where surface machining imperfections, chemical surface damage and corrosion, and material discontinuities (such as produced by voids, slag inclusions, or chemical impurities) can produce high, local stress concentrations. All of these stresses combine to produce a local stress field that changes periodically. The end result can produce a small, local region where stresses exceed the maximum that the material can withstand, and a microcrack will form in the material.

Rotor bending tends to produce the highest stresses at the outer surface of the shaft; rotor cracks often start at or near the outer surface. Cracks can also be initiated on the surface of steam turbine rotors as a result of quenching and thermal shock due to boiler upsets. However, rapid heating of a rotor can produce high tensile stresses in the center or at the bore hole surface of a rotor (See Chapter 19). Sometimes, because of chemical or other processing problems in the rotor billet, a microcrack may already exist inside the rotor before it is put into service.

If the cyclic stresses are sufficiently high, the leading edge of the crack (the crack tip) will slowly propagate so that the plane of the crack is perpendicular to the orientation of the tensile stress field. The orientation of this stress field is affected by the type of stress (bending or torsional) and by geometric factors. If a rotor is subjected only to simple bending stresses, then the stress field will be oriented along the long axis of the rotor, and the crack will propagate directly into and across the rotor section, forming a *transverse crack* (Figure 24-2, top).

Pure torsional stress will produce a tensile stress field that is oriented at 45° relative to the shaft axis. A crack in this stress field will propagate into the rotor and tend to form a spiral on the shaft surface (Figure 24-2, bottom). In most rotor systems, the stress field contains a mixture of bending and torsional stress. Bending stress is usually the dominant component; thus, the crack will usually

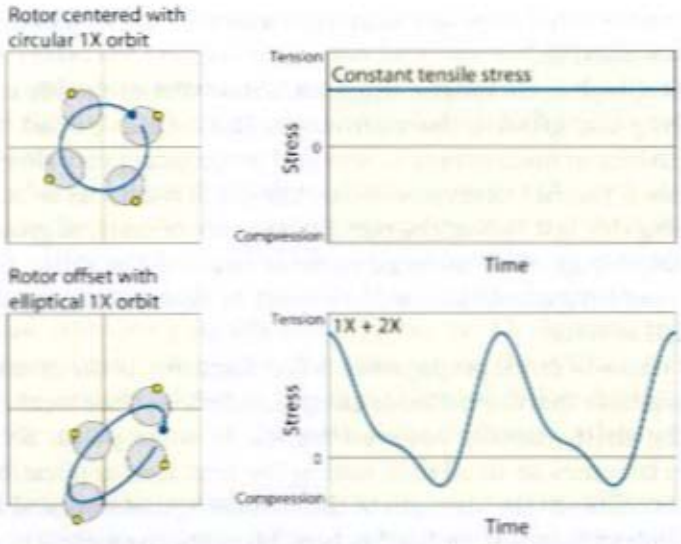


Figure 24-1. Stress cycling in a rotating rotor. A circular, 1X orbit, centered on the rotor system axis, produces a constant stress level (top). When the rotor is subjected to a static radial load, and is deflected in bending, the constant stress due to the circular orbit is added to any alternating stress due to the static radial load deflection. If the orbit is elliptical, an additional 2X component of stress may appear (bottom). Sub- and supersynchronous vibration frequencies further complicate the stress picture.

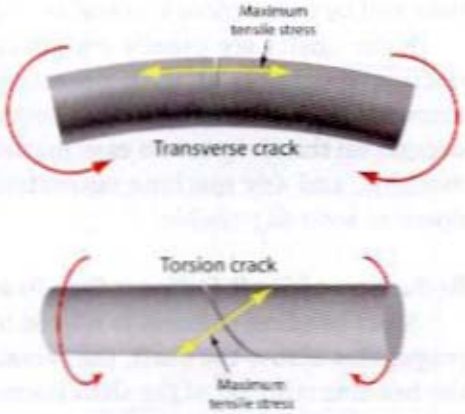


Figure 24-2. Transverse (top) and torsional (bottom) cracks. A transverse crack that results from bending stress in the shaft propagates directly into the shaft. A crack that results from torsional stress forms a spiral at 45° to the long axis of the shaft. Most shafts contain a mixture of bending and torsion stress. The local stress field at the crack tip, which can be influenced by local geometry, determines crack propagation direction. The crack tip propagates perpendicular to the direction of the tensile stress field (yellow).

propagate into the rotor more or less as a transverse crack. However, other crack geometries are possible.

In a typical high-cycle fatigue situation, about 90% of the life of the part is spent initiating and growing the microcrack. During this period, the crack is either nonexistent or microscopic in size and propagating very slowly.

As the crack reaches measurable size, the crack enters its second phase of growth. During this last 10% of the part life, the rate of crack of growth increases rapidly, but enough shaft material remains to resist the static and dynamic loads. In an operating machine, the crack must be detected during this relatively short period of time.

The last phase of crack propagation is fast fracture. As the crack grows, less material is available to transmit loads along the shaft, and the local stress across the remaining shaft material becomes higher. At some point, the remaining cross section becomes so small that, during the next load application, the local stress intensity exceeds the strength of the remaining material, and the remaining section undergoes a fast, brittle fracture, breaking the rotor.

The *fracture toughness* is a measure of the material's resistance to fast fracture and is a function of the alloy, heat treatment, the material temperature, and the rate of loading. A material with high fracture toughness will be able to withstand a higher stress intensity without fracture than a material with low fracture toughness. Also, the fracture toughness of steel is higher when the rotor temperature is above the ductile/brittle transition (nil ductility) temperature, which, for most rotor steels, is near or above room temperature. Cold rotors may be below the ductile/brittle transition and have low fracture toughness. This is one reason why steam turbine rotors must be heated very carefully; a rotor that is below its ductile/brittle transition temperature will have low fracture toughness and be much more vulnerable to cracking due to thermal shock.

Rotor shafts are usually manufactured out of materials that possess high fracture toughness at their operating temperature. Rotor shaft cracks have exceeded 90% of the shaft cross sectional area before final fracture, but don't depend on this. It is not an easy matter to determine crack size in an operating machine, and any machine suspected of having a shaft crack should be shut down as soon as possible.

Reduction of Shaft Stiffness Due To a Crack

Shaft bending stiffness is related to the shaft cross-section area. As a crack propagates across the shaft, the remaining cross section becomes smaller, and the bending stiffness of the shaft decreases. For a given radial force on the rotor, if the stiffness decreases, the rotor deflection increases.

This reduction in shaft stiffness causes the rotor to bow more in response to

a static or dynamic load, such as a rotating unbalance. Depending on the orientation of the crack, the original heavy spot location, and the relationship of running speed relative to resonance, the rotor could bow in virtually any direction. Thus, it is likely that crack propagation will produce a change in *both* 1X amplitude and phase.

The 1X vibration vector is likely to change slowly in the early stages of crack growth. But, as the crack continues to propagate across the shaft, further reducing the shaft stiffness, 1X amplitude is likely to increase more rapidly. 1X phase can also change dramatically in the later stages of crack growth.

Rotor bow will have a significant influence on 1X slow roll vectors. As the crack propagates across the shaft, the slow roll vector is likely to change dramatically from one shutdown to the next. High vibration during a resonance may cause the rotor to yield at the weak point and cause a dramatic change in 1X response.

Natural frequencies are affected by changes in rotor stiffness. As a crack propagates across the rotor, a balance resonance speed may shift downward (Figure 24-3). Depending on the rotor mode shape and the location of the crack, some resonances (modes) could be affected more than others. If a crack devel-

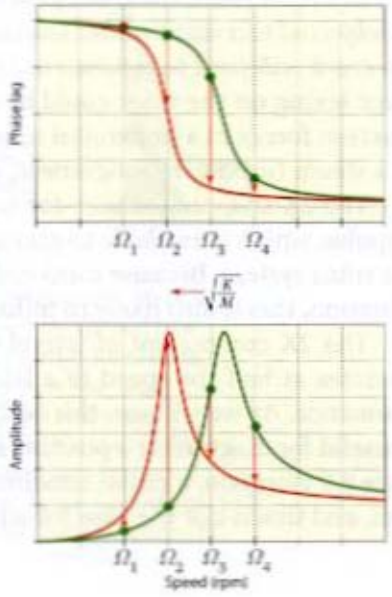


Figure 24-3. Effect of shaft crack on 1X vibration. A shaft crack reduces the spring stiffness of the system, K, shifting the resonance to a lower speed. The amplitude and phase of steady state, 1X vibration changes from the "no crack" condition (green) to the cracked condition (red). The amplitude of vibration increases or decreases, depending on the relationship of operating speed to the resonance frequency. Rotor bow due to a crack may also change the magnitude and location of the heavy spot, adding to the effect shown here.

ops in an area near an inflection point of the rotor (where no significant bending exists for that mode), then that mode may see little or no change in resonance frequency. A crack in an area of significant curvature (bending) for a particular mode is likely to have a larger effect on that mode's resonance frequency. Thus, rotor mode shapes may change, depending on the location of the crack and how it affects the stiffness distribution of the shaft.

Shaft Asymmetry and 2X Vibration

As a transverse crack propagates across the rotor, the remaining, uncracked material will become increasingly asymmetric (Figure 24-4). This shape produces a rotating, anisotropic stiffness in the rotor.

This can be demonstrated with a flexible ruler (Figure 24-5). The ruler is a beam with an asymmetric cross section, and the stiffness in one direction is very much larger than the stiffness in the perpendicular direction. If the ruler is supported horizontally at its ends, and a steady, unidirectional force is applied at midspan, the resulting deflection will be very different for different orientations of the ruler; the deflection will be maximum when the force is applied to the long side, and minimum when the force is applied to the short side. Furthermore, when the ruler is between these orientations, the force produces a response that has a quadrature component. As the ruler rotates, the weak/strong/weak transition will produce a "snapping" action two times per revolution: a 2X vibration component.

In a rotor shaft, the crack produces a similar anisotropic stiffness. *If the rotor is subjected to a static radial load that produces shaft bending near a crack, then the crack will tend to produce a 2X component in the rotor response.* The radial force acting on the rotor could be due to gravity in a horizontal machine, fluid reaction forces in a horizontal or vertical machine, partial arc steam admission in a steam turbine, misalignment, etc.

The 2X snapping action due to the crack also produces a strong 2X torque impulse, which is available to excite any torsional resonance at that frequency in the rotor system. Because torsional vibration can easily cross couple into lateral vibration, this is also likely to influence the 2X lateral vibration.

The 2X component of lateral vibration can be greatly amplified if a rotor operates at half the speed of a lateral or (because of cross coupling) torsional resonance. As we will see, this behavior can provide a piece of information that is useful for diagnosing a possible shaft crack. But remember, for a crack to produce 2X vibration, a radial, unidirectional load must be present in the rotor system, and this is not the case for all machines.

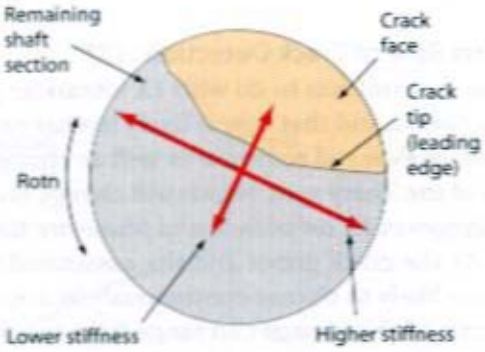


Figure 24-4. Crack growth and shaft asymmetry. As a crack grows, the remaining shaft cross section becomes asymmetric. This produces an anisotropic shaft stiffness (red arrows), which rotates with the rotor.

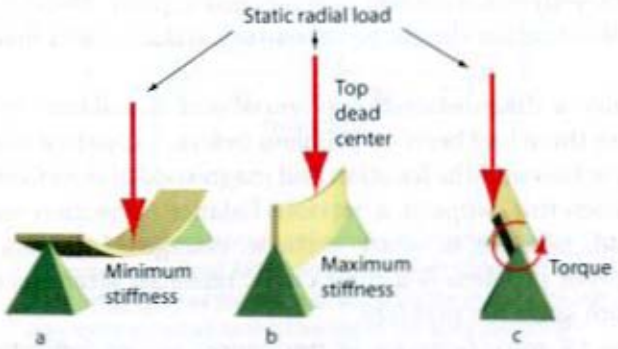


Figure 24-5. Shaft crack, anisotropic stiffness, and 2X vibration. Identical, vertical loads (red arrows) are applied to a simply supported, asymmetric beam (ruler) in three positions. In (a), the ruler has minimum stiffness and largest deflection. In (b), it has maximum stiffness and smallest deflection at top dead center. In (c), when the load is applied between the two stiffness axes, the ruler has a horizontal (quadrature) component of deflection. As the ruler rotates, a snapping motion will be seen two times per revolution (2X).

When it occurs, the 2X vibration due to a crack is likely to cause an increase in stress cycling, promoting more rapid crack growth and hastening the failure of the rotor.

The First Rule of Crack Detection (1X)

The first rule has to do with 1X vibration: *If a rotor is cracked, it is very likely to be bowed*, and that bow is likely to change over time. A change in rotor bow will change slow roll response as well as change the effective location and magnitude of the heavy spot, which will change the 1X rotor response.

Changes in 1X amplitude and phase are the best primary indicator of a shaft crack. As the crack grows and the associated bow develops, 1X amplitude and phase are likely to change continuously in a nonrepeating way (Figure 24-6). The time scale of this change can range from months to weeks in the early stages of crack growth, to weeks to days as the rotor begins to seriously weaken, and to hours as the rotor nears catastrophic failure.

As the crack propagates and the bow of the rotor changes, the amplitude and phase of the 1X slow roll vector will change.

As failure nears, 1X vibration amplitude will usually increase rapidly. At this point, 1X vibration is likely to be the dominant source of vibration in the system, so direct (unfiltered) vibration will also increase rapidly. Thus, steady increases in peak-to-peak vibration should be taken very seriously and investigated thoroughly.

Occasionally, a diagnostician may encounter a machine with a "balance problem" where there had been no problem before. Growth of a shaft crack will change the rotor bow and the location and magnitude of the effective heavy spot of the rotor. When this happens, a previous balance correction may be rendered ineffective, and, because of shaft stiffness changes, influence vectors may change. If the root problem is a shaft crack, repeated attempts to balance the machine will not solve the problem.

Changes in 1X rotor behavior in resonances are an indication that something has changed in the rotor system. A significant downward shift in a balance resonance speed is a clear indication that the stiffness of the rotor system has decreased. Always ask yourself why this has happened. A weakened shaft due to a crack is a possibility.

1X vibration is usually very sensitive to the presence of a shaft crack because of the relationship between crack growth and rotor bow, and it can provide significant early warning of a crack.

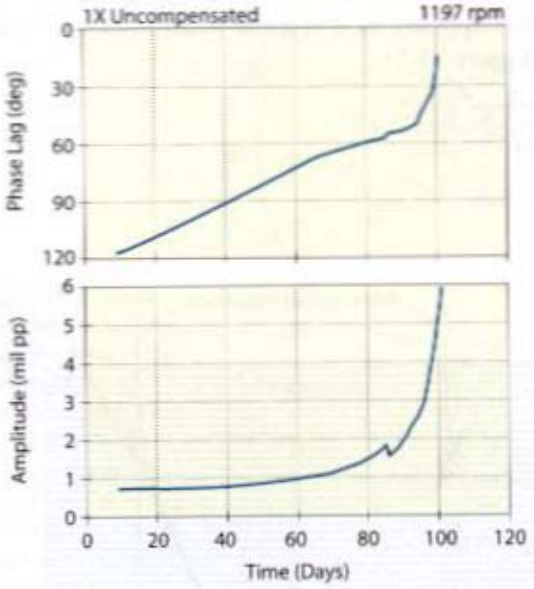


Figure 24-6. Development of a crack in a vertical pump. The 1X, uncompensated APHT plot shows the changes in amplitude and phase as the crack develops. Note that the phase was more sensitive to the change in the early period of crack growth. Vibration amplitude began to increase very rapidly in the last 20 days. The data ends when the pump was shut down for inspection; a 120° crack was found.

The Second Rule of Crack Detection (2X)

The second rule has to do with 2X vibration: *If a rotor with a crack has a steady, unidirectional radial load, then a strong 2X response may appear when the rotor is turning at half of any balance resonance speed.* The 2X snapping action of the rotor produces lateral and torsional impulses in the system. Because these impulses occur twice per revolution, the rotor will respond at the 2X frequency. If a resonance exists at twice running speed, then the 2X vibration will be amplified. This appears as a vertical relationship on a half or full spectrum cascade plot (Figure 24-7).

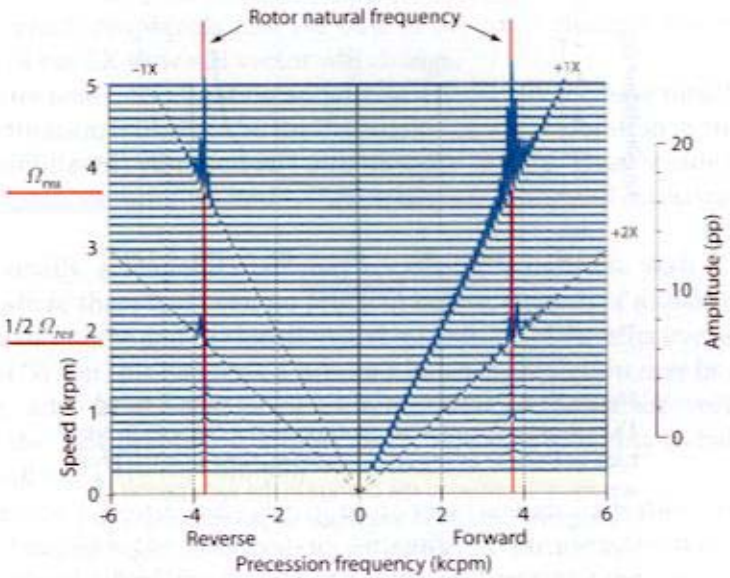


Figure 24-7. Amplification of 2X at half a resonance speed. The data is from an experimental machine with a simulated crack and a static radial load. When the rotor speed is one-half the balance resonance speed, Ω_{res}, the 2X response excites the natural frequency of the rotor, amplifying the 2X vibration. This forms a vertical relationship on the full spectrum cascade plot.

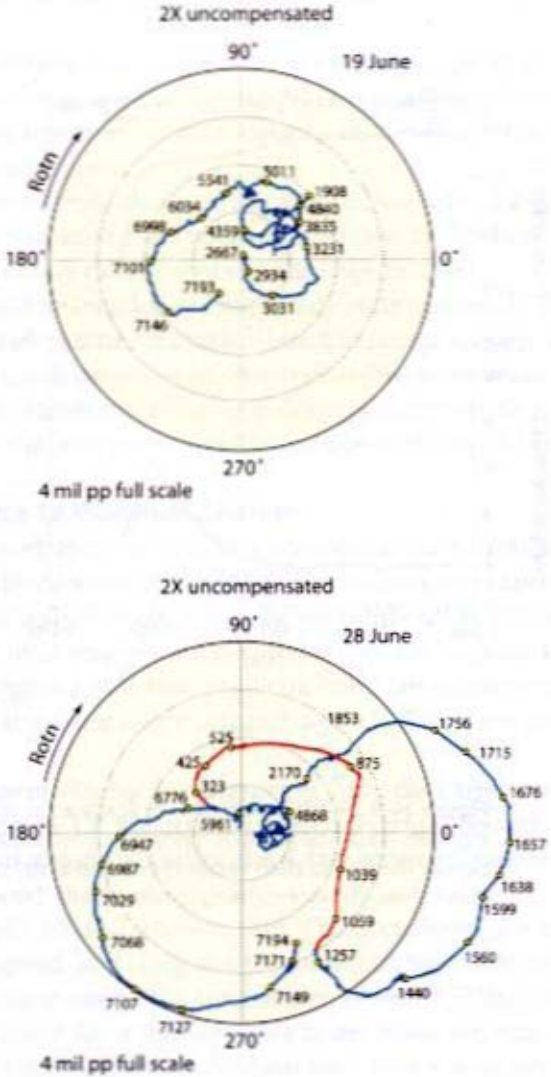


Figure 24-8. 2X polar plots showing evidence of a shaft crack. The dual flow LP turbine had markedly different behavior on 28 June (bottom) compared to its behavior nine days earlier (top). While the earlier data looks normal, the later data shows large changes in amplitude and anomalous phase behavior; the phase lag decreases with increasing speed (red).

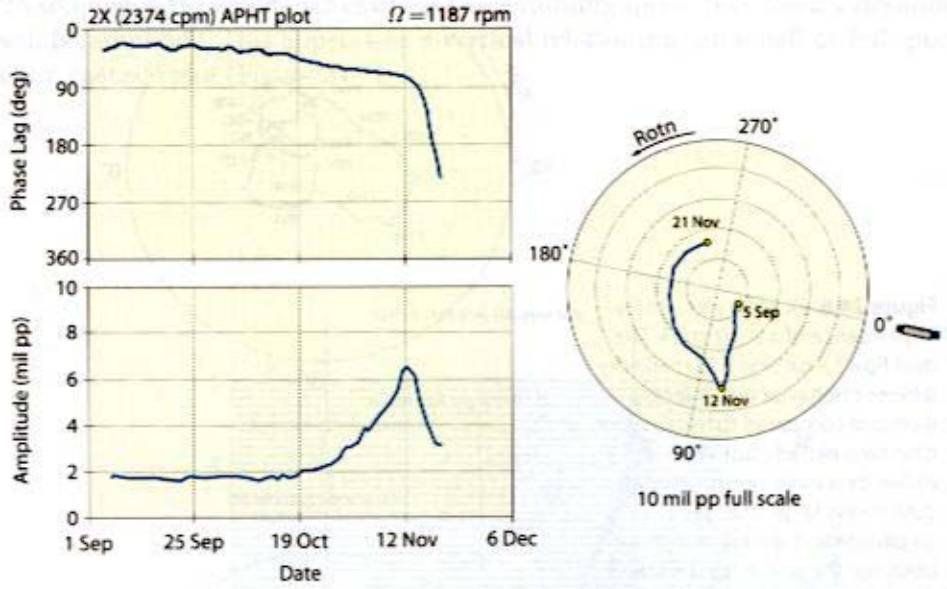


Figure 24-9. 2X APHT plots of a vertical reactor-coolant pump with a developing shaft crack. The pump is running at a constant speed during this entire period. The 2X vibration vector changes dramatically as a resonance moves down through twice running speed.

Many machines pass through such a 2X speed relationship during startup or shutdown. It is less likely that such a relationship will exist at normal operating speed (although this has happened). For this reason, and because a unidirectional radial load must be present, a crack may not produce significant 2X vibration at running speed.

In fact, in most cases, 2X vibration does not appear at operating speed. For this reason, *2X vibration should never be used as the only tool* for crack detection. 2X vibration data should be used together with 1X data to ensure the earliest and most reliable detection of a shaft crack.

Like 1X vibration, 2X vibration amplitude and/or phase is likely to change as the crack propagates through the rotor. Startup and shutdown 2X Bode and polar plots should be examined for any evidence of change (Figure 24-8).

2X amplitude and phase should be trended during steady state operation. In one case, a reactor coolant pump developed a crack while the pump was operating at a constant speed. As the crack propagated, the shaft stiffness decreased so much that a resonance that was above twice the operating speed moved down through the 2X frequency before the pump was finally shut down (Figure 24-9).

Other Malfunctions That Produce 1X Vibration Changes

A loose bearing support or soft foot can cause a change in 1X vibration. Usually, but not always, this manifests as an increase in 1X vibration amplitude. Because it mimics the behavior of a shaft crack, it can be very difficult to determine the root cause. An increase in casing vibration with little or no increase in shaft relative vibration might suggest a soft foot problem, while an increase in shaft relative with little increase in casing might suggest a crack. But, there are no firm rules here.

Thermal growth and subsequent changes in alignment can affect the bearing stiffness and produce changes in 1X vibration, as can a thermal bow of a rotor. These changes in vibration should stabilize once the machine reaches thermal equilibrium at steady speed and load.

Rub can cause changes in both 1X and 2X vibration. These changes can be sudden, occurring at operating speed or during startup or shutdown. Rub can disappear if the parts in contact wear away (this can happen in seals). If the rub is severe, contact may be maintained for a considerable time. However, rub is not as likely to produce a steadily increasing 1X vibration level over a long period of time.

A loose rotating part can also produce changes in 1X response. If a part moves to a different angular or axial position on the rotor, the resulting total unbalance of the rotor is likely to change, and the 1X amplitude and phase will change accordingly. Loose parts can shift occasionally, producing stepwise

changes in 1X response, or they can shift continuously, producing a continuously changing response. Continuously moving parts will produce cyclic behavior on a polar or APHT plot. A loose part is not likely to produce a steady, long-term increase in 1X vibration amplitude.

Shifting debris or liquid inside a rotor can produce significant changes in the heavy spot location in a machine. This will produce corresponding changes in 1X vibration response and cause a machine to go out of balance frequently.

A locked gear coupling can also produce a sudden step change in 1X vibration.

The key to crack identification is to realize that a developing crack is likely to produce a steady and accelerating increase in 1X vibration amplitude over time as the shaft stiffness decreases. While some malfunctions will produce periodic changes in 1X vibration amplitude and/or phase, shaft cracks will tend to produce nonrepeating patterns on polar and APHT trend plots, and ever higher levels of 1X vibration amplitude.

Other Malfunctions That Produce 2X Vibration

Nonlinearities in rotor system stiffness can cause harmonics (2X, 3X, etc.) of running speed to appear in spectra. Nonlinear stiffness can be caused by high shaft eccentricity ratios in fluid-film bearings or by rub. Misalignment, high radial load, and coupling problems can produce 2X vibration.

If any source of 2X vibration exists in a machine, it will be available to excite a resonance at half of a balance resonance speed. Thus, the presence of 2X at half a resonance, while suspicious, is not, in and of itself, confirmation of a crack.

Design and Operating Recommendations

Shaft cracks usually result from a combination of design shortcomings, improper machine installation, or poor operating practice. At the design level, any stress concentration can lead to trouble if it is not properly accounted for in the fatigue life of the design. Keyways, notches, drilled holes, threads, and step changes in shaft diameter are all capable of producing large stress concentrations in the shaft. Shrink fits produce high tensile stresses at the surface of the rotor in the vicinity of the fit. Retaining-pin holes and grooves near or under shrink fits can lead to very high local stresses because of the combination of shrink fit stresses and geometric stress concentrations.

Estimated vibration mode shapes can be helpful in the prediction of deflections and stresses, but the dynamic behavior of individual machines can change when several machines are coupled together. For this reason, the dynamic characteristics of the entire coupled machine train should be taken into account when estimating mode shapes, deflections, and stresses.

Good machine train alignment will help minimize the chance of a shaft crack. Severe misalignment between adjacent machines in a train can cause bending of the rotor between machines. This can result in high, 1X cyclic stresses that may exceed the limits of the design.

Startup and shutdown of high and low temperature machines should be done carefully to minimize the possibility of thermal shock and microcracking of the rotor surface or interior. A boiler upset can cause low quality steam to contact the hot steam turbine shaft surface and produce a severe quench and thermal shock. The low temperature condensate of steam turbines operated in cold climates may cause thermal shock to the LP turbine rotor. These types of thermal shock produce tensile stresses that can cause a microcrack on the rotor surface. Hard rubs have also caused local microcracking, due to extreme frictional heating, local compressive yielding, and subsequent cooling.

Vibration trends are important to diagnosing a crack, but vibration amplitude is a poor indicator of stress levels important to crack generation; low vibration can be due to high radial loads and severe misalignment, both associated with high stress levels. High vibration can produce higher cyclic stresses than the designer may have anticipated in the design. High, dynamically induced stresses have combined with geometric factors to produce catastrophic cracks.

Operation of machines on or near a resonance should be avoided. It can produce high sensitivity to unbalance and high vibration, with the potential for high stresses. Operation of a machine at half of any lateral or torsional resonance should also be avoided. If 2X vibration is present, it is likely to be amplified by the resonance at twice running speed. The 2X vibration will produce cyclic stresses and increase the risk of a shaft crack. 2X Bode or polar plots should be examined for any evidence of a resonance at or near twice operating speed.

Monitoring Recommendations

During a startup, a steam turbine that encountered high 1X vibration was shut down. The problem was assumed to be an unbalance problem. Successive attempts to restart the unit encountered increasing 1X vibration levels and the appearance of some 2X vibration. A decision was made to disassemble the unit, and a partial shaft crack was found in the rotor. Subsequent examination of all of the vibration data showed that there had been, in addition to the high 1X vibration amplitude, a significant 1X phase shift of nearly 40° during the attempted startup. But the phase had not been monitored, even though the equipment had been available to do so. Remember: if you don't look for signs of a shaft crack, you may not detect one in time to prevent a catastrophe.

Unfortunately, it is rarely possible to look at a single piece of data and positively diagnose a shaft crack. Many times it is difficult to make a diagnosis with

all the data at hand. However, if you apply the following monitoring recommendations, you will have a much better chance of detecting a shaft crack in a timely manner.

1. At operating speed, monitor and trend direct (unfiltered) vibration levels and 1X and 2X amplitude and phase. The trend can be as simple as a list of hand-logged data (1X and 2X amplitude and phase) or, better, as computer-generated APHT and polar plots. In addition, 1X and 2X acceptance regions should be defined so significant changes in vibration can be detected. Different acceptance regions may need to be established for different load conditions. Steady state monitoring of this kind has provided warning as early as a 25% crack.

2. Every time a machine is started up or shut down, data (in the form of 1X and 2X Bode and polar plots) should be taken and compared to earlier data. Look for significant changes in 1X and 2X behavior through resonances, a decrease in one or more resonance speeds, or other abnormal behavior. Significant changes in startup or shutdown machine response should be a cause for concern and investigated. This kind of monitoring has provided warning as early as a 20% crack.

3. 1X slow roll vectors should be logged and compared to earlier data. As a crack propagates, the bow of the rotor is likely to change, and the amplitude and phase of the 1X slow roll vectors will also change.

While there are no guarantees, careful and thoughtful machine monitoring gives you a very good chance of detecting a crack before a catastrophic failure occurs.

Summary

A shaft crack is a slowly growing fatigue fracture of the rotor. When the reduced cross section of the rotor is unable to withstand the static or dynamic loads that are applied to it, the rotor fails in a fast brittle fracture mode, a catastrophic failure.

Cyclic stresses cause shaft cracks to start in high local stress regions of the shaft, such as near step changes in shaft diameter, shrink fits, keyways, drilled holes, or other discontinuities. Cracks start at the microstructure level, where

surface machining imperfections, chemical surface damage, or material discontinuities (produced by voids, slag inclusions, or chemical impurities) can produce high, local stress concentrations.

Once initiated, the crack tip will slowly propagate in a direction perpendicular to the orientation of the tensile stress field that exists at the crack tip. If a rotor is subjected only to simple bending stresses, then the stress field will be oriented along the long axis of the rotor, and the crack will propagate directly into the rotor, a transverse crack. Torsional stress will produce a spiral crack that is oriented at 45° relative to the shaft axis.

Shaft cracks reduce the bending stiffness of the shaft, due to the reduced available cross-section area, and the stiffness often becomes anisotropic. These characteristics lead to the two rules of crack detection:

1. *If a shaft is cracked, then it is most likely bowed.* This produces 1X vibration behavior that changes as the crack evolves. 1X vibration changes are the most reliable indicator of a shaft crack.

2. *If a cracked shaft is subjected to a static radial load and is operated at one-half of a resonance, then 2X vibration can appear.* This effect is due to the crack-induced, rotating anisotropic stiffness of the shaft.

Both 1X and 2X vibration can be caused by a large number of other malfunctions that must be excluded before making a diagnosis.

Shaft cracks usually result from a combination of design shortcomings, improper machine installation, or poor operating practice. There are several things that can contribute to the initiation and propagation of a shaft crack: stress concentrations, such as keyways, notches, drilled holes, and step changes in shaft diameter; shrink fits; retaining-pin holes and grooves; misalignment; high radial load; thermal shock; high vibration; and operation of machines on or near resonances, or at half of any lateral or torsional resonance.

For crack detection, 1X and 2X vibration vectors should be trended at slow roll, during startup and shutdown, and at steady state operation, with Bode, polar, and APHT plots.

Case Histories

Chapter 25

High Vibration in a Syngas Compressor Train

THIS CASE HISTORY PRESENTS AN EXAMPLE of an "unbalance" problem that wasn't unbalance. As is often the case, as the problem was investigated, the real root cause turned out to be something completely unexpected. Transient data provided the additional information needed to correctly diagnose the problem.

A new methanol plant was being commissioned on an island in the Caribbean. The plant produced liquid methanol from local natural gas, and the process depended upon a synthesis gas (syngas) compressor train, which moved partially reformed product to the plant's final process reactor. The machine train consisted of a single-case, HP/LP steam turbine and tandem LP and HP barrel compressors (Figure 25-1). Table 24-1 lists the nameplate and operating data for this machine train.

The turbine and LP compressor were connected with a shim-pack, spool-type coupling that was not the coupling that had shipped with the unit. During final assembly, mechanics had discovered that, because of a foundation error, the original coupling was too short. On short notice, the coupling manufacturer sent another coupling that had been high-speed balanced at the factory.

Plant construction engineers tested the compressor train at low speed, then continued low-speed operation while the entire plant was started up. It took several days for the reactor beds upstream of the compressor to reach operating temperatures and pressures.

Production began, and operators started to bring the compressor up to its 10 480 rpm operating speed. The operators noticed that, as speed increased, so did vibration at bearing 3 on the LP compressor. On the vibration monitor, direct vibration increased dramatically as the compressor neared its operating speed,